

Country Risk Premiums

To estimate the equity risk premium for a country, I start with a mature market. Use the look up table in the next worksheet, to look up the statistics for a

Step 1: Estimating mature market risk premium

To estimate the mature market risk premium, I start with the implied equity risk premium. Link to site: <http://www.damodaran.com>

Historical monthly ERP: <https://pages.stern.nyu.edu/~adamodar/pc/implpr>

Until the January 2025 update, I used the S&P's equity risk premium as the

On May 16, 2025, Moody's downgraded the US from Aaa to Aa1. To get the

Mature market premium = ERP for the S&P 500 minus (Default spread for

In January 2026, for example, when the implied ERP for the S&P 500 was

Mature market premium = $4.61\% - (0.23\% \times 1.52) = 4.23\%$

Step 2: Estimate the default spread for the country in question. I

Moody's ratings: <http://www.moody's> (You will have to register, but

Ratings to spreads: Based upon my estimates of typical spreads for each

CDS spreads: Bloomberg

If you cannot find a country on this list, it is because that country does not

Step 3: Convert the default spread into a country risk premium

With sovereign ratings default spreads, you have two choices:

Choice 1: Use the default spread as the measure of the additional country

Choice 2: Scale the default spread up to reflect the higher risk of equity in

With CDS spreads, I compute the base number in two steps

Substep 1: Since the base equity premium is for a mature market (assume

Substep 2: I apply the scaling factor that you chose for the default spread. I

Step 4: Compute a total equity risk premium

Add the mature market premium from step 1 to the country risk premium

Step 5: Compute regional averages and regional weighted averages

For the regional averages, I use a simple average of the total and country

For the weighted averages, I use the World Bank GDP estimates from the

If you are interested in a fuller explanation of these concepts, try these re

My paper on equity risk premiums:

My paper on country risk premiums:

Watch my lectures on country risk premiums:

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=344444

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=344444

<https://www.youtube.com/watch?v=344444>

<https://www.youtube.com/watch?v=344444>

market premium and add an additional country risk premium, based upon the risk in individual country or region.

ity risk premium for the S&P 500. To see the latest estimate for this number, go to

[em/ERPbymonth.xls](#)

e mature market premium, drawing on the Moody's Aaa rating for the United States. The mature market premium, I now subtract the US's country risk premium (based on US sovereign rating $\times$ Relative Equity Volatility)

4.59% (over the adjusted dollar riskfree rate), the US had a rating of Aa1 and the

offer two choices, one based upon the local currency sovereign rating if it is free. Look under sovereign ratings)

ich ratings class. I compute these by averaging CDS spreads and sovereign US\$ I

if it have a sovereign rating or a sovereign CDS spread. Try the PRS worksheet in the

risk premium. To make this choice, go into the ERP worksheet and set cell E5 to the market, relative to the default spread. I used the ratio of the S&P Emerging

ed to be default free), I net out the CDS spread for Switzerland (the lowest sovereign rating) to this number to get a country risk premium. The default scaling is set at the r

from step 3 to get a total equity risk premium.

yes

risk premiums by region
most recent year.

<https://data.worldbank.org/indicator/NY.GDP.MKTP.CD>

ferences:

[l3/papers.cfm?abstract_id=4398884](#)

[l3/papers.cfm?abstract_id=4509578](#)

[e.com/watch?v=aIRPvY2SQ94](#)

[e.com/watch?v=D3IGn6tH03c](#)

of the country in question.

to my website and you can download the excel spreadsheet containing the implied

tes.

on its rating and the relative equity market volatility) from the S&P's eRP

default spread for that rating was 0.23% and the relative equity market volatility

for the country from Moody's and the other is the CDS spread for the country

bond spreads by ratings class, at the start of every year.

this spreadsheet for an alternate estimate.

1.00.

Market Equity Index std deviation to the iShares Emerging Market Bond Index std

sign CDS) from each country's CDS to get a net spread.

my most recent year's estimate, but you can change it to 1, if you would

ed premium

y was 1.52

entry (if one exists)

andard deviation

<i>Mature Marlet Premium</i>
<i>US ERP</i>
<i>Mature market ERP</i>
<i>Sovereign Ratings</i>
<i>Default Spreads</i>
<i>Relative Equity Market Volatility</i>
Bottom Line

Update Notes

The starting point for the mature market premiums is the implied equity risk premium for the S&P 500. You can review the calculation in this spreadsheet. (<https://pages.stern.nyu.edu/~adamodar/pc/implprem/ERPJan26.xlsx>). I net out a riskfree rate in US dollars, which is the treasury bond rate adjusted for the default spread for the US, to get an equity risk premium for the US. I then back out the US default spread to get a mature market premium.

ERP for US = Implied Expected Return on US stocks - (US 10-yr T.Bond - Default spread for US)

Mature market ERP = ERP for US - Default spread for US

The ratings (Moody's and S&P) have been updated to reflect the most recent ratings. (See **Ratings Worksheet**)

The change in default spreads is computed by looking at percentage changes in sovereign CDS spreads over the period. (See **Default Spreads for Ratings** worksheet)

The biggest change in this update is in this measure. For the last few years, I have used the coefficient of variation in emerging market bond yields and the standard deviation of returns in emerging market equities. In this iteration, I have replaced the former with the standard deviation in emerging market sovereign bond ETF returns, thus creating more consistency in the calculations. While this change does not create a substantive shift in the numbers in this iteration, I think it is a more solid basis for computing this value in future ones.

The combination of a lower base mature market premium and lower sovereign default spreads has led to lower equity risk premiums across the board, a climb down from historically high numbers at the start of the year.

To look up the equity risk premium for a country, use this

Country	Turkey	<i>If you cannot find a country, use the sovereign rating of the country's local currency.</i>
Moody's sovereign rating	Ba3	Local currency
S&P sovereign rating	BB-	Local currency
CDS spread	2.99%	
Excess CDS spread (over Swiss CDS)	2.85%	
Country Default Spread (based on CDS)	3.06%	
Country Risk Premium (Rating)	4.66%	
Equity Risk Premium (Rating)	8.89%	
Country Risk Premium (CDS)	4.34%	
Equity Risk Premium (CDS)	8.57%	

To look up the equity risk premium for a region, use this

Region	Asia
Country Risk Premium (simple average)	4.83%
Total Equity Risk Premium (simple average)	9.06%
Country Risk Premium (GDP weighted)	1.49%
Total Equity Risk Premium (GDP weighted)	5.72%

To construct your own regional ERP, use the data on GDP and ERP for countries in the region.

For countries not on the list above, try this frontier mark

Country	Zimbabwe
PRS Score	58.50
ERP based on PRS Score	15.89%

s worksheet

*· find a country on this list, it is because that country does not have a sovereign rating
n CDS spread. Try the PRS worksheet in this spreadsheet for an alternate estimate.*

worksheet

tries in the Regional Weighted Averages Worksheet

et list

Country and Equity Risk Premiums

Date of update:

1-Jan-26

(Updated Feb 16 for sovereign ratings)

Enter the current risk premium for a mature equity market

Enter the current risk premium for the US =

Do you want to adjust the country default spread for the additional volatility of the equity market to get to a country premium?

If yes, enter the multiplier to use on the default spread (See worksheet for volatility numbers for selected emerging markets)

Country		Moody's rating	Rating-based Default Spread
Abu Dhabi	Africa	Aa2	0.42%
Albania	Middle East	Ba3	3.06%
Andorra (Principality of)	Eastern Europe & Russia	Baa1	1.36%
Angola	Western Europe	B3	5.52%
Argentina	Africa	B3	5.52%
Armenia	Central and South America	Caa1	6.37%
Aruba	Eastern Europe & Russia	Ba3	3.06%
Australia	Caribbean	Baa3	1.87%
Austria	Australia & New Zealand	Aaa	0.00%
Azerbaijan	Western Europe	Aa1	0.23%
Bahamas	Eastern Europe & Russia	Baa3	1.87%
Bahrain	Caribbean	B1	3.83%
Bangladesh	Middle East	B2	4.67%
Barbados	Asia	B2	4.67%
Belarus	Caribbean	B2	4.67%
Belgium	Eastern Europe & Russia	C	17.50%
Belize	Western Europe	Aa3	0.51%
Benin	Central and South America	Caa1	6.37%
Bermuda	Africa	B1	3.83%
Bolivia	Caribbean	A2	0.72%
Bosnia and Herzegovina	Central and South America	Ca	10.20%
Botswana	Eastern Europe & Russia	B3	5.52%
Brazil	Africa	Baa1	1.36%
	Central and South America	Ba1	2.13%

Bulgaria	Eastern Europe & Russia	Baa1	1.36%
Burkina Faso	Africa	Caa1	6.37%
Cambodia	Asia	B2	4.67%
Cameroon	Africa	Caa1	6.37%
Canada	North America	Aaa	0.00%
Cape Verde	Africa	B2	4.67%
Cayman Islands	Caribbean	Aa3	0.51%
Chile	Central and South America	A2	0.72%
China	Asia	A1	0.60%
Colombia	Central and South America	Baa3	1.87%
Congo (Democratic Republic of)	Africa	B3	5.52%
Congo (Republic of)	Africa	Caa2	7.65%
Cook Islands	Australia & New Zealand	B1	3.83%
Costa Rica	Central and South America	Ba2	2.56%
Côte d'Ivoire	Africa	Ba2	2.56%
Croatia	Eastern Europe & Russia	A3	1.02%
Cuba	Caribbean	Ca	10.20%
Curacao	Caribbean	Baa3	1.87%
Cyprus	Western Europe	A3	1.02%
Czech Republic	Eastern Europe & Russia	Aa3	0.51%
Denmark	Western Europe	Aaa	0.00%
Dominican Republic	Caribbean	Ba2	2.56%
Ecuador	Central and South America	Caa3	8.50%
Egypt	Africa	Caa1	6.37%
El Salvador	Central and South America	B3	5.52%
Estonia	Eastern Europe & Russia	A1	0.60%
Ethiopia	Africa	Caa2	7.65%
Fiji	Asia	B1	3.83%
Finland	Western Europe	Aa1	0.23%
France	Western Europe	Aa3	0.51%
Gabon	Africa	Caa2	7.65%

Georgia	Eastern Europe & Russia	Ba2	2.56%
Germany	Western Europe	Aaa	0.00%
Ghana	Africa	Caa1	6.37%
Greece	Western Europe	Baa3	1.87%
Guatemala	Central and South America	Ba1	2.13%
Guernsey (States of)	Western Europe	A1	0.60%
Honduras	Central and South America	B1	3.83%
Hong Kong	Asia	Aa3	0.51%
Hungary	Eastern Europe & Russia	Baa2	1.62%
Iceland	Western Europe	A1	0.60%
India	Asia	Baa3	1.87%
Indonesia	Asia	Baa2	1.62%
Iraq	Middle East	Caa1	6.37%
Ireland	Western Europe	Aa3	0.51%
Isle of Man	Western Europe	Aa3	0.51%
Israel	Middle East	Baa1	1.36%
Italy	Western Europe	Baa2	1.62%
Jamaica	Caribbean	Ba3	3.06%
Japan	Asia	A1	0.60%
Jersey (States of)	Western Europe	Aa3	0.51%
Jordan	Middle East	Ba3	3.06%
Kazakhstan	Eastern Europe & Russia	Baa1	1.36%
Kenya	Africa	B3	5.52%
Korea	Asia	Aa2	0.42%
Kuwait	Middle East	A1	0.60%
Kyrgyzstan	Eastern Europe & Russia	B3	5.52%
Laos	Asia	Caa2	7.65%
Latvia	Eastern Europe & Russia	A3	1.02%
Lebanon	Middle East	C	17.50%
Liechtenstein	Western Europe	Aaa	0.00%
Lithuania	Eastern Europe & Russia	A2	0.72%

Luxembourg	Western Europe	Aaa	0.00%
Macao	Asia	Aa3	0.51%
Macedonia	Eastern Europe & Russia	Ba3	3.06%
Malaysia	Asia	A3	1.02%
Maldives	Asia	Caa2	7.65%
Mali	Africa	Caa2	7.65%
Malta	Western Europe	A2	0.72%
Mauritius	Africa	Baa3	1.87%
Mexico	Central and South America	Baa2	1.62%
Moldova	Eastern Europe & Russia	B3	5.52%
Mongolia	Asia	B1	3.83%
Montenegro	Eastern Europe & Russia	B1	3.83%
Montserrat	Caribbean	Baa3	1.87%
Morocco	Africa	Ba1	2.13%
Mozambique	Africa	Caa3	8.50%
Namibia	Africa	B1	3.83%
Nepal	Asia	Ba3	3.06%
Netherlands	Western Europe	Aaa	0.00%
New Zealand	Australia & New Zealand	Aaa	0.00%
Nicaragua	Central and South America	B2	4.67%
Niger	Africa	Caa3	8.50%
Nigeria	Africa	B3	5.52%
Norway	Western Europe	Aaa	0.00%
Oman	Middle East	Baa3	1.87%
Pakistan	Asia	Caa1	6.37%
Panama	Central and South America	Baa3	1.87%
Papua New Guinea	Asia	B2	4.67%
Paraguay	Central and South America	Baa3	1.87%
Peru	Central and South America	Baa1	1.36%
Philippines	Asia	Baa2	1.62%
Poland	Eastern Europe & Russia	A2	0.72%

Portugal	Western Europe	A3	1.02%
Qatar	Middle East	Aa2	0.42%
Ras Al Khaimah (Emirate of)	Middle East	A3	1.02%
Romania	Eastern Europe & Russia	Baa3	1.87%
Rwanda	Africa	B2	4.67%
Saudi Arabia	Middle East	Aa3	0.51%
Senegal	Africa	Caa1	6.37%
Serbia	Eastern Europe & Russia	Ba2	2.56%
Sharjah	Middle East	Ba1	2.13%
Singapore	Asia	Aaa	0.00%
Slovakia	Eastern Europe & Russia	A3	1.02%
Slovenia	Eastern Europe & Russia	A3	1.02%
Solomon Islands	Asia	Caa1	6.37%
South Africa	Africa	Ba2	2.56%
Spain	Western Europe	A3	1.02%
Sri Lanka	Asia	Ca	10.20%
St. Maarten	Caribbean	Ba2	2.56%
St. Vincent & the Grenadines	Caribbean	B3	5.52%
Suriname	Central and South America	Caa1	6.37%
Swaziland	Africa	B2	4.67%
Sweden	Western Europe	Aaa	0.00%
Switzerland	Western Europe	Aaa	0.00%
Taiwan	Asia	Aa3	0.51%
Tajikistan	Eastern Europe & Russia	B3	5.52%
Tanzania	Africa	B1	3.83%
Thailand	Asia	Baa1	1.36%
Togo	Africa	B3	5.52%
Trinidad and Tobago	Caribbean	Ba2	2.56%
Tunisia	Africa	Caa1	6.37%
Turkey	Western Europe	Ba3	3.06%
Turks and Caicos Islands	Caribbean	Baa1	1.36%

Uganda	Africa	B3	5.52%
Ukraine	Eastern Europe & Russia	Ca	10.20%
United Arab Emirates	Middle East	Aa2	0.42%
United Kingdom	Western Europe	Aa3	0.51%
United States	North America	Aa1	0.23%
Uruguay	Central and South America	Baa1	1.36%
Uzbekistan	Eastern Europe & Russia	Ba3	3.06%
Venezuela	Central and South America	C	17.50%
Vietnam	Asia	Ba2	2.56%
Zambia	Africa	Caa2	7.65%

Frontier Markets (no sovereign ratings)

<i>Country</i>	<i>PRS Composite Risk Score</i>	<i>ERP</i>	<i>CRP</i>
Algeria	67	10.06%	5.83%
Brunei	80.75	5.01%	0.78%
Gambia	65.5	11.35%	7.12%
Guinea	59.5	15.89%	11.66%
Guinea-Bissau	60.75	13.94%	9.71%
Guyana	75.5	6.30%	2.07%
Haiti	56.75	17.18%	12.95%
Iran	61.25	13.94%	9.71%
Korea, D.P.R.	51	19.77%	15.54%
Liberia	57.75	15.89%	11.66%
Libya	71.5	8.13%	3.90%
Madagascar	62.5	12.64%	8.41%
Malawi	56	17.18%	12.95%
Myanmar	54.75	19.77%	15.54%
Russia	70.25	8.13%	3.90%
Sierra Leone	62.75	12.64%	8.41%
Somalia	55.5	17.18%	12.95%
Sudan	47.75	30.89%	26.66%
Syria	51.5	19.77%	15.54%

Yemen, Republic	51.75	19.77%	15.54%
Zimbabwe	58.5	15.89%	11.66%

<i>Rating</i>	<i>Default spread in basis points</i>
A1	60
A2	72
A3	102
Aa1	23
Aa2	42
Aa3	51
Aaa	0
B1	383
B2	467
B3	552
Ba1	213
Ba2	256
Ba3	306
Baa1	136
Baa2	162
Baa3	187
C	1750
Ca	1020
Caa1	637
Caa2	765
Caa3	850
NR	NA

4.23%	<i>Implied ERP for S&P 500 with adjusted dollar riskfree rate (for US Aa1 rating</i>
4.46%	<i>US equity risk premium computed using implied ERP approach</i>
Yes	
1.52	<i>See "Relative Equity Volatility" worksheet</i>

Total Equity Risk Premium	Country Risk Premium	Sovereign CDS, net of Swiss CDS	Total Equity Risk Premium2
4.87%	0.64%	0.46%	4.93%
8.89%	4.66%	NA	NA
6.30%	2.07%	NA	NA
12.64%	8.41%	6.10%	13.52%
13.94%	9.71%	NA	NA
8.89%	4.66%	NA	NA
7.08%	2.85%	NA	NA
4.23%	0.00%	0.05%	4.31%
4.59%	0.36%	0.14%	4.44%
7.08%	2.85%	NA	NA
10.06%	5.83%	NA	NA
11.35%	7.12%	2.33%	7.78%
11.35%	7.12%	NA	NA
11.35%	7.12%	NA	NA
30.89%	26.66%	NA	NA
5.01%	0.78%	0.26%	4.63%
13.94%	9.71%	NA	NA
10.06%	5.83%	NA	NA
5.33%	1.10%	NA	NA
19.77%	15.54%	NA	NA
12.64%	8.41%	NA	NA
6.30%	2.07%	NA	NA
7.47%	3.24%	2.21%	7.60%

6.30%	2.07%	0.63%	5.19%
13.94%	9.71%	NA	NA
11.35%	7.12%	NA	NA
13.94%	9.71%	6.73%	14.48%
4.23%	0.00%	0.19%	4.52%
11.35%	7.12%	NA	NA
5.01%	0.78%	NA	NA
5.33%	1.10%	0.73%	5.34%
5.14%	0.91%	0.50%	4.99%
7.08%	2.85%	3.20%	9.10%
12.64%	8.41%	NA	NA
15.89%	11.66%	NA	NA
10.06%	5.83%	NA	NA
8.13%	3.90%	1.61%	6.68%
8.13%	3.90%	NA	NA
5.78%	1.55%	0.88%	5.57%
19.77%	15.54%	NA	NA
7.08%	2.85%	NA	NA
5.78%	1.55%	0.61%	5.16%
5.01%	0.78%	0.34%	4.75%
4.23%	0.00%	0.04%	4.29%
8.13%	3.90%	NA	NA
17.18%	12.95%	5.53%	12.65%
13.94%	9.71%	3.41%	9.42%
12.64%	8.41%	3.26%	9.20%
5.14%	0.91%	0.86%	5.54%
15.89%	11.66%	NA	NA
10.06%	5.83%	NA	NA
4.59%	0.36%	0.12%	4.41%
5.01%	0.78%	0.50%	4.99%
15.89%	11.66%	NA	NA

8.13%	3.90%	NA	NA
4.23%	0.00%	0.07%	4.34%
13.94%	9.71%	NA	NA
7.08%	2.85%	0.61%	5.16%
7.47%	3.24%	NA	NA
5.14%	0.91%	NA	NA
10.06%	5.83%	NA	NA
5.01%	0.78%	0.30%	4.69%
6.69%	2.46%	1.46%	6.45%
5.14%	0.91%	0.31%	4.70%
7.08%	2.85%	0.66%	5.24%
6.69%	2.46%	1.05%	5.83%
13.94%	9.71%	2.75%	8.42%
5.01%	0.78%	0.20%	4.53%
5.01%	0.78%	NA	NA
6.30%	2.07%	0.99%	5.74%
6.69%	2.46%	0.47%	4.95%
8.89%	4.66%	NA	NA
5.14%	0.91%	0.30%	4.69%
5.01%	0.78%	NA	NA
8.89%	4.66%	NA	NA
6.30%	2.07%	1.17%	6.01%
12.64%	8.41%	4.37%	10.89%
4.87%	0.64%	0.20%	4.53%
5.14%	0.91%	0.73%	5.34%
12.64%	8.41%	NA	NA
15.89%	11.66%	NA	NA
5.78%	1.55%	0.68%	5.27%
30.89%	26.66%	NA	NA
4.23%	0.00%	NA	NA
5.33%	1.10%	0.68%	5.27%

4.23%	0.00%	NA	NA
5.01%	0.78%	NA	NA
8.89%	4.66%	NA	NA
5.78%	1.55%	0.53%	5.04%
15.89%	11.66%	NA	NA
15.89%	11.66%	NA	NA
5.33%	1.10%	NA	NA
7.08%	2.85%	NA	NA
6.69%	2.46%	1.52%	6.55%
12.64%	8.41%	NA	NA
10.06%	5.83%	NA	NA
10.06%	5.83%	NA	NA
7.08%	2.85%	NA	NA
7.47%	3.24%	1.16%	6.00%
17.18%	12.95%	NA	NA
10.06%	5.83%	NA	NA
8.89%	4.66%	NA	NA
4.23%	0.00%	0.06%	4.32%
4.23%	0.00%	0.09%	4.37%
11.35%	7.12%	5.15%	12.08%
17.18%	12.95%	NA	NA
12.64%	8.41%	3.77%	9.97%
4.23%	0.00%	0.04%	4.29%
7.08%	2.85%	1.16%	6.00%
13.94%	9.71%	5.09%	11.98%
7.08%	2.85%	2.08%	7.40%
11.35%	7.12%	NA	NA
7.08%	2.85%	NA	NA
6.30%	2.07%	1.12%	5.94%
6.69%	2.46%	0.88%	5.57%
5.33%	1.10%	0.86%	5.54%

5.78%	1.55%	0.31%	4.70%
4.87%	0.64%	0.47%	4.95%
5.78%	1.55%	NA	NA
7.08%	2.85%	1.97%	7.23%
11.35%	7.12%	3.89%	10.16%
5.01%	0.78%	0.98%	5.72%
13.94%	9.71%	9.79%	19.14%
8.13%	3.90%	1.82%	7.00%
7.47%	3.24%	#N/A	#N/A
4.23%	0.00%	#N/A	#N/A
5.78%	1.55%	0.43%	4.89%
5.78%	1.55%	0.53%	5.04%
13.94%	9.71%	#N/A	#N/A
8.13%	3.90%	2.26%	7.67%
5.78%	1.55%	0.31%	4.70%
19.77%	15.54%	NA	NA
8.13%	3.90%	#N/A	#N/A
12.64%	8.41%	#N/A	#N/A
13.94%	9.71%	#N/A	#N/A
11.35%	7.12%	#N/A	#N/A
4.23%	0.00%	0.06%	4.32%
4.23%	0.00%	0.00%	4.23%
5.01%	0.78%	#N/A	#N/A
12.64%	8.41%	#N/A	#N/A
10.06%	5.83%	#N/A	#N/A
6.30%	2.07%	0.52%	5.02%
12.64%	8.41%	#N/A	#N/A
8.13%	3.90%	#N/A	#N/A
13.94%	9.71%	6.87%	14.70%
8.89%	4.66%	2.85%	8.57%
6.30%	2.07%	#N/A	#N/A

12.64%	8.41%	#N/A	#N/A
19.77%	15.54%	NA	NA
4.87%	0.64%	#N/A	#N/A
5.01%	0.78%	0.23%	4.58%
4.46%	0.23%	0.30%	4.69%
6.30%	2.07%	0.63%	5.19%
8.89%	4.66%	#N/A	#N/A
30.89%	26.66%	9.15%	18.17%
8.13%	3.90%	1.37%	6.32%
15.89%	11.66%	3.94%	10.23%

Default Spread

3.83%
0.51%
4.67%
7.65%
6.37%
1.36%
8.50%
6.37%
10.20%
7.65%
2.56%
5.52%
8.50%
10.20%
2.56%
5.52%
8.50%
17.50%
10.20%

10.20%
7.65%

Country Risk Premium3 *Has to be sorted in ascending order*

	<i>Rating</i>	<i>spread in basis points</i>
0.70%		
NA	A1	60
NA	A2	72
9.29%	A3	102
NA	Aa1	23
NA	Aa2	42
NA	Aa3	51
0.08%	Aaa	0
0.21%	B1	383
NA	B2	467
NA	B3	552
3.55%	Ba1	213
NA	Ba2	256
NA	Ba3	306
NA	Baa1	136
0.40%	Baa2	162
NA	Baa3	187
NA	C	1750
NA	Ca	1020
NA	Caa1	637
NA	Caa2	765
NA	Caa3	850
3.37%	NR	NA

0.96%
NA
NA
10.25%
0.29%
NA
NA
1.11%
0.76%
4.87%
NA
NA
NA
2.45%
NA
1.34%
NA
NA
0.93%
0.52%
0.06%
NA
8.42%
5.19%
4.97%
1.31%
NA
NA
0.18%
0.76%
NA

NA
0.11%
NA
0.93%
NA
NA
NA
0.46%
2.22%
0.47%
1.01%
1.60%
4.19%
0.30%
NA
1.51%
0.72%
NA
0.46%
NA
NA
1.78%
6.66%
0.30%
1.11%
NA
NA
1.04%
NA
NA
1.04%

NA
NA
NA
0.81%
NA
NA
NA
NA
2.32%
NA
NA
NA
NA
1.77%
NA
NA
NA
0.09%
0.14%
7.85%
NA
5.74%
0.06%
1.77%
7.75%
3.17%
NA
NA
1.71%
1.34%
1.31%

0.47%
0.72%
NA
3.00%
5.93%
1.49%
14.91%
2.77%
#N/A
#N/A
0.66%
0.81%
#N/A
3.44%
0.47%
NA
#N/A
#N/A
#N/A
#N/A
0.09%
0.00%
#N/A
#N/A
#N/A
0.79%
#N/A
#N/A
10.47%
4.34%
#N/A

#N/A
NA
#N/A
0.35%
0.46%
0.96%
#N/A
13.94%
2.09%
6.00%

Year	Std Dev (BMI)	d Dev (JPM Sov Bon	REL VOL
2021	14.32%	6.87%	2.08
2022	18.63%	15.38%	1.21
2023	11.14%	9.87%	1.13
2024	11.87%	7.41%	1.60
2025	14.61%	6.79%	2.15
Average Relative Volatility	14.11%	9.26%	1.52

Index: iShares JP Morgan USD Emerging Markets Bond ETF
Source: Yahoo!

<https://finance.yahoo.com/quote/EMB/>

Period Five years (Dec 31, 2020 - Dec 31, 2025)
Interval Daily

Date

Close

Return

31-Dec-20	115.91	
4-Jan-21	115.27	-0.55%
5-Jan-21	115.35	0.07%
6-Jan-21	114.52	-0.72%
7-Jan-21	114.48	-0.03%
8-Jan-21	114.54	0.05%
11-Jan-21	113.64	-0.79%
12-Jan-21	113.1	-0.48%
13-Jan-21	113.63	0.47%
14-Jan-21	113.3	-0.29%
15-Jan-21	113.43	0.11%
19-Jan-21	113.59	0.14%
20-Jan-21	113.72	0.11%
21-Jan-21	113.8	0.07%
22-Jan-21	113.85	0.04%
25-Jan-21	114.3	0.40%
26-Jan-21	114.15	-0.13%
27-Jan-21	113.62	-0.46%
28-Jan-21	113.86	0.21%
29-Jan-21	113.85	-0.01%
1-Feb-21	114.05	0.18%
2-Feb-21	114.05	0.00%
3-Feb-21	113.76	-0.25%
4-Feb-21	114.09	0.29%
5-Feb-21	114.22	0.11%

8-Feb-21	114.22	0.00%
9-Feb-21	113.81	-0.36%
10-Feb-21	114.04	0.20%
11-Feb-21	114.11	0.06%
12-Feb-21	113.48	-0.55%
16-Feb-21	112.09	-1.22%
17-Feb-21	112.71	0.55%
18-Feb-21	112.59	-0.11%
19-Feb-21	111.99	-0.53%
22-Feb-21	111.23	-0.68%
23-Feb-21	111.36	0.12%
24-Feb-21	111.53	0.15%
25-Feb-21	109.54	-1.78%
26-Feb-21	110.03	0.45%
1-Mar-21	110.29	0.24%
2-Mar-21	110.22	-0.06%
3-Mar-21	109.56	-0.60%
4-Mar-21	108.62	-0.86%
5-Mar-21	108.6	-0.02%
8-Mar-21	106.72	-1.73%
9-Mar-21	107.98	1.18%
10-Mar-21	109.03	0.97%
11-Mar-21	109.84	0.74%
12-Mar-21	108.46	-1.26%
15-Mar-21	109.17	0.65%
16-Mar-21	109.55	0.35%
17-Mar-21	110.47	0.84%
18-Mar-21	108.97	-1.36%
19-Mar-21	109.29	0.29%
22-Mar-21	109.53	0.22%
23-Mar-21	109.14	-0.36%
24-Mar-21	109.38	0.22%
25-Mar-21	109.22	-0.15%
26-Mar-21	109.26	0.04%
29-Mar-21	108.69	-0.52%
30-Mar-21	108.74	0.05%
31-Mar-21	108.88	0.13%

1-Apr-21	109.39	0.47%
5-Apr-21	109.26	-0.12%
6-Apr-21	109.63	0.34%
7-Apr-21	109.25	-0.35%
8-Apr-21	109.88	0.58%
9-Apr-21	109.58	-0.27%
12-Apr-21	109.76	0.16%
13-Apr-21	110.34	0.53%
14-Apr-21	110.24	-0.09%
15-Apr-21	111.58	1.22%
16-Apr-21	111.22	-0.32%
19-Apr-21	111.14	-0.07%
20-Apr-21	110.86	-0.25%
21-Apr-21	111.07	0.19%
22-Apr-21	111.24	0.15%
23-Apr-21	111.43	0.17%
26-Apr-21	110.92	-0.46%
27-Apr-21	110.32	-0.54%
28-Apr-21	110.69	0.34%
29-Apr-21	110.82	0.12%
30-Apr-21	111.14	0.29%
3-May-21	110.64	-0.45%
4-May-21	110.74	0.09%
5-May-21	110.89	0.14%
6-May-21	111.1	0.19%
7-May-21	111.6	0.45%
10-May-21	111.35	-0.22%
11-May-21	111.09	-0.23%
12-May-21	110.08	-0.91%
13-May-21	110.65	0.52%
14-May-21	111.23	0.52%
17-May-21	111.17	-0.05%
18-May-21	111.16	-0.01%
19-May-21	110.64	-0.47%
20-May-21	111.41	0.70%
21-May-21	111.19	-0.20%
24-May-21	111.31	0.11%

25-May-21	111.85	0.49%
26-May-21	111.84	-0.01%
27-May-21	111.63	-0.19%
28-May-21	111.86	0.21%
1-Jun-21	111.65	-0.19%
2-Jun-21	111.82	0.15%
3-Jun-21	111.27	-0.49%
4-Jun-21	112.01	0.67%
7-Jun-21	111.76	-0.22%
8-Jun-21	112.2	0.39%
9-Jun-21	112.6	0.36%
10-Jun-21	112.77	0.15%
11-Jun-21	112.9	0.12%
14-Jun-21	112.22	-0.60%
15-Jun-21	112.47	0.22%
16-Jun-21	111.51	-0.85%
17-Jun-21	111.95	0.39%
18-Jun-21	112.27	0.29%
21-Jun-21	112.2	-0.06%
22-Jun-21	112.15	-0.04%
23-Jun-21	112.28	0.12%
24-Jun-21	112.32	0.04%
25-Jun-21	112.13	-0.17%
28-Jun-21	112.34	0.19%
29-Jun-21	112.31	-0.03%
30-Jun-21	112.46	0.13%
1-Jul-21	111.99	-0.42%
2-Jul-21	112.12	0.12%
6-Jul-21	112.25	0.12%
7-Jul-21	112.41	0.14%
8-Jul-21	112.18	-0.20%
9-Jul-21	112.15	-0.03%
12-Jul-21	112.22	0.06%
13-Jul-21	111.82	-0.36%
14-Jul-21	112.3	0.43%
15-Jul-21	112.65	0.31%
16-Jul-21	112.5	-0.13%

19-Jul-21	112.52	0.02%
20-Jul-21	112.83	0.28%
21-Jul-21	112.4	-0.38%
22-Jul-21	112.65	0.22%
23-Jul-21	112.66	0.01%
26-Jul-21	112.35	-0.28%
27-Jul-21	112.05	-0.27%
28-Jul-21	112.33	0.25%
29-Jul-21	112.43	0.09%
30-Jul-21	112.73	0.27%
2-Aug-21	112.53	-0.18%
3-Aug-21	112.58	0.04%
4-Aug-21	112.63	0.04%
5-Aug-21	112.56	-0.06%
6-Aug-21	111.96	-0.53%
9-Aug-21	111.59	-0.33%
10-Aug-21	111.57	-0.02%
11-Aug-21	111.64	0.06%
12-Aug-21	111.79	0.13%
13-Aug-21	112.32	0.47%
16-Aug-21	112.34	0.02%
17-Aug-21	112.24	-0.09%
18-Aug-21	112.15	-0.08%
19-Aug-21	112.26	0.10%
20-Aug-21	112.1	-0.14%
23-Aug-21	112.52	0.37%
24-Aug-21	112.59	0.06%
25-Aug-21	112.51	-0.07%
26-Aug-21	112.33	-0.16%
27-Aug-21	113.05	0.64%
30-Aug-21	113.35	0.27%
31-Aug-21	113.39	0.04%
1-Sep-21	113.24	-0.13%
2-Sep-21	113.55	0.27%
3-Sep-21	113.18	-0.33%
7-Sep-21	112.87	-0.27%
8-Sep-21	113.07	0.18%

9-Sep-21	113.26	0.17%
10-Sep-21	112.9	-0.32%
13-Sep-21	113.05	0.13%
14-Sep-21	113.27	0.19%
15-Sep-21	113.25	-0.02%
16-Sep-21	112.87	-0.34%
17-Sep-21	112.5	-0.33%
20-Sep-21	111.98	-0.46%
21-Sep-21	112.19	0.19%
22-Sep-21	112.26	0.06%
23-Sep-21	111.4	-0.77%
24-Sep-21	110.94	-0.41%
27-Sep-21	110.61	-0.30%
28-Sep-21	109.98	-0.57%
29-Sep-21	110.08	0.09%
30-Sep-21	110.06	-0.02%
1-Oct-21	109.69	-0.34%
4-Oct-21	109.16	-0.48%
5-Oct-21	109.23	0.06%
6-Oct-21	109	-0.21%
7-Oct-21	109.12	0.11%
8-Oct-21	108.85	-0.25%
11-Oct-21	108.55	-0.28%
12-Oct-21	108.9	0.32%
13-Oct-21	109.59	0.63%
14-Oct-21	110.06	0.43%
15-Oct-21	109.93	-0.12%
18-Oct-21	109.57	-0.33%
19-Oct-21	109.49	-0.07%
20-Oct-21	109.55	0.05%
21-Oct-21	109.08	-0.43%
22-Oct-21	109.51	0.39%
25-Oct-21	109.59	0.07%
26-Oct-21	109.82	0.21%
27-Oct-21	110.47	0.59%
28-Oct-21	110.26	-0.19%
29-Oct-21	109.89	-0.34%

1-Nov-21	109.23	-0.60%
2-Nov-21	109.37	0.13%
3-Nov-21	109.2	-0.16%
4-Nov-21	109.99	0.72%
5-Nov-21	110.82	0.75%
8-Nov-21	110.72	-0.09%
9-Nov-21	110.92	0.18%
10-Nov-21	109.3	-1.46%
11-Nov-21	109.3	0.00%
12-Nov-21	109.58	0.26%
15-Nov-21	109.29	-0.26%
16-Nov-21	109.04	-0.23%
17-Nov-21	109.16	0.11%
18-Nov-21	109.44	0.26%
19-Nov-21	109.41	-0.03%
22-Nov-21	108.28	-1.03%
23-Nov-21	107.6	-0.63%
24-Nov-21	107.77	0.16%
26-Nov-21	106.57	-1.11%
29-Nov-21	107.49	0.86%
30-Nov-21	107.47	-0.02%
1-Dec-21	107.86	0.36%
2-Dec-21	108.5	0.59%
3-Dec-21	108.64	0.13%
6-Dec-21	108.74	0.09%
7-Dec-21	109.32	0.53%
8-Dec-21	109.16	-0.15%
9-Dec-21	109.09	-0.06%
10-Dec-21	109.11	0.02%
13-Dec-21	109.56	0.41%
14-Dec-21	109.28	-0.26%
15-Dec-21	108.93	-0.32%
16-Dec-21	108.67	-0.24%
17-Dec-21	108.48	-0.17%
20-Dec-21	107.9	-0.53%
21-Dec-21	108.22	0.30%
22-Dec-21	108.34	0.11%

23-Dec-21	108.5	0.15%
27-Dec-21	108.9	0.37%
28-Dec-21	108.8	-0.09%
29-Dec-21	108.69	-0.10%
30-Dec-21	109.2	0.47%
31-Dec-21	109.06	-0.13%
3-Jan-22	107.98	-0.99%
4-Jan-22	107.92	-0.06%
5-Jan-22	106.89	-0.95%
6-Jan-22	106.79	-0.09%
7-Jan-22	106.58	-0.20%
10-Jan-22	106.22	-0.34%
11-Jan-22	106.44	0.21%
12-Jan-22	106.22	-0.21%
13-Jan-22	105.86	-0.34%
14-Jan-22	104.93	-0.88%
18-Jan-22	103.96	-0.92%
19-Jan-22	104.66	0.67%
20-Jan-22	105.03	0.35%
21-Jan-22	105.59	0.53%
24-Jan-22	104.88	-0.67%
25-Jan-22	105.04	0.15%
26-Jan-22	104.49	-0.52%
27-Jan-22	105.08	0.56%
28-Jan-22	105.41	0.31%
31-Jan-22	105.38	-0.03%
1-Feb-22	105.28	-0.09%
2-Feb-22	105.76	0.46%
3-Feb-22	104.85	-0.86%
4-Feb-22	104.35	-0.48%
7-Feb-22	104.23	-0.11%
8-Feb-22	103.96	-0.26%
9-Feb-22	104.33	0.36%
10-Feb-22	102.96	-1.31%
11-Feb-22	102.36	-0.58%
14-Feb-22	102.75	0.38%
15-Feb-22	103.14	0.38%

16-Feb-22	103.44	0.29%
17-Feb-22	102.96	-0.46%
18-Feb-22	103	0.04%
22-Feb-22	101.7	-1.26%
23-Feb-22	99.99	-1.68%
24-Feb-22	98.98	-1.01%
25-Feb-22	100.17	1.20%
28-Feb-22	99.2	-0.97%
1-Mar-22	97.32	-1.90%
2-Mar-22	97.11	-0.22%
3-Mar-22	96.69	-0.43%
4-Mar-22	94.61	-2.15%
7-Mar-22	93.87	-0.78%
8-Mar-22	94.69	0.87%
9-Mar-22	95.61	0.97%
10-Mar-22	95.75	0.15%
11-Mar-22	95.01	-0.77%
14-Mar-22	94.81	-0.21%
15-Mar-22	95.38	0.60%
16-Mar-22	97.33	2.04%
17-Mar-22	97.95	0.64%
18-Mar-22	97.55	-0.41%
21-Mar-22	95.84	-1.75%
22-Mar-22	95.67	-0.18%
23-Mar-22	95.75	0.08%
24-Mar-22	96.13	0.40%
25-Mar-22	95.91	-0.23%
28-Mar-22	96.84	0.97%
29-Mar-22	97.79	0.98%
30-Mar-22	97.86	0.07%
31-Mar-22	97.76	-0.10%
1-Apr-22	97.73	-0.03%
4-Apr-22	98.11	0.39%
5-Apr-22	96.53	-1.61%
6-Apr-22	96.24	-0.30%
7-Apr-22	95.96	-0.29%
8-Apr-22	95.13	-0.86%

11-Apr-22	94	-1.19%
12-Apr-22	94.35	0.37%
13-Apr-22	94.59	0.25%
14-Apr-22	93.71	-0.93%
18-Apr-22	93.66	-0.05%
19-Apr-22	93.29	-0.40%
20-Apr-22	93.93	0.69%
21-Apr-22	93.08	-0.90%
22-Apr-22	92.09	-1.06%
25-Apr-22	93.17	1.17%
26-Apr-22	92.53	-0.69%
27-Apr-22	92.28	-0.27%
28-Apr-22	92.45	0.18%
29-Apr-22	90.88	-1.70%
2-May-22	90.02	-0.95%
3-May-22	90.4	0.42%
4-May-22	91.95	1.71%
5-May-22	90.57	-1.50%
6-May-22	89.35	-1.35%
9-May-22	88.56	-0.88%
10-May-22	89	0.50%
11-May-22	89.39	0.44%
12-May-22	89.45	0.07%
13-May-22	89.3	-0.17%
16-May-22	88.82	-0.54%
17-May-22	89.14	0.36%
18-May-22	88.57	-0.64%
19-May-22	89.05	0.54%
20-May-22	89.63	0.65%
23-May-22	89.69	0.07%
24-May-22	90.47	0.87%
25-May-22	91.27	0.88%
26-May-22	91.57	0.33%
27-May-22	91.72	0.16%
31-May-22	91.25	-0.51%
1-Jun-22	90.75	-0.55%
2-Jun-22	91.63	0.97%

3-Jun-22	91.01	-0.68%
6-Jun-22	89.81	-1.32%
7-Jun-22	90.34	0.59%
8-Jun-22	89.45	-0.99%
9-Jun-22	88.71	-0.83%
10-Jun-22	87.22	-1.68%
13-Jun-22	84.91	-2.65%
14-Jun-22	84.87	-0.05%
15-Jun-22	87.41	2.99%
16-Jun-22	85.47	-2.22%
17-Jun-22	85.93	0.54%
21-Jun-22	85.68	-0.29%
22-Jun-22	85.81	0.15%
23-Jun-22	86.38	0.66%
24-Jun-22	86.49	0.13%
27-Jun-22	85.31	-1.36%
28-Jun-22	84.66	-0.76%
29-Jun-22	84.97	0.37%
30-Jun-22	85.32	0.41%
1-Jul-22	86.15	0.97%
5-Jul-22	85.47	-0.79%
6-Jul-22	84.78	-0.81%
7-Jul-22	85.03	0.29%
8-Jul-22	84.63	-0.47%
11-Jul-22	83.92	-0.84%
12-Jul-22	83.68	-0.29%
13-Jul-22	83.34	-0.41%
14-Jul-22	81.88	-1.75%
15-Jul-22	82.7	1.00%
18-Jul-22	82.74	0.05%
19-Jul-22	83.95	1.46%
20-Jul-22	84.61	0.79%
21-Jul-22	85.08	0.56%
22-Jul-22	86.01	1.09%
25-Jul-22	86.07	0.07%
26-Jul-22	85.27	-0.93%
27-Jul-22	86.56	1.51%

28-Jul-22	87.68	1.29%
29-Jul-22	88.02	0.39%
1-Aug-22	88.07	0.06%
2-Aug-22	86.79	-1.45%
3-Aug-22	88.46	1.92%
4-Aug-22	88.65	0.21%
5-Aug-22	88.47	-0.20%
8-Aug-22	89.78	1.48%
9-Aug-22	88.28	-1.67%
10-Aug-22	89.86	1.79%
11-Aug-22	89.49	-0.41%
12-Aug-22	90.17	0.76%
15-Aug-22	90	-0.19%
16-Aug-22	89.22	-0.87%
17-Aug-22	88.24	-1.10%
18-Aug-22	88.14	-0.11%
19-Aug-22	86.8	-1.52%
22-Aug-22	86.16	-0.74%
23-Aug-22	87.07	1.06%
24-Aug-22	86.99	-0.09%
25-Aug-22	87.91	1.06%
26-Aug-22	86.54	-1.56%
29-Aug-22	86.02	-0.60%
30-Aug-22	85.99	-0.03%
31-Aug-22	85.3	-0.80%
1-Sep-22	84.78	-0.61%
2-Sep-22	85.08	0.35%
6-Sep-22	84.3	-0.92%
7-Sep-22	85.58	1.52%
8-Sep-22	85.61	0.04%
9-Sep-22	86.13	0.61%
12-Sep-22	86.1	-0.03%
13-Sep-22	84.78	-1.53%
14-Sep-22	84.72	-0.07%
15-Sep-22	84.66	-0.07%
16-Sep-22	84.2	-0.54%
19-Sep-22	84.5	0.36%

20-Sep-22	83.72	-0.92%
21-Sep-22	83.48	-0.29%
22-Sep-22	82.92	-0.67%
23-Sep-22	81.92	-1.21%
26-Sep-22	79.99	-2.36%
27-Sep-22	78.71	-1.60%
28-Sep-22	80.6	2.40%
29-Sep-22	79.58	-1.27%
30-Sep-22	79.4	-0.23%
3-Oct-22	80.23	1.05%
4-Oct-22	81.74	1.88%
5-Oct-22	80.58	-1.42%
6-Oct-22	80.27	-0.38%
7-Oct-22	79.19	-1.35%
10-Oct-22	78.96	-0.29%
11-Oct-22	78.9	-0.08%
12-Oct-22	78.62	-0.35%
13-Oct-22	78.38	-0.31%
14-Oct-22	77.8	-0.74%
17-Oct-22	78.23	0.55%
18-Oct-22	79.06	1.06%
19-Oct-22	77.88	-1.49%
20-Oct-22	77.09	-1.01%
21-Oct-22	77.94	1.10%
24-Oct-22	77.89	-0.06%
25-Oct-22	79.24	1.73%
26-Oct-22	79.76	0.66%
27-Oct-22	79.94	0.23%
28-Oct-22	80.14	0.25%
31-Oct-22	78.9	-1.55%
1-Nov-22	80.03	1.43%
2-Nov-22	79.42	-0.76%
3-Nov-22	79.14	-0.35%
4-Nov-22	80.41	1.60%
7-Nov-22	80.38	-0.04%
8-Nov-22	80.94	0.70%
9-Nov-22	79.85	-1.35%

10-Nov-22	83.52	4.60%
11-Nov-22	83.5	-0.02%
14-Nov-22	82.79	-0.85%
15-Nov-22	84.35	1.88%
16-Nov-22	84.35	0.00%
17-Nov-22	83.59	-0.90%
18-Nov-22	83.59	0.00%
21-Nov-22	83.6	0.01%
22-Nov-22	84.47	1.04%
23-Nov-22	85.02	0.65%
25-Nov-22	85.41	0.46%
28-Nov-22	85.07	-0.40%
29-Nov-22	85.46	0.46%
30-Nov-22	86.43	1.14%
1-Dec-22	87.8	1.59%
2-Dec-22	87.43	-0.42%
5-Dec-22	86.09	-1.53%
6-Dec-22	85.62	-0.55%
7-Dec-22	86.86	1.45%
8-Dec-22	86.7	-0.18%
9-Dec-22	86.1	-0.69%
12-Dec-22	86.42	0.37%
13-Dec-22	87.25	0.96%
14-Dec-22	87.12	-0.15%
15-Dec-22	86.32	-0.92%
16-Dec-22	85.87	-0.52%
19-Dec-22	85.4	-0.55%
20-Dec-22	84.97	-0.50%
21-Dec-22	86.08	1.31%
22-Dec-22	85.64	-0.51%
23-Dec-22	85.18	-0.54%
27-Dec-22	84.67	-0.60%
28-Dec-22	84.36	-0.37%
29-Dec-22	85.01	0.77%
30-Dec-22	84.59	-0.49%
3-Jan-23	84.96	0.44%
4-Jan-23	86	1.22%

5-Jan-23	85.22	-0.91%
6-Jan-23	86.73	1.77%
9-Jan-23	86.56	-0.20%
10-Jan-23	86.02	-0.62%
11-Jan-23	87	1.14%
12-Jan-23	88.03	1.18%
13-Jan-23	87.82	-0.24%
17-Jan-23	87.91	0.10%
18-Jan-23	88.95	1.18%
19-Jan-23	88.7	-0.28%
20-Jan-23	88.5	-0.23%
23-Jan-23	88.45	-0.06%
24-Jan-23	88.84	0.44%
25-Jan-23	88.78	-0.07%
26-Jan-23	88.78	0.00%
27-Jan-23	88.61	-0.19%
30-Jan-23	87.48	-1.28%
31-Jan-23	87.91	0.49%
1-Feb-23	89.26	1.54%
2-Feb-23	89.55	0.32%
3-Feb-23	88.32	-1.37%
6-Feb-23	87.07	-1.42%
7-Feb-23	87.15	0.09%
8-Feb-23	87.1	-0.06%
9-Feb-23	86.53	-0.65%
10-Feb-23	85.86	-0.77%
13-Feb-23	86.1	0.28%
14-Feb-23	86.15	0.06%
15-Feb-23	85.9	-0.29%
16-Feb-23	85.34	-0.65%
17-Feb-23	85.37	0.04%
21-Feb-23	84.08	-1.51%
22-Feb-23	84.41	0.39%
23-Feb-23	85.76	1.60%
24-Feb-23	85.17	-0.69%
27-Feb-23	85.6	0.50%
28-Feb-23	85.24	-0.42%

1-Mar-23	84.3	-1.10%
2-Mar-23	83.99	-0.37%
3-Mar-23	85.28	1.54%
6-Mar-23	84.94	-0.40%
7-Mar-23	84.59	-0.41%
8-Mar-23	84.19	-0.47%
9-Mar-23	84.21	0.02%
10-Mar-23	85	0.94%
13-Mar-23	84.86	-0.16%
14-Mar-23	84.63	-0.27%
15-Mar-23	84.94	0.37%
16-Mar-23	84.58	-0.42%
17-Mar-23	84.2	-0.45%
20-Mar-23	84.21	0.01%
21-Mar-23	84.81	0.71%
22-Mar-23	85.55	0.87%
23-Mar-23	85.86	0.36%
24-Mar-23	85.39	-0.55%
27-Mar-23	85.03	-0.42%
28-Mar-23	84.73	-0.35%
29-Mar-23	85.14	0.48%
30-Mar-23	85.73	0.69%
31-Mar-23	86.28	0.64%
3-Apr-23	86.24	-0.05%
4-Apr-23	86.36	0.14%
5-Apr-23	86.18	-0.21%
6-Apr-23	86.09	-0.10%
10-Apr-23	85.64	-0.52%
11-Apr-23	85.73	0.11%
12-Apr-23	85.98	0.29%
13-Apr-23	86.07	0.10%
14-Apr-23	85.89	-0.21%
17-Apr-23	85.34	-0.64%
18-Apr-23	85.32	-0.02%
19-Apr-23	84.91	-0.48%
20-Apr-23	84.81	-0.12%
21-Apr-23	84.82	0.01%

24-Apr-23	85.05	0.27%
25-Apr-23	85.57	0.61%
26-Apr-23	85.34	-0.27%
27-Apr-23	85.36	0.02%
28-Apr-23	86.15	0.93%
1-May-23	84.95	-1.39%
2-May-23	85.31	0.42%
3-May-23	85.58	0.32%
4-May-23	85.15	-0.50%
5-May-23	85.39	0.28%
8-May-23	85.05	-0.40%
9-May-23	85.01	-0.05%
10-May-23	85.48	0.55%
11-May-23	85.75	0.32%
12-May-23	85.21	-0.63%
15-May-23	84.84	-0.43%
16-May-23	84.51	-0.39%
17-May-23	84.52	0.01%
18-May-23	84.18	-0.40%
19-May-23	84.17	-0.01%
22-May-23	84.16	-0.01%
23-May-23	84.26	0.12%
24-May-23	84.12	-0.17%
25-May-23	83.9	-0.26%
26-May-23	84.25	0.42%
30-May-23	84.89	0.76%
31-May-23	84.78	-0.13%
1-Jun-23	84.99	0.25%
2-Jun-23	84.86	-0.15%
5-Jun-23	85.14	0.33%
6-Jun-23	85.29	0.18%
7-Jun-23	84.79	-0.59%
8-Jun-23	85.27	0.57%
9-Jun-23	85.17	-0.12%
12-Jun-23	85.64	0.55%
13-Jun-23	85.51	-0.15%
14-Jun-23	85.69	0.21%

15-Jun-23	86.13	0.51%
16-Jun-23	85.79	-0.39%
20-Jun-23	85.83	0.05%
21-Jun-23	86.28	0.52%
22-Jun-23	85.95	-0.38%
23-Jun-23	86.05	0.12%
26-Jun-23	86.32	0.31%
27-Jun-23	86.32	0.00%
28-Jun-23	86.38	0.07%
29-Jun-23	85.92	-0.53%
30-Jun-23	86.54	0.72%
3-Jul-23	86.36	-0.21%
5-Jul-23	85.92	-0.51%
6-Jul-23	84.69	-1.43%
7-Jul-23	84.59	-0.12%
10-Jul-23	84.95	0.43%
11-Jul-23	85.49	0.64%
12-Jul-23	86.51	1.19%
13-Jul-23	87.45	1.09%
14-Jul-23	86.94	-0.58%
17-Jul-23	87.07	0.15%
18-Jul-23	87.5	0.49%
19-Jul-23	87.26	-0.27%
20-Jul-23	86.73	-0.61%
21-Jul-23	87	0.31%
24-Jul-23	87.17	0.20%
25-Jul-23	87.06	-0.13%
26-Jul-23	87.34	0.32%
27-Jul-23	86.51	-0.95%
28-Jul-23	87.56	1.21%
31-Jul-23	87.68	0.14%
1-Aug-23	86.46	-1.39%
2-Aug-23	85.84	-0.72%
3-Aug-23	85.11	-0.85%
4-Aug-23	86.24	1.33%
7-Aug-23	85.96	-0.32%
8-Aug-23	86.18	0.26%

9-Aug-23	86.36	0.21%
10-Aug-23	86.03	-0.38%
11-Aug-23	85.83	-0.23%
14-Aug-23	85.14	-0.80%
15-Aug-23	84.42	-0.85%
16-Aug-23	84.26	-0.19%
17-Aug-23	84.19	-0.08%
18-Aug-23	84.17	-0.02%
21-Aug-23	83.75	-0.50%
22-Aug-23	83.98	0.27%
23-Aug-23	85.33	1.61%
24-Aug-23	84.75	-0.68%
25-Aug-23	84.88	0.15%
28-Aug-23	85.15	0.32%
29-Aug-23	86.03	1.03%
30-Aug-23	85.8	-0.27%
31-Aug-23	85.72	-0.09%
1-Sep-23	85.15	-0.66%
5-Sep-23	84.32	-0.97%
6-Sep-23	84.19	-0.15%
7-Sep-23	84.64	0.53%
8-Sep-23	84.71	0.08%
11-Sep-23	84.58	-0.15%
12-Sep-23	84.64	0.07%
13-Sep-23	84.69	0.06%
14-Sep-23	84.72	0.04%
15-Sep-23	84.44	-0.33%
18-Sep-23	84.49	0.06%
19-Sep-23	84.39	-0.12%
20-Sep-23	84.32	-0.08%
21-Sep-23	83.47	-1.01%
22-Sep-23	83.81	0.41%
25-Sep-23	83.33	-0.57%
26-Sep-23	82.78	-0.66%
27-Sep-23	82.42	-0.43%
28-Sep-23	82.61	0.23%
29-Sep-23	82.52	-0.11%

2-Oct-23	81.27	-1.51%
3-Oct-23	80.36	-1.12%
4-Oct-23	80.74	0.47%
5-Oct-23	80.71	-0.04%
6-Oct-23	80.74	0.04%
9-Oct-23	81.14	0.50%
10-Oct-23	81.27	0.16%
11-Oct-23	81.93	0.81%
12-Oct-23	81.07	-1.05%
13-Oct-23	81.3	0.28%
16-Oct-23	81.21	-0.11%
17-Oct-23	80.82	-0.48%
18-Oct-23	80.3	-0.64%
19-Oct-23	79.71	-0.73%
20-Oct-23	80.15	0.55%
23-Oct-23	80.61	0.57%
24-Oct-23	81.25	0.79%
25-Oct-23	80.64	-0.75%
26-Oct-23	80.96	0.40%
27-Oct-23	81	0.05%
30-Oct-23	81.21	0.26%
31-Oct-23	81.19	-0.02%
1-Nov-23	81.89	0.86%
2-Nov-23	82.71	1.00%
3-Nov-23	83.52	0.98%
6-Nov-23	82.83	-0.83%
7-Nov-23	83.11	0.34%
8-Nov-23	83.27	0.19%
9-Nov-23	82.42	-1.02%
10-Nov-23	82.78	0.44%
13-Nov-23	82.34	-0.53%
14-Nov-23	83.77	1.74%
15-Nov-23	83.25	-0.62%
16-Nov-23	83.94	0.83%
17-Nov-23	84.21	0.32%
20-Nov-23	84.74	0.63%
21-Nov-23	84.74	0.00%

22-Nov-23	85.16	0.50%
24-Nov-23	84.99	-0.20%
27-Nov-23	85.36	0.44%
28-Nov-23	85.8	0.52%
29-Nov-23	86.48	0.79%
30-Nov-23	85.87	-0.71%
1-Dec-23	86.85	1.14%
4-Dec-23	86.25	-0.69%
5-Dec-23	86.75	0.58%
6-Dec-23	87.23	0.55%
7-Dec-23	87.14	-0.10%
8-Dec-23	86.84	-0.34%
11-Dec-23	86.7	-0.16%
12-Dec-23	86.88	0.21%
13-Dec-23	88.32	1.66%
14-Dec-23	88.87	0.62%
15-Dec-23	88.75	-0.14%
18-Dec-23	88.77	0.02%
19-Dec-23	88.99	0.25%
20-Dec-23	89	0.01%
21-Dec-23	89.16	0.18%
22-Dec-23	89.04	-0.13%
26-Dec-23	89.36	0.36%
27-Dec-23	89.68	0.36%
28-Dec-23	89.43	-0.28%
29-Dec-23	89.06	-0.41%
2-Jan-24	88.18	-0.99%
3-Jan-24	87.84	-0.39%
4-Jan-24	87.44	-0.46%
5-Jan-24	87.17	-0.31%
8-Jan-24	87.28	0.13%
9-Jan-24	86.94	-0.39%
10-Jan-24	87.56	0.71%
11-Jan-24	88.35	0.90%
12-Jan-24	88.51	0.18%
16-Jan-24	87.69	-0.93%
17-Jan-24	87.36	-0.38%

18-Jan-24	87.45	0.10%
19-Jan-24	87.49	0.05%
22-Jan-24	87.55	0.07%
23-Jan-24	87.1	-0.51%
24-Jan-24	86.86	-0.28%
25-Jan-24	87.48	0.71%
26-Jan-24	87.61	0.15%
29-Jan-24	87.94	0.38%
30-Jan-24	88.14	0.23%
31-Jan-24	87.98	-0.18%
1-Feb-24	88.35	0.42%
2-Feb-24	87.64	-0.80%
5-Feb-24	86.9	-0.84%
6-Feb-24	87.69	0.91%
7-Feb-24	87.57	-0.14%
8-Feb-24	87.46	-0.13%
9-Feb-24	87.49	0.03%
12-Feb-24	87.57	0.09%
13-Feb-24	86.49	-1.23%
14-Feb-24	87.08	0.68%
15-Feb-24	87.79	0.82%
16-Feb-24	87.19	-0.68%
20-Feb-24	87.18	-0.01%
21-Feb-24	87.11	-0.08%
22-Feb-24	87.66	0.63%
23-Feb-24	88.15	0.56%
26-Feb-24	87.9	-0.28%
27-Feb-24	87.82	-0.09%
28-Feb-24	88	0.20%
29-Feb-24	88.3	0.34%
1-Mar-24	88.43	0.15%
4-Mar-24	88.35	-0.09%
5-Mar-24	88.54	0.22%
6-Mar-24	88.87	0.37%
7-Mar-24	89.11	0.27%
8-Mar-24	89.21	0.11%
11-Mar-24	89.07	-0.16%

12-Mar-24	88.95	-0.13%
13-Mar-24	89.04	0.10%
14-Mar-24	88.59	-0.51%
15-Mar-24	88.31	-0.32%
18-Mar-24	88.36	0.06%
19-Mar-24	88.77	0.46%
20-Mar-24	89.19	0.47%
21-Mar-24	89.5	0.35%
22-Mar-24	89.79	0.32%
25-Mar-24	89.72	-0.08%
26-Mar-24	89.71	-0.01%
27-Mar-24	90.08	0.41%
28-Mar-24	89.67	-0.46%
1-Apr-24	88.92	-0.84%
2-Apr-24	88.96	0.04%
3-Apr-24	89.15	0.21%
4-Apr-24	89.11	-0.04%
5-Apr-24	89.25	0.16%
8-Apr-24	89.28	0.03%
9-Apr-24	89.88	0.67%
10-Apr-24	88.46	-1.58%
11-Apr-24	88.2	-0.29%
12-Apr-24	87.95	-0.28%
15-Apr-24	86.93	-1.16%
16-Apr-24	86.66	-0.31%
17-Apr-24	87.34	0.78%
18-Apr-24	87.27	-0.08%
19-Apr-24	87.52	0.29%
22-Apr-24	87.99	0.54%
23-Apr-24	88.11	0.14%
24-Apr-24	87.48	-0.72%
25-Apr-24	87.23	-0.29%
26-Apr-24	87.61	0.44%
29-Apr-24	88.04	0.49%
30-Apr-24	87.15	-1.01%
1-May-24	87.15	0.00%
2-May-24	87.86	0.81%

3-May-24	88.58	0.82%
6-May-24	88.83	0.28%
7-May-24	88.9	0.08%
8-May-24	88.57	-0.37%
9-May-24	88.87	0.34%
10-May-24	88.66	-0.24%
13-May-24	88.75	0.10%
14-May-24	88.95	0.23%
15-May-24	89.88	1.05%
16-May-24	89.58	-0.33%
17-May-24	89.41	-0.19%
20-May-24	89.42	0.01%
21-May-24	89.49	0.08%
22-May-24	89.2	-0.32%
23-May-24	88.59	-0.68%
24-May-24	88.94	0.40%
28-May-24	88.41	-0.60%
29-May-24	88.09	-0.36%
30-May-24	88.62	0.60%
31-May-24	89.05	0.49%
3-Jun-24	89.09	0.04%
4-Jun-24	89.15	0.07%
5-Jun-24	89.39	0.27%
6-Jun-24	89.18	-0.23%
7-Jun-24	88.48	-0.78%
10-Jun-24	88.46	-0.02%
11-Jun-24	88.8	0.38%
12-Jun-24	89.15	0.39%
13-Jun-24	89.39	0.27%
14-Jun-24	89.28	-0.12%
17-Jun-24	89.01	-0.30%
18-Jun-24	89.6	0.66%
20-Jun-24	89.27	-0.37%
21-Jun-24	89.38	0.12%
24-Jun-24	89.42	0.04%
25-Jun-24	89.41	-0.01%
26-Jun-24	89.11	-0.34%

27-Jun-24	89.19	0.09%
28-Jun-24	88.48	-0.80%
1-Jul-24	87.75	-0.83%
2-Jul-24	88.18	0.49%
3-Jul-24	88.87	0.78%
5-Jul-24	89.45	0.65%
8-Jul-24	89.43	-0.02%
9-Jul-24	89.15	-0.31%
10-Jul-24	89.57	0.47%
11-Jul-24	90.09	0.58%
12-Jul-24	90.26	0.19%
15-Jul-24	89.89	-0.41%
16-Jul-24	90.29	0.44%
17-Jul-24	89.97	-0.35%
18-Jul-24	89.5	-0.52%
19-Jul-24	89.26	-0.27%
22-Jul-24	89.66	0.45%
23-Jul-24	89.69	0.03%
24-Jul-24	89.11	-0.65%
25-Jul-24	89.26	0.17%
26-Jul-24	89.76	0.56%
29-Jul-24	89.91	0.17%
30-Jul-24	90.03	0.13%
31-Jul-24	90.45	0.47%
1-Aug-24	90.01	-0.49%
2-Aug-24	90.61	0.67%
5-Aug-24	89.69	-1.02%
6-Aug-24	89.39	-0.33%
7-Aug-24	89.58	0.21%
8-Aug-24	90	0.47%
9-Aug-24	90.4	0.44%
12-Aug-24	90.5	0.11%
13-Aug-24	90.99	0.54%
14-Aug-24	91.19	0.22%
15-Aug-24	91.16	-0.03%
16-Aug-24	91.53	0.41%
19-Aug-24	91.81	0.31%

20-Aug-24	92.03	0.24%
21-Aug-24	92.3	0.29%
22-Aug-24	91.53	-0.83%
23-Aug-24	92.52	1.08%
26-Aug-24	92.33	-0.21%
27-Aug-24	92.21	-0.13%
28-Aug-24	92.06	-0.16%
29-Aug-24	92.02	-0.04%
30-Aug-24	92.1	0.09%
3-Sep-24	91.37	-0.79%
4-Sep-24	91.78	0.45%
5-Sep-24	92.13	0.38%
6-Sep-24	91.92	-0.23%
9-Sep-24	92.17	0.27%
10-Sep-24	92.24	0.08%
11-Sep-24	92.48	0.26%
12-Sep-24	92.63	0.16%
13-Sep-24	93.05	0.45%
16-Sep-24	93.62	0.61%
17-Sep-24	93.6	-0.02%
18-Sep-24	93.33	-0.29%
19-Sep-24	93.74	0.44%
20-Sep-24	93.56	-0.19%
23-Sep-24	93.29	-0.29%
24-Sep-24	93.41	0.13%
25-Sep-24	93.13	-0.30%
26-Sep-24	93.26	0.14%
27-Sep-24	93.42	0.17%
30-Sep-24	93.58	0.17%
1-Oct-24	93.46	-0.13%
2-Oct-24	93.25	-0.22%
3-Oct-24	92.86	-0.42%
4-Oct-24	92.51	-0.38%
7-Oct-24	92.09	-0.45%
8-Oct-24	92.24	0.16%
9-Oct-24	92.09	-0.16%
10-Oct-24	91.94	-0.16%

11-Oct-24	92.06	0.13%
14-Oct-24	92.25	0.21%
15-Oct-24	92.5	0.27%
16-Oct-24	92.73	0.25%
17-Oct-24	92.14	-0.64%
18-Oct-24	92.35	0.23%
21-Oct-24	91.44	-0.99%
22-Oct-24	91.23	-0.23%
23-Oct-24	90.89	-0.37%
24-Oct-24	91.26	0.41%
25-Oct-24	91.26	0.00%
28-Oct-24	91.35	0.10%
29-Oct-24	91.56	0.23%
30-Oct-24	91.46	-0.11%
31-Oct-24	90.92	-0.59%
1-Nov-24	90.14	-0.86%
4-Nov-24	90.53	0.43%
5-Nov-24	90.81	0.31%
6-Nov-24	90.52	-0.32%
7-Nov-24	91.51	1.09%
8-Nov-24	91.76	0.27%
11-Nov-24	91.57	-0.21%
12-Nov-24	90.86	-0.78%
13-Nov-24	90.73	-0.14%
14-Nov-24	90.5	-0.25%
15-Nov-24	90.37	-0.14%
18-Nov-24	90.48	0.12%
19-Nov-24	90.91	0.48%
20-Nov-24	90.92	0.01%
21-Nov-24	90.96	0.04%
22-Nov-24	90.85	-0.12%
25-Nov-24	91.7	0.94%
26-Nov-24	91.57	-0.14%
27-Nov-24	91.89	0.35%
29-Nov-24	92.15	0.28%
2-Dec-24	91.66	-0.53%
3-Dec-24	91.68	0.02%

4-Dec-24	92.02	0.37%
5-Dec-24	92.14	0.13%
6-Dec-24	92.45	0.34%
9-Dec-24	92.24	-0.23%
10-Dec-24	92.09	-0.16%
11-Dec-24	92.04	-0.05%
12-Dec-24	91.56	-0.52%
13-Dec-24	91.09	-0.51%
16-Dec-24	91.17	0.09%
17-Dec-24	91.12	-0.05%
18-Dec-24	89.33	-1.96%
19-Dec-24	88.94	-0.44%
20-Dec-24	89.56	0.70%
23-Dec-24	89.33	-0.26%
24-Dec-24	89.73	0.45%
26-Dec-24	89.64	-0.10%
27-Dec-24	89.3	-0.38%
30-Dec-24	89.21	-0.10%
31-Dec-24	89.04	-0.19%
2-Jan-25	89.34	0.34%
3-Jan-25	89.5	0.18%
6-Jan-25	89.54	0.04%
7-Jan-25	89.18	-0.40%
8-Jan-25	89.25	0.08%
10-Jan-25	88.77	-0.54%
13-Jan-25	88.64	-0.15%
14-Jan-25	88.77	0.15%
15-Jan-25	89.79	1.15%
16-Jan-25	89.68	-0.12%
17-Jan-25	89.85	0.19%
21-Jan-25	90.46	0.68%
22-Jan-25	90.21	-0.28%
23-Jan-25	90.19	-0.02%
24-Jan-25	90.34	0.17%
27-Jan-25	90.6	0.29%
28-Jan-25	90.56	-0.04%
29-Jan-25	90.48	-0.09%

30-Jan-25	90.86	0.42%
31-Jan-25	90.55	-0.34%
3-Feb-25	90.25	-0.33%
4-Feb-25	90.57	0.35%
5-Feb-25	91.08	0.56%
6-Feb-25	91	-0.09%
7-Feb-25	90.65	-0.38%
10-Feb-25	90.65	0.00%
11-Feb-25	90.26	-0.43%
12-Feb-25	89.97	-0.32%
13-Feb-25	90.71	0.82%
14-Feb-25	90.99	0.31%
18-Feb-25	90.6	-0.43%
19-Feb-25	90.62	0.02%
20-Feb-25	90.79	0.19%
21-Feb-25	90.82	0.03%
24-Feb-25	90.91	0.10%
25-Feb-25	91.48	0.63%
26-Feb-25	91.78	0.33%
27-Feb-25	91.47	-0.34%
28-Feb-25	92.03	0.61%
3-Mar-25	91.45	-0.63%
4-Mar-25	91.31	-0.15%
5-Mar-25	91.28	-0.03%
6-Mar-25	90.74	-0.59%
7-Mar-25	90.99	0.28%
10-Mar-25	90.87	-0.13%
11-Mar-25	90.69	-0.20%
12-Mar-25	90.67	-0.02%
13-Mar-25	90.63	-0.04%
14-Mar-25	90.72	0.10%
17-Mar-25	90.96	0.26%
18-Mar-25	90.93	-0.03%
19-Mar-25	91.43	0.55%
20-Mar-25	91.25	-0.20%
21-Mar-25	90.98	-0.30%
24-Mar-25	90.92	-0.07%

25-Mar-25	91.01	0.10%
26-Mar-25	90.49	-0.57%
27-Mar-25	90.29	-0.22%
28-Mar-25	90.39	0.11%
31-Mar-25	90.59	0.22%
1-Apr-25	90.48	-0.12%
2-Apr-25	90.44	-0.04%
3-Apr-25	89.9	-0.60%
4-Apr-25	88.59	-1.46%
7-Apr-25	87.16	-1.61%
8-Apr-25	86.54	-0.71%
9-Apr-25	88.77	2.58%
10-Apr-25	87.05	-1.94%
11-Apr-25	87.36	0.36%
14-Apr-25	88.27	1.04%
15-Apr-25	88.54	0.31%
16-Apr-25	88.57	0.03%
17-Apr-25	88.8	0.26%
21-Apr-25	87.77	-1.16%
22-Apr-25	88.68	1.04%
23-Apr-25	89.07	0.44%
24-Apr-25	90.02	1.07%
25-Apr-25	90.38	0.40%
28-Apr-25	90.48	0.11%
29-Apr-25	90.6	0.13%
30-Apr-25	89.96	-0.71%
1-May-25	89.6	-0.40%
2-May-25	89.36	-0.27%
5-May-25	89.18	-0.20%
6-May-25	89.55	0.41%
7-May-25	89.9	0.39%
8-May-25	89.54	-0.40%
9-May-25	89.68	0.16%
12-May-25	90.2	0.58%
13-May-25	90.28	0.09%
14-May-25	89.98	-0.33%
15-May-25	90.28	0.33%

16-May-25	90.45	0.19%
19-May-25	90.51	0.07%
20-May-25	90.37	-0.15%
21-May-25	89.54	-0.92%
22-May-25	89.84	0.34%
23-May-25	89.93	0.10%
27-May-25	90.48	0.61%
28-May-25	90.29	-0.21%
29-May-25	90.6	0.34%
30-May-25	90.6	0.00%
2-Jun-25	90.16	-0.49%
3-Jun-25	90.45	0.32%
4-Jun-25	90.79	0.38%
5-Jun-25	90.57	-0.24%
6-Jun-25	90.43	-0.15%
9-Jun-25	90.66	0.25%
10-Jun-25	91.06	0.44%
11-Jun-25	91.33	0.30%
12-Jun-25	91.49	0.18%
13-Jun-25	90.91	-0.63%
16-Jun-25	91.01	0.11%
17-Jun-25	91.08	0.08%
18-Jun-25	91.02	-0.07%
20-Jun-25	91.14	0.13%
23-Jun-25	91.4	0.29%
24-Jun-25	91.95	0.60%
25-Jun-25	91.9	-0.05%
26-Jun-25	92.18	0.30%
27-Jun-25	92.1	-0.09%
30-Jun-25	92.62	0.56%
1-Jul-25	92.42	-0.22%
2-Jul-25	92.33	-0.10%
3-Jul-25	92.49	0.17%
7-Jul-25	92.02	-0.51%
8-Jul-25	91.73	-0.32%
9-Jul-25	92.38	0.71%
10-Jul-25	92.24	-0.15%

11-Jul-25	91.76	-0.52%
14-Jul-25	91.77	0.01%
15-Jul-25	91.52	-0.27%
16-Jul-25	91.59	0.08%
17-Jul-25	91.69	0.11%
18-Jul-25	91.89	0.22%
21-Jul-25	92.24	0.38%
22-Jul-25	92.4	0.17%
23-Jul-25	92.44	0.04%
24-Jul-25	92.42	-0.02%
25-Jul-25	92.72	0.32%
28-Jul-25	92.66	-0.06%
29-Jul-25	93.01	0.38%
30-Jul-25	92.85	-0.17%
31-Jul-25	92.87	0.02%
1-Aug-25	92.72	-0.16%
4-Aug-25	93.18	0.50%
5-Aug-25	93.18	0.00%
6-Aug-25	93.24	0.06%
7-Aug-25	93.38	0.15%
8-Aug-25	93.38	0.00%
11-Aug-25	93.5	0.13%
12-Aug-25	93.68	0.19%
13-Aug-25	94.14	0.49%
14-Aug-25	93.93	-0.22%
15-Aug-25	93.85	-0.09%
18-Aug-25	93.94	0.10%
19-Aug-25	93.98	0.04%
20-Aug-25	93.83	-0.16%
21-Aug-25	93.53	-0.32%
22-Aug-25	94.21	0.73%
25-Aug-25	93.84	-0.39%
26-Aug-25	93.82	-0.02%
27-Aug-25	93.71	-0.12%
28-Aug-25	94.26	0.59%
29-Aug-25	94.12	-0.15%
2-Sep-25	93.36	-0.81%

3-Sep-25	93.76	0.43%
4-Sep-25	94.13	0.39%
5-Sep-25	94.76	0.67%
8-Sep-25	94.75	-0.01%
9-Sep-25	94.66	-0.09%
10-Sep-25	95.18	0.55%
11-Sep-25	95.53	0.37%
12-Sep-25	95.25	-0.29%
15-Sep-25	95.59	0.36%
16-Sep-25	95.53	-0.06%
17-Sep-25	95.48	-0.05%
18-Sep-25	95.08	-0.42%
19-Sep-25	94.98	-0.11%
22-Sep-25	95.19	0.22%
23-Sep-25	95.56	0.39%
24-Sep-25	95.5	-0.06%
25-Sep-25	95.26	-0.25%
26-Sep-25	95.23	-0.03%
29-Sep-25	95.39	0.17%
30-Sep-25	95.19	-0.21%
1-Oct-25	95.12	-0.07%
2-Oct-25	95.17	0.05%
3-Oct-25	95.21	0.04%
6-Oct-25	95.09	-0.13%
7-Oct-25	95.05	-0.04%
8-Oct-25	95.1	0.05%
9-Oct-25	95.05	-0.05%
10-Oct-25	94.63	-0.44%
13-Oct-25	95.28	0.69%
14-Oct-25	95.24	-0.04%
15-Oct-25	95.57	0.35%
16-Oct-25	95.71	0.15%
17-Oct-25	95.74	0.03%
20-Oct-25	96.16	0.44%
21-Oct-25	96.28	0.12%
22-Oct-25	96.08	-0.21%
23-Oct-25	96.01	-0.07%

24-Oct-25	96.29	0.29%
27-Oct-25	97.03	0.77%
28-Oct-25	97.03	0.00%
29-Oct-25	96.6	-0.44%
30-Oct-25	96.49	-0.11%
31-Oct-25	96.67	0.19%
3-Nov-25	95.97	-0.72%
4-Nov-25	95.86	-0.11%
5-Nov-25	95.65	-0.22%
6-Nov-25	95.97	0.33%
7-Nov-25	96.02	0.05%
10-Nov-25	96.24	0.23%
11-Nov-25	96.46	0.23%
12-Nov-25	96.5	0.04%
13-Nov-25	96.03	-0.49%
14-Nov-25	95.97	-0.06%
17-Nov-25	95.99	0.02%
18-Nov-25	95.96	-0.03%
19-Nov-25	96.12	0.17%
20-Nov-25	95.89	-0.24%
21-Nov-25	96.15	0.27%
24-Nov-25	96.35	0.21%
25-Nov-25	96.6	0.26%
26-Nov-25	96.7	0.10%
28-Nov-25	96.73	0.03%
1-Dec-25	96.02	-0.73%
2-Dec-25	96.42	0.42%
3-Dec-25	96.54	0.12%
4-Dec-25	96.46	-0.08%
5-Dec-25	96.33	-0.13%
8-Dec-25	95.93	-0.42%
9-Dec-25	95.85	-0.08%
10-Dec-25	96.24	0.41%
11-Dec-25	96.31	0.07%
12-Dec-25	96.26	-0.05%
15-Dec-25	96.39	0.14%
16-Dec-25	96.65	0.27%

17-Dec-25	96.28	-0.38%
18-Dec-25	96.7	0.44%
19-Dec-25	96.27	-0.44%
22-Dec-25	96.31	0.04%
23-Dec-25	96.26	-0.05%
24-Dec-25	96.55	0.30%
26-Dec-25	96.54	-0.01%
29-Dec-25	96.59	0.05%
30-Dec-25	96.47	-0.12%
31-Dec-25	96.28	-0.20%

Estimation note: Those of you who have with getting a proxy for returns on emergi estimate. I have finally replaced that estir deviations are now computed consistently approach would have yielded a ratio of 1.

Index: S&P Emerging BMI Index
Source: S&P
<https://www.spglobal.com/spdji/en/indices/equity/sp-emerging-bmi>

Period **Five years**(Dec 31, 2020 - Dec 31, 2025)

Interval Daily
S&P
Emerging

Effective date BMI (USD) Return

12/31/2020	331.34	
1/1/2021	331.57	0.07%
1/4/2021	334.34	0.84%
1/5/2021	337.8	1.03%
1/6/2021	336.87	-0.28%
1/7/2021	337.98	0.33%
1/8/2021	343.78	1.72%
1/11/2021	342.48	-0.38%
1/12/2021	344.33	0.54%
1/13/2021	346.61	0.66%
1/14/2021	348.89	0.66%
1/15/2021	346.19	-0.77%
1/18/2021	347.78	0.46%
1/19/2021	352.56	1.37%
1/20/2021	357.12	1.29%
1/21/2021	357.52	0.11%
1/22/2021	354.69	-0.79%
1/25/2021	357.84	0.89%
1/26/2021	353.68	-1.16%
1/27/2021	349.28	-1.24%
1/28/2021	344.95	-1.24%
1/29/2021	340.73	-1.22%
2/1/2021	347.93	2.11%
2/2/2021	353.35	1.56%
2/3/2021	355.98	0.74%

2/4/2021	355.5	-0.13%
2/5/2021	357.29	0.50%
2/8/2021	359.51	0.62%
2/9/2021	362.25	0.76%
2/10/2021	365.49	0.89%
2/11/2021	367.04	0.42%
2/12/2021	367.4	0.10%
2/15/2021	368.84	0.39%
2/16/2021	369.97	0.31%
2/17/2021	371.45	0.40%
2/18/2021	367.1	-1.17%
2/19/2021	368.17	0.29%
2/22/2021	360.27	-2.15%
2/23/2021	360.55	0.08%
2/24/2021	355.64	-1.36%
2/25/2021	355.86	0.06%
2/26/2021	345.68	-2.86%
3/1/2021	352.43	1.95%
3/2/2021	350.92	-0.43%
3/3/2021	355.27	1.24%
3/4/2021	347.71	-2.13%
3/5/2021	345.69	-0.58%
3/8/2021	337.39	-2.40%
3/9/2021	340.26	0.85%
3/10/2021	342.84	0.76%
3/11/2021	351.1	2.41%
3/12/2021	348.12	-0.85%
3/15/2021	346.23	-0.54%
3/16/2021	348.11	0.54%
3/17/2021	346.81	-0.37%
3/18/2021	347.35	0.16%
3/19/2021	345.31	-0.59%
3/22/2021	345.15	-0.05%
3/23/2021	341.91	-0.94%
3/24/2021	335.12	-1.99%
3/25/2021	332.33	-0.83%
3/26/2021	337	1.41%

3/29/2021	337.71	0.21%
3/30/2021	340.1	0.71%
3/31/2021	339.62	-0.14%
4/1/2021	344.11	1.32%
4/2/2021	344.2	0.03%
4/5/2021	343.95	-0.07%
4/6/2021	345.97	0.59%
4/7/2021	344.33	-0.47%
4/8/2021	345.8	0.43%
4/9/2021	342.87	-0.85%
4/12/2021	340.05	-0.82%
4/13/2021	339.76	-0.09%
4/14/2021	342.53	0.82%
4/15/2021	343.64	0.32%
4/16/2021	346.25	0.76%
4/19/2021	346.6	0.10%
4/20/2021	346.06	-0.16%
4/21/2021	344.58	-0.43%
4/22/2021	345.69	0.32%
4/23/2021	348.69	0.87%
4/26/2021	350.67	0.57%
4/27/2021	351.2	0.15%
4/28/2021	353.08	0.54%
4/29/2021	353.02	-0.02%
4/30/2021	349	-1.14%
5/3/2021	347.17	-0.52%
5/4/2021	344.99	-0.63%
5/5/2021	344.66	-0.10%
5/6/2021	346.65	0.58%
5/7/2021	348.66	0.58%
5/10/2021	347.55	-0.32%
5/11/2021	343.59	-1.14%
5/12/2021	340.16	-1.00%
5/13/2021	335.04	-1.51%
5/14/2021	338.29	0.97%
5/17/2021	339.63	0.40%
5/18/2021	345.37	1.69%

5/19/2021	343.94	-0.41%
5/20/2021	344.45	0.15%
5/21/2021	344.78	0.10%
5/24/2021	344.65	-0.04%
5/25/2021	348.92	1.24%
5/26/2021	350.79	0.54%
5/27/2021	352.03	0.35%
5/28/2021	353.55	0.43%
5/31/2021	357.07	1.00%
6/1/2021	360.55	0.97%
6/2/2021	360.82	0.07%
6/3/2021	359.36	-0.40%
6/4/2021	359.35	0.00%
6/7/2021	359	-0.10%
6/8/2021	358.41	-0.16%
6/9/2021	357.69	-0.20%
6/10/2021	359.54	0.52%
6/11/2021	359.7	0.04%
6/14/2021	360.42	0.20%
6/15/2021	358.56	-0.52%
6/16/2021	355.87	-0.75%
6/17/2021	354.57	-0.37%
6/18/2021	354.09	-0.14%
6/21/2021	351.85	-0.63%
6/22/2021	350.98	-0.25%
6/23/2021	354.53	1.01%
6/24/2021	356.13	0.45%
6/25/2021	359.35	0.90%
6/28/2021	359.43	0.02%
6/29/2021	358.53	-0.25%
6/30/2021	357.51	-0.28%
7/1/2021	355.92	-0.44%
7/2/2021	353.12	-0.79%
7/5/2021	353.41	0.08%
7/6/2021	350.16	-0.92%
7/7/2021	349.27	-0.25%
7/8/2021	343.35	-1.69%

7/9/2021	345.29	0.57%
7/12/2021	347.23	0.56%
7/13/2021	349.69	0.71%
7/14/2021	349.57	-0.03%
7/15/2021	351.76	0.63%
7/16/2021	349.79	-0.56%
7/19/2021	344.53	-1.50%
7/20/2021	342.95	-0.46%
7/21/2021	343.74	0.23%
7/22/2021	347.03	0.96%
7/23/2021	342.79	-1.22%
7/26/2021	334.85	-2.32%
7/27/2021	327.16	-2.30%
7/28/2021	331.47	1.32%
7/29/2021	338.17	2.02%
7/30/2021	334.62	-1.05%
8/2/2021	338.84	1.26%
8/3/2021	338.05	-0.23%
8/4/2021	340.17	0.63%
8/5/2021	339.06	-0.33%
8/6/2021	336.9	-0.64%
8/9/2021	337.54	0.19%
8/10/2021	339.36	0.54%
8/11/2021	339.28	-0.02%
8/12/2021	338.21	-0.32%
8/13/2021	336.45	-0.52%
8/16/2021	333.54	-0.86%
8/17/2021	329.49	-1.21%
8/18/2021	330.87	0.42%
8/19/2021	323.88	-2.11%
8/20/2021	321.23	-0.82%
8/23/2021	326.25	1.56%
8/24/2021	333.81	2.32%
8/25/2021	335.47	0.50%
8/26/2021	333.09	-0.71%
8/27/2021	335.25	0.65%
8/30/2021	338.88	1.08%

8/31/2021	343.62	1.40%
9/1/2021	345.15	0.45%
9/2/2021	345.18	0.01%
9/3/2021	346.12	0.27%
9/6/2021	347.97	0.53%
9/7/2021	348.89	0.26%
9/8/2021	345.54	-0.96%
9/9/2021	343.68	-0.54%
9/10/2021	345.63	0.57%
9/13/2021	344.47	-0.34%
9/14/2021	342.42	-0.60%
9/15/2021	340.71	-0.50%
9/16/2021	337.88	-0.83%
9/17/2021	337.68	-0.06%
9/20/2021	331.06	-1.96%
9/21/2021	332.2	0.34%
9/22/2021	333.07	0.26%
9/23/2021	336.36	0.99%
9/24/2021	334.03	-0.69%
9/27/2021	334.7	0.20%
9/28/2021	332.91	-0.53%
9/29/2021	330.45	-0.74%
9/30/2021	331.89	0.44%
10/1/2021	330.75	-0.34%
10/4/2021	328.07	-0.81%
10/5/2021	329.39	0.40%
10/6/2021	327.22	-0.66%
10/7/2021	333.88	2.04%
10/8/2021	334.74	0.26%
10/11/2021	335.98	0.37%
10/12/2021	333.96	-0.60%
10/13/2021	335.39	0.43%
10/14/2021	336.43	0.31%
10/15/2021	340.73	1.28%
10/18/2021	340.99	0.08%
10/19/2021	343.88	0.85%
10/20/2021	344.52	0.19%

10/21/2021	342.55	-0.57%
10/22/2021	342.18	-0.11%
10/25/2021	342.85	0.20%
10/26/2021	342.14	-0.21%
10/27/2021	339.39	-0.80%
10/28/2021	337.29	-0.62%
10/29/2021	334.9	-0.71%
11/1/2021	335.97	0.32%
11/2/2021	334.32	-0.49%
11/3/2021	335.12	0.24%
11/4/2021	336.3	0.35%
11/5/2021	335.24	-0.32%
11/8/2021	337.32	0.62%
11/9/2021	338.32	0.30%
11/10/2021	338.97	0.19%
11/11/2021	341.22	0.66%
11/12/2021	341.98	0.22%
11/15/2021	342.06	0.02%
11/16/2021	342.54	0.14%
11/17/2021	341.62	-0.27%
11/18/2021	337.38	-1.24%
11/19/2021	337.02	-0.11%
11/22/2021	333.63	-1.01%
11/23/2021	332.5	-0.34%
11/24/2021	332.53	0.01%
11/25/2021	333.41	0.26%
11/26/2021	325.05	-2.51%
11/29/2021	323.91	-0.35%
11/30/2021	322.84	-0.33%
12/1/2021	324.96	0.66%
12/2/2021	326.52	0.48%
12/3/2021	323.44	-0.94%
12/6/2021	322.01	-0.44%
12/7/2021	326.63	1.43%
12/8/2021	328.96	0.71%
12/9/2021	329.87	0.28%
12/10/2021	328.36	-0.46%

12/13/2021	326.16	-0.67%
12/14/2021	324.17	-0.61%
12/15/2021	321.59	-0.80%
12/16/2021	323.85	0.70%
12/17/2021	321.5	-0.73%
12/20/2021	314.64	-2.13%
12/21/2021	319.6	1.58%
12/22/2021	320.72	0.35%
12/23/2021	323.51	0.87%
12/24/2021	323.61	0.03%
12/27/2021	324.19	0.18%
12/28/2021	324.95	0.23%
12/29/2021	322.9	-0.63%
12/30/2021	325.86	0.92%
12/31/2021	327.39	0.47%
1/3/2022	328.07	0.21%
1/4/2022	327.93	-0.04%
1/5/2022	325.46	-0.75%
1/6/2022	324.43	-0.32%
1/7/2022	325.81	0.43%
1/10/2022	328.12	0.71%
1/11/2022	330.92	0.85%
1/12/2022	336.3	1.63%
1/13/2022	334.62	-0.50%
1/14/2022	333.92	-0.21%
1/17/2022	334.06	0.04%
1/18/2022	330.6	-1.04%
1/19/2022	330.67	0.02%
1/20/2022	334.27	1.09%
1/21/2022	330.67	-1.08%
1/24/2022	325.07	-1.69%
1/25/2022	322.73	-0.72%
1/26/2022	322.88	0.05%
1/27/2022	319.26	-1.12%
1/28/2022	318.61	-0.20%
1/31/2022	324.16	1.74%
2/1/2022	325.96	0.56%

2/2/2022	325.51	-0.14%
2/3/2022	324.19	-0.41%
2/4/2022	325.6	0.43%
2/7/2022	325.87	0.08%
2/8/2022	327.07	0.37%
2/9/2022	331.92	1.48%
2/10/2022	333.71	0.54%
2/11/2022	330.42	-0.99%
2/14/2022	325.08	-1.62%
2/15/2022	329.02	1.21%
2/16/2022	332.03	0.91%
2/17/2022	331.18	-0.26%
2/18/2022	328.52	-0.80%
2/21/2022	325.37	-0.96%
2/22/2022	321.67	-1.14%
2/23/2022	321.9	0.07%
2/24/2022	308.29	-4.23%
2/25/2022	313.45	1.67%
2/28/2022	312.85	-0.19%
3/1/2022	314.65	0.58%
3/2/2022	312.51	-0.68%
3/3/2022	312.6	0.03%
3/4/2022	305.89	-2.15%
3/7/2022	296.24	-3.15%
3/8/2022	288.91	-2.47%
3/9/2022	292.6	1.28%
3/10/2022	294.03	0.49%
3/11/2022	290.22	-1.30%
3/14/2022	282.39	-2.70%
3/15/2022	275.45	-2.46%
3/16/2022	290.45	5.45%
3/17/2022	298.56	2.79%
3/18/2022	300.37	0.61%
3/21/2022	298.5	-0.62%
3/22/2022	302.64	1.39%
3/23/2022	304.24	0.53%
3/24/2022	303.88	-0.12%

3/25/2022	301.17	-0.89%
3/28/2022	301.21	0.01%
3/29/2022	304.05	0.94%
3/30/2022	307.36	1.09%
3/31/2022	304.87	-0.81%
4/1/2022	307.12	0.74%
4/4/2022	311.65	1.47%
4/5/2022	309.92	-0.56%
4/6/2022	307.1	-0.91%
4/7/2022	302.59	-1.47%
4/8/2022	303.61	0.34%
4/11/2022	299.58	-1.33%
4/12/2022	299.17	-0.14%
4/13/2022	300.61	0.48%
4/14/2022	299.93	-0.23%
4/15/2022	299.17	-0.25%
4/18/2022	297.59	-0.53%
4/19/2022	294.92	-0.90%
4/20/2022	294.34	-0.20%
4/21/2022	291.91	-0.83%
4/22/2022	288.94	-1.02%
4/25/2022	281.29	-2.65%
4/26/2022	281.32	0.01%
4/27/2022	281.36	0.01%
4/28/2022	283.63	0.81%
4/29/2022	287.7	1.43%
5/2/2022	287.11	-0.21%
5/3/2022	286.72	-0.14%
5/4/2022	285.45	-0.44%
5/5/2022	283.43	-0.71%
5/6/2022	276.92	-2.30%
5/9/2022	271.36	-2.01%
5/10/2022	269.69	-0.62%
5/11/2022	270.1	0.15%
5/12/2022	264.84	-1.95%
5/13/2022	269.28	1.68%
5/16/2022	269.9	0.23%

5/17/2022	275.8	2.19%
5/18/2022	275.46	-0.12%
5/19/2022	272.17	-1.19%
5/20/2022	276.56	1.61%
5/23/2022	276.66	0.04%
5/24/2022	272.01	-1.68%
5/25/2022	272.47	0.17%
5/26/2022	274.97	0.92%
5/27/2022	278.9	1.43%
5/30/2022	283.68	1.71%
5/31/2022	286.8	1.10%
6/1/2022	284.32	-0.86%
6/2/2022	284.52	0.07%
6/3/2022	283.41	-0.39%
6/6/2022	286.03	0.92%
6/7/2022	284.61	-0.50%
6/8/2022	287.78	1.11%
6/9/2022	284.72	-1.06%
6/10/2022	281.85	-1.01%
6/13/2022	272.14	-3.45%
6/14/2022	273.59	0.53%
6/15/2022	274.91	0.48%
6/16/2022	270.71	-1.53%
6/17/2022	269.87	-0.31%
6/20/2022	268.97	-0.33%
6/21/2022	273.79	1.79%
6/22/2022	268.32	-2.00%
6/23/2022	269.44	0.42%
6/24/2022	273.15	1.38%
6/27/2022	276.75	1.32%
6/28/2022	276.66	-0.03%
6/29/2022	273.39	-1.18%
6/30/2022	270.16	-1.18%
7/1/2022	268.42	-0.64%
7/4/2022	268.43	0.00%
7/5/2022	267.56	-0.32%
7/6/2022	264.97	-0.97%

7/7/2022	268.51	1.34%
7/8/2022	269.23	0.27%
7/11/2022	264.19	-1.87%
7/12/2022	261.17	-1.14%
7/13/2022	261.8	0.24%
7/14/2022	259.8	-0.76%
7/15/2022	259.3	-0.19%
7/18/2022	263.72	1.70%
7/19/2022	264.4	0.26%
7/20/2022	265.78	0.52%
7/21/2022	266.7	0.35%
7/22/2022	266.48	-0.08%
7/25/2022	265.98	-0.19%
7/26/2022	265.78	-0.08%
7/27/2022	266.66	0.33%
7/28/2022	268.45	0.67%
7/29/2022	267.32	-0.42%
8/1/2022	268.36	0.39%
8/2/2022	265.89	-0.92%
8/3/2022	265.88	0.00%
8/4/2022	267.97	0.79%
8/5/2022	269.67	0.63%
8/8/2022	269.98	0.11%
8/9/2022	269.94	-0.01%
8/10/2022	269.96	0.01%
8/11/2022	273.68	1.38%
8/12/2022	274.43	0.27%
8/15/2022	274.16	-0.10%
8/16/2022	274.03	-0.05%
8/17/2022	274.42	0.14%
8/18/2022	273.38	-0.38%
8/19/2022	271.13	-0.82%
8/22/2022	269.26	-0.69%
8/23/2022	269.58	0.12%
8/24/2022	268.7	-0.33%
8/25/2022	273.05	1.62%
8/26/2022	273.3	0.09%

8/29/2022	270.16	-1.15%
8/30/2022	270.04	-0.04%
8/31/2022	270.04	0.00%
9/1/2022	266.43	-1.34%
9/2/2022	265.78	-0.24%
9/5/2022	265.18	-0.23%
9/6/2022	264.02	-0.44%
9/7/2022	262.8	-0.46%
9/8/2022	263.1	0.11%
9/9/2022	266.71	1.37%
9/12/2022	269.52	1.05%
9/13/2022	267.82	-0.63%
9/14/2022	264.52	-1.23%
9/15/2022	263.36	-0.44%
9/16/2022	258.71	-1.77%
9/19/2022	257.75	-0.37%
9/20/2022	259.27	0.59%
9/21/2022	255.58	-1.42%
9/22/2022	253.71	-0.73%
9/23/2022	249.03	-1.84%
9/26/2022	244.28	-1.91%
9/27/2022	245.01	0.30%
9/28/2022	241.76	-1.33%
9/29/2022	240.73	-0.43%
9/30/2022	241.86	0.47%
10/3/2022	242.13	0.11%
10/4/2022	247.15	2.07%
10/5/2022	249.7	1.03%
10/6/2022	249.86	0.06%
10/7/2022	246.52	-1.34%
10/10/2022	242.76	-1.53%
10/11/2022	238.05	-1.94%
10/12/2022	238.2	0.06%
10/13/2022	235.74	-1.03%
10/14/2022	237.27	0.65%
10/17/2022	238.83	0.66%
10/18/2022	241.48	1.11%

10/19/2022	237.94	-1.47%
10/20/2022	238.47	0.22%
10/21/2022	238.58	0.05%
10/24/2022	232.55	-2.53%
10/25/2022	232.64	0.04%
10/26/2022	235.08	1.05%
10/27/2022	236.08	0.43%
10/28/2022	232.7	-1.43%
10/31/2022	233.14	0.19%
11/1/2022	237.78	1.99%
11/2/2022	239.09	0.55%
11/3/2022	237.17	-0.80%
11/4/2022	243.36	2.61%
11/7/2022	245.9	1.04%
11/8/2022	245.9	0.00%
11/9/2022	244.96	-0.38%
11/10/2022	243.72	-0.51%
11/11/2022	252.99	3.80%
11/14/2022	254.5	0.60%
11/15/2022	259.58	2.00%
11/16/2022	257.61	-0.76%
11/17/2022	255.93	-0.65%
11/18/2022	255.38	-0.21%
11/21/2022	252.71	-1.05%
11/22/2022	252.28	-0.17%
11/23/2022	253.28	0.40%
11/24/2022	256.19	1.15%
11/25/2022	254.73	-0.57%
11/28/2022	253.35	-0.54%
11/29/2022	258.19	1.91%
11/30/2022	263.56	2.08%
12/1/2022	264.58	0.39%
12/2/2022	264.84	0.10%
12/5/2022	266.39	0.59%
12/6/2022	264.55	-0.69%
12/7/2022	261.37	-1.20%
12/8/2022	264.12	1.05%

12/9/2022	265	0.33%
12/12/2022	261.68	-1.25%
12/13/2022	262.49	0.31%
12/14/2022	264.1	0.61%
12/15/2022	260.86	-1.23%
12/16/2022	259.95	-0.35%
12/19/2022	259.83	-0.05%
12/20/2022	258.15	-0.65%
12/21/2022	258.75	0.23%
12/22/2022	260.16	0.54%
12/23/2022	258.07	-0.80%
12/26/2022	258.76	0.27%
12/27/2022	259.57	0.31%
12/28/2022	259.97	0.15%
12/29/2022	260.64	0.26%
12/30/2022	260.42	-0.08%
1/2/2023	260.49	0.03%
1/3/2023	262.48	0.76%
1/4/2023	265.35	1.09%
1/5/2023	267.16	0.68%
1/6/2023	268.1	0.35%
1/9/2023	272.96	1.81%
1/10/2023	273.58	0.23%
1/11/2023	274.13	0.20%
1/12/2023	274.59	0.17%
1/13/2023	277.44	1.04%
1/16/2023	277.59	0.05%
1/17/2023	276.54	-0.38%
1/18/2023	277.35	0.29%
1/19/2023	277.03	-0.12%
1/20/2023	279.04	0.73%
1/23/2023	279.8	0.27%
1/24/2023	279.85	0.02%
1/25/2023	279.63	-0.08%
1/26/2023	281.96	0.83%
1/27/2023	280.89	-0.38%
1/30/2023	279.37	-0.54%

1/31/2023	277.65	-0.62%
2/1/2023	280.16	0.90%
2/2/2023	280.39	0.08%
2/3/2023	278.47	-0.68%
2/6/2023	273.5	-1.78%
2/7/2023	273.51	0.00%
2/8/2023	274.32	0.30%
2/9/2023	275.18	0.31%
2/10/2023	272.36	-1.02%
2/13/2023	273.02	0.24%
2/14/2023	273	-0.01%
2/15/2023	271.49	-0.55%
2/16/2023	272.37	0.32%
2/17/2023	269.87	-0.92%
2/20/2023	271.38	0.56%
2/21/2023	268.9	-0.91%
2/22/2023	266.1	-1.04%
2/23/2023	266.82	0.27%
2/24/2023	263.04	-1.42%
2/27/2023	262.44	-0.23%
2/28/2023	261.96	-0.18%
3/1/2023	266.91	1.89%
3/2/2023	266.11	-0.30%
3/3/2023	268.01	0.71%
3/6/2023	269.11	0.41%
3/7/2023	267.35	-0.65%
3/8/2023	266.03	-0.49%
3/9/2023	263.09	-1.11%
3/10/2023	260.03	-1.16%
3/13/2023	260.31	0.11%
3/14/2023	257.08	-1.24%
3/15/2023	256.34	-0.29%
3/16/2023	255.29	-0.41%
3/17/2023	257.3	0.79%
3/20/2023	255.27	-0.79%
3/21/2023	257.87	1.02%
3/22/2023	259.75	0.73%

3/23/2023	263.1	1.29%
3/24/2023	262.02	-0.41%
3/27/2023	260.29	-0.66%
3/28/2023	261.61	0.51%
3/29/2023	264.2	0.99%
3/30/2023	265.65	0.55%
3/31/2023	266.8	0.43%
4/3/2023	267.08	0.10%
4/4/2023	266.65	-0.16%
4/5/2023	266.48	-0.06%
4/6/2023	266.43	-0.02%
4/7/2023	266.53	0.04%
4/10/2023	266.7	0.06%
4/11/2023	268.79	0.78%
4/12/2023	268.49	-0.11%
4/13/2023	269.14	0.24%
4/14/2023	269.76	0.23%
4/17/2023	270.72	0.36%
4/18/2023	270.08	-0.24%
4/19/2023	267.46	-0.97%
4/20/2023	267.21	-0.09%
4/21/2023	264.67	-0.95%
4/24/2023	264.33	-0.13%
4/25/2023	261.2	-1.18%
4/26/2023	262.14	0.36%
4/27/2023	263.25	0.42%
4/28/2023	264.77	0.58%
5/1/2023	264.69	-0.03%
5/2/2023	263.9	-0.30%
5/3/2023	263.07	-0.31%
5/4/2023	264.75	0.64%
5/5/2023	265.84	0.41%
5/8/2023	267.91	0.78%
5/9/2023	265.78	-0.80%
5/10/2023	265.53	-0.09%
5/11/2023	265.05	-0.18%
5/12/2023	264.03	-0.38%

5/15/2023	265.19	0.44%
5/16/2023	265.34	0.06%
5/17/2023	264.23	-0.42%
5/18/2023	264.45	0.08%
5/19/2023	263.67	-0.29%
5/22/2023	264.95	0.49%
5/23/2023	263.52	-0.54%
5/24/2023	261.59	-0.73%
5/25/2023	259.92	-0.64%
5/26/2023	261.86	0.75%
5/29/2023	261.94	0.03%
5/30/2023	261.22	-0.27%
5/31/2023	258.61	-1.00%
6/1/2023	259.74	0.44%
6/2/2023	264.87	1.98%
6/5/2023	265.62	0.28%
6/6/2023	265.96	0.13%
6/7/2023	267.9	0.73%
6/8/2023	267.49	-0.15%
6/9/2023	269.1	0.60%
6/12/2023	269.38	0.10%
6/13/2023	271.41	0.75%
6/14/2023	272.49	0.40%
6/15/2023	274.97	0.91%
6/16/2023	276.52	0.56%
6/19/2023	275.3	-0.44%
6/20/2023	272.67	-0.96%
6/21/2023	271.08	-0.58%
6/22/2023	270	-0.40%
6/23/2023	267.77	-0.83%
6/26/2023	266.78	-0.37%
6/27/2023	268.27	0.56%
6/28/2023	267.87	-0.15%
6/29/2023	267.24	-0.24%
6/30/2023	268.13	0.33%
7/3/2023	271.83	1.38%
7/4/2023	272.62	0.29%

7/5/2023	270.79	-0.67%
7/6/2023	266.88	-1.44%
7/7/2023	266.27	-0.23%
7/10/2023	266.65	0.14%
7/11/2023	269.35	1.01%
7/12/2023	271.79	0.91%
7/13/2023	275.27	1.28%
7/14/2023	276.8	0.56%
7/17/2023	277.06	0.09%
7/18/2023	275.45	-0.58%
7/19/2023	274.85	-0.22%
7/20/2023	274.93	0.03%
7/21/2023	274.46	-0.17%
7/24/2023	273.87	-0.21%
7/25/2023	278.67	1.75%
7/26/2023	278.96	0.10%
7/27/2023	279.7	0.27%
7/28/2023	281.96	0.81%
7/31/2023	283.04	0.38%
8/1/2023	281.66	-0.49%
8/2/2023	276.19	-1.94%
8/3/2023	275.31	-0.32%
8/4/2023	276.54	0.45%
8/7/2023	276.24	-0.11%
8/8/2023	273.3	-1.06%
8/9/2023	274.14	0.31%
8/10/2023	274.06	-0.03%
8/11/2023	271.55	-0.92%
8/14/2023	268.61	-1.08%
8/15/2023	267.47	-0.42%
8/16/2023	266.51	-0.36%
8/17/2023	266.16	-0.13%
8/18/2023	263.7	-0.92%
8/21/2023	262.63	-0.41%
8/22/2023	264.44	0.69%
8/23/2023	265.56	0.42%
8/24/2023	267.78	0.84%

8/25/2023	265.19	-0.97%
8/28/2023	266.92	0.65%
8/29/2023	269.68	1.03%
8/30/2023	269.99	0.11%
8/31/2023	267.94	-0.76%
9/1/2023	269.08	0.43%
9/4/2023	271.71	0.98%
9/5/2023	269.5	-0.81%
9/6/2023	268.59	-0.34%
9/7/2023	266.77	-0.68%
9/8/2023	267.02	0.09%
9/11/2023	268.09	0.40%
9/12/2023	267.32	-0.29%
9/13/2023	267.22	-0.04%
9/14/2023	268.83	0.60%
9/15/2023	269.11	0.10%
9/18/2023	267.07	-0.76%
9/19/2023	266.76	-0.12%
9/20/2023	265.62	-0.43%
9/21/2023	262.12	-1.32%
9/22/2023	264.29	0.83%
9/25/2023	262.67	-0.61%
9/26/2023	260.52	-0.82%
9/27/2023	260.96	0.17%
9/28/2023	260.06	-0.34%
9/29/2023	262.4	0.90%
10/2/2023	262	-0.15%
10/3/2023	258.64	-1.28%
10/4/2023	256.56	-0.80%
10/5/2023	257.05	0.19%
10/6/2023	258.93	0.73%
10/9/2023	258.01	-0.36%
10/10/2023	260.68	1.03%
10/11/2023	263.04	0.91%
10/12/2023	264.26	0.46%
10/13/2023	261.7	-0.97%
10/16/2023	260.54	-0.44%

10/17/2023	261.52	0.38%
10/18/2023	258.59	-1.12%
10/19/2023	256.02	-0.99%
10/20/2023	254.71	-0.51%
10/23/2023	252.58	-0.84%
10/24/2023	252.8	0.09%
10/25/2023	253.1	0.12%
10/26/2023	251.36	-0.69%
10/27/2023	253.85	0.99%
10/30/2023	254.42	0.22%
10/31/2023	252.89	-0.60%
11/1/2023	252.89	0.00%
11/2/2023	256.15	1.29%
11/3/2023	260.38	1.65%
11/6/2023	264.01	1.39%
11/7/2023	262.93	-0.41%
11/8/2023	262.82	-0.04%
11/9/2023	262.14	-0.26%
11/10/2023	260.39	-0.67%
11/13/2023	261.82	0.55%
11/14/2023	263.62	0.69%
11/15/2023	268.76	1.95%
11/16/2023	268.39	-0.14%
11/17/2023	267.7	-0.26%
11/20/2023	269.95	0.84%
11/21/2023	270.63	0.25%
11/22/2023	269.36	-0.47%
11/23/2023	270.63	0.47%
11/24/2023	269.23	-0.52%
11/27/2023	268.42	-0.30%
11/28/2023	270.3	0.70%
11/29/2023	269.84	-0.17%
11/30/2023	270.67	0.31%
12/1/2023	270.45	-0.08%
12/4/2023	270	-0.17%
12/5/2023	268.32	-0.62%
12/6/2023	269.23	0.34%

12/7/2023	268.43	-0.30%
12/8/2023	268.78	0.13%
12/11/2023	268.4	-0.14%
12/12/2023	268.92	0.19%
12/13/2023	268.64	-0.10%
12/14/2023	273.18	1.69%
12/15/2023	275.06	0.69%
12/18/2023	274.09	-0.35%
12/19/2023	274.46	0.13%
12/20/2023	272.72	-0.63%
12/21/2023	273.62	0.33%
12/22/2023	272.38	-0.45%
12/25/2023	272.5	0.04%
12/26/2023	273.88	0.51%
12/27/2023	276.49	0.95%
12/28/2023	279.58	1.12%
12/29/2023	280.09	0.18%
1/1/2024	280.45	0.13%
1/2/2024	278.12	-0.83%
1/3/2024	275.66	-0.88%
1/4/2024	276.35	0.25%
1/5/2024	276.12	-0.08%
1/8/2024	274.74	-0.50%
1/9/2024	273.95	-0.29%
1/10/2024	273.37	-0.21%
1/11/2024	274.96	0.58%
1/12/2024	275.6	0.23%
1/15/2024	275.55	-0.02%
1/16/2024	271.96	-1.30%
1/17/2024	266.71	-1.93%
1/18/2024	267.09	0.14%
1/19/2024	269.14	0.77%
1/22/2024	267.35	-0.67%
1/23/2024	268.06	0.27%
1/24/2024	271.92	1.44%
1/25/2024	273.65	0.64%
1/26/2024	272.8	-0.31%

1/29/2024	273.84	0.38%
1/30/2024	271.08	-1.01%
1/31/2024	270.26	-0.30%
2/1/2024	271.18	0.34%
2/2/2024	271.36	0.07%
2/5/2024	269.99	-0.50%
2/6/2024	275.04	1.87%
2/7/2024	275.66	0.23%
2/8/2024	274.5	-0.42%
2/9/2024	273.78	-0.26%
2/12/2024	273.49	-0.11%
2/13/2024	273.26	-0.08%
2/14/2024	274.52	0.46%
2/15/2024	277.24	0.99%
2/16/2024	279.36	0.76%
2/19/2024	279.29	-0.03%
2/20/2024	280.32	0.37%
2/21/2024	280.67	0.12%
2/22/2024	282.97	0.82%
2/23/2024	282.69	-0.10%
2/26/2024	282.06	-0.22%
2/27/2024	283.41	0.48%
2/28/2024	280.09	-1.17%
2/29/2024	281.25	0.41%
3/1/2024	282.53	0.46%
3/4/2024	283.43	0.32%
3/5/2024	281.72	-0.60%
3/6/2024	283.2	0.53%
3/7/2024	283.71	0.18%
3/8/2024	284.6	0.31%
3/11/2024	285.17	0.20%
3/12/2024	287.26	0.73%
3/13/2024	285.71	-0.54%
3/14/2024	286.34	0.22%
3/15/2024	283.88	-0.86%
3/18/2024	284.82	0.33%
3/19/2024	282.51	-0.81%

3/20/2024	283.09	0.21%
3/21/2024	286.22	1.11%
3/22/2024	284.14	-0.73%
3/25/2024	283.47	-0.24%
3/26/2024	283.75	0.10%
3/27/2024	283.13	-0.22%
3/28/2024	284.14	0.36%
3/29/2024	284.68	0.19%
4/1/2024	284.88	0.07%
4/2/2024	287.31	0.85%
4/3/2024	286.38	-0.32%
4/4/2024	287.41	0.36%
4/5/2024	287.29	-0.04%
4/8/2024	288.21	0.32%
4/9/2024	290.18	0.68%
4/10/2024	290.59	0.14%
4/11/2024	289.87	-0.25%
4/12/2024	286.72	-1.09%
4/15/2024	283.89	-0.99%
4/16/2024	278.69	-1.83%
4/17/2024	280.21	0.55%
4/18/2024	281.02	0.29%
4/19/2024	277.82	-1.14%
4/22/2024	279.23	0.51%
4/23/2024	281.38	0.77%
4/24/2024	284.66	1.17%
4/25/2024	283.96	-0.25%
4/26/2024	287.24	1.16%
4/29/2024	290.11	1.00%
4/30/2024	288.71	-0.48%
5/1/2024	288.62	-0.03%
5/2/2024	290.66	0.71%
5/3/2024	292.85	0.75%
5/6/2024	294.23	0.47%
5/7/2024	293.47	-0.26%
5/8/2024	292.92	-0.19%
5/9/2024	292.31	-0.21%

5/10/2024	294.28	0.67%
5/13/2024	295.85	0.53%
5/14/2024	296.77	0.31%
5/15/2024	298.22	0.49%
5/16/2024	301.06	0.95%
5/17/2024	302.56	0.50%
5/20/2024	302.94	0.13%
5/21/2024	301.33	-0.53%
5/22/2024	301.77	0.15%
5/23/2024	300.47	-0.43%
5/24/2024	299.11	-0.45%
5/27/2024	300.54	0.48%
5/28/2024	299.63	-0.30%
5/29/2024	296.22	-1.14%
5/30/2024	293.13	-1.04%
5/31/2024	291.24	-0.64%
6/3/2024	295.94	1.61%
6/4/2024	290.33	-1.90%
6/5/2024	292.82	0.86%
6/6/2024	295.66	0.97%
6/7/2024	295.47	-0.06%
6/10/2024	295.45	-0.01%
6/11/2024	294.31	-0.39%
6/12/2024	294.97	0.22%
6/13/2024	296.44	0.50%
6/14/2024	297.02	0.20%
6/17/2024	296.55	-0.16%
6/18/2024	298.47	0.65%
6/19/2024	301.07	0.87%
6/20/2024	301.24	0.06%
6/21/2024	299.6	-0.54%
6/24/2024	298.68	-0.31%
6/25/2024	298.75	0.02%
6/26/2024	298.88	0.04%
6/27/2024	297.45	-0.48%
6/28/2024	298.63	0.40%
7/1/2024	298.98	0.12%

7/2/2024	297.6	-0.46%
7/3/2024	300.83	1.09%
7/4/2024	303.18	0.78%
7/5/2024	302.88	-0.10%
7/8/2024	303.25	0.12%
7/9/2024	304.38	0.37%
7/10/2024	304.31	-0.02%
7/11/2024	307.86	1.17%
7/12/2024	308.16	0.10%
7/15/2024	307.23	-0.30%
7/16/2024	306.58	-0.21%
7/17/2024	305.28	-0.42%
7/18/2024	303.8	-0.48%
7/19/2024	299.57	-1.39%
7/22/2024	298.73	-0.28%
7/23/2024	298.71	-0.01%
7/24/2024	297.89	-0.27%
7/25/2024	296.36	-0.51%
7/26/2024	295.99	-0.12%
7/29/2024	296.47	0.16%
7/30/2024	295.65	-0.28%
7/31/2024	298.63	1.01%
8/1/2024	299.15	0.17%
8/2/2024	293.28	-1.96%
8/5/2024	282.46	-3.69%
8/6/2024	285.2	0.97%
8/7/2024	290.67	1.92%
8/8/2024	289.94	-0.25%
8/9/2024	293.97	1.39%
8/12/2024	295.38	0.48%
8/13/2024	295.65	0.09%
8/14/2024	296.8	0.39%
8/15/2024	296.84	0.01%
8/16/2024	300.81	1.34%
8/19/2024	303.88	1.02%
8/20/2024	303.81	-0.02%
8/21/2024	303.13	-0.22%

8/22/2024	302.77	-0.12%
8/23/2024	303.12	0.12%
8/26/2024	303.95	0.27%
8/27/2024	303.19	-0.25%
8/28/2024	302.46	-0.24%
8/29/2024	302.17	-0.10%
8/30/2024	303.82	0.55%
9/2/2024	302.55	-0.42%
9/3/2024	301.38	-0.39%
9/4/2024	297.86	-1.17%
9/5/2024	298.76	0.30%
9/6/2024	298.63	-0.04%
9/9/2024	295.65	-1.00%
9/10/2024	295.77	0.04%
9/11/2024	294.67	-0.37%
9/12/2024	298.54	1.31%
9/13/2024	300.61	0.69%
9/16/2024	301.71	0.37%
9/17/2024	302.93	0.40%
9/18/2024	301.92	-0.33%
9/19/2024	305.64	1.23%
9/20/2024	306.99	0.44%
9/23/2024	308.14	0.37%
9/24/2024	313.72	1.81%
9/25/2024	315.52	0.57%
9/26/2024	321.59	1.92%
9/27/2024	325.35	1.17%
9/30/2024	326.56	0.37%
10/1/2024	326.78	0.07%
10/2/2024	331.34	1.40%
10/3/2024	326.84	-1.36%
10/4/2024	328.42	0.48%
10/7/2024	329.27	0.26%
10/8/2024	323.21	-1.84%
10/9/2024	319.9	-1.02%
10/10/2024	322.12	0.69%
10/11/2024	323.19	0.33%

10/14/2024	322.75	-0.14%
10/15/2024	319.76	-0.93%
10/16/2024	319.03	-0.23%
10/17/2024	316.54	-0.78%
10/18/2024	322.13	1.77%
10/21/2024	320.47	-0.52%
10/22/2024	318.94	-0.48%
10/23/2024	318.5	-0.14%
10/24/2024	316.71	-0.56%
10/25/2024	316.75	0.01%
10/28/2024	317.5	0.24%
10/29/2024	316.75	-0.24%
10/30/2024	314.75	-0.63%
10/31/2024	313.97	-0.25%
11/1/2024	314.74	0.25%
11/4/2024	315.72	0.31%
11/5/2024	319.02	1.05%
11/6/2024	317.91	-0.35%
11/7/2024	320.57	0.84%
11/8/2024	318.7	-0.58%
11/11/2024	316.68	-0.63%
11/12/2024	310.99	-1.80%
11/13/2024	308.84	-0.69%
11/14/2024	306	-0.92%
11/15/2024	305.55	-0.15%
11/18/2024	305.42	-0.04%
11/19/2024	307.5	0.68%
11/20/2024	307.39	-0.04%
11/21/2024	305.15	-0.73%
11/22/2024	305.15	0.00%
11/25/2024	306.66	0.49%
11/26/2024	305.33	-0.43%
11/27/2024	306.08	0.25%
11/28/2024	303.97	-0.69%
11/29/2024	304.97	0.33%
12/2/2024	307.16	0.72%
12/3/2024	309.83	0.87%

12/4/2024	311.33	0.48%
12/5/2024	312.24	0.29%
12/6/2024	313.48	0.40%
12/9/2024	316.63	1.00%
12/10/2024	315.27	-0.43%
12/11/2024	314.52	-0.24%
12/12/2024	315.19	0.21%
12/13/2024	313.1	-0.66%
12/16/2024	311.84	-0.40%
12/17/2024	309.48	-0.76%
12/18/2024	309.44	-0.01%
12/19/2024	306.42	-0.98%
12/20/2024	304.12	-0.75%
12/23/2024	305.92	0.59%
12/24/2024	306.89	0.32%
12/25/2024	307.25	0.12%
12/26/2024	307.3	0.02%
12/27/2024	307.06	-0.08%
12/30/2024	306.06	-0.33%
12/31/2024	305.14	-0.30%
1/1/2025	305.55	0.13%
1/2/2025	303.61	-0.63%
1/3/2025	303.2	-0.14%
1/6/2025	303.61	0.14%
1/7/2025	304.38	0.25%
1/8/2025	301.21	-1.04%
1/9/2025	299.64	-0.52%
1/10/2025	296.73	-0.97%
1/13/2025	291.35	-1.81%
1/14/2025	295.64	1.47%
1/15/2025	296.27	0.21%
1/16/2025	298.95	0.90%
1/17/2025	299.97	0.34%
1/20/2025	303.1	1.04%
1/21/2025	302.9	-0.07%
1/22/2025	302.75	-0.05%
1/23/2025	302.9	0.05%

1/24/2025	304.86	0.65%
1/27/2025	303.46	-0.46%
1/28/2025	303.43	-0.01%
1/29/2025	304.7	0.42%
1/30/2025	305.57	0.29%
1/31/2025	305.64	0.02%
2/3/2025	300.8	-1.58%
2/4/2025	305.59	1.59%
2/5/2025	306.05	0.15%
2/6/2025	307.83	0.58%
2/7/2025	309.75	0.62%
2/10/2025	309.4	-0.11%
2/11/2025	307.91	-0.48%
2/12/2025	309.55	0.53%
2/13/2025	309.86	0.10%
2/14/2025	312.55	0.87%
2/17/2025	313.86	0.42%
2/18/2025	315.27	0.45%
2/19/2025	314.64	-0.20%
2/20/2025	314.12	-0.17%
2/21/2025	317.8	1.17%
2/24/2025	314.64	-0.99%
2/25/2025	311.02	-1.15%
2/26/2025	314.5	1.12%
2/27/2025	311.56	-0.93%
2/28/2025	305.21	-2.04%
3/3/2025	304.94	-0.09%
3/4/2025	304.04	-0.30%
3/5/2025	310.35	2.08%
3/6/2025	314.26	1.26%
3/7/2025	313.8	-0.15%
3/10/2025	309.35	-1.42%
3/11/2025	308.67	-0.22%
3/12/2025	308.53	-0.05%
3/13/2025	307.02	-0.49%
3/14/2025	310.98	1.29%
3/17/2025	313.53	0.82%

3/18/2025	317.4	1.23%
3/19/2025	316.7	-0.22%
3/20/2025	315.98	-0.23%
3/21/2025	313.55	-0.77%
3/24/2025	314.91	0.43%
3/25/2025	313.23	-0.53%
3/26/2025	313.31	0.03%
3/27/2025	313.56	0.08%
3/28/2025	311.11	-0.78%
3/31/2025	306.47	-1.49%
4/1/2025	308.62	0.70%
4/2/2025	309.19	0.18%
4/3/2025	307.12	-0.67%
4/4/2025	302.13	-1.62%
4/7/2025	278.73	-7.75%
4/8/2025	278.6	-0.05%
4/9/2025	276.02	-0.93%
4/10/2025	284.54	3.09%
4/11/2025	289.65	1.80%
4/14/2025	293.49	1.33%
4/15/2025	296.61	1.06%
4/16/2025	294.01	-0.88%
4/17/2025	295.97	0.67%
4/18/2025	296.23	0.09%
4/21/2025	297.09	0.29%
4/22/2025	297.98	0.30%
4/23/2025	303.69	1.92%
4/24/2025	303.45	-0.08%
4/25/2025	304.07	0.20%
4/28/2025	305.74	0.55%
4/29/2025	306.79	0.34%
4/30/2025	308.17	0.45%
5/1/2025	307.92	-0.08%
5/2/2025	313.79	1.91%
5/5/2025	315.8	0.64%
5/6/2025	315.11	-0.22%
5/7/2025	314.85	-0.08%

5/8/2025	313.66	-0.38%
5/9/2025	315	0.43%
5/12/2025	321.47	2.05%
5/13/2025	320.79	-0.21%
5/14/2025	325.37	1.43%
5/15/2025	324.98	-0.12%
5/16/2025	324.91	-0.02%
5/19/2025	323.91	-0.31%
5/20/2025	323.92	0.00%
5/21/2025	325.76	0.57%
5/22/2025	323.55	-0.68%
5/23/2025	325.01	0.45%
5/26/2025	324.26	-0.23%
5/27/2025	322.84	-0.44%
5/28/2025	322.45	-0.12%
5/29/2025	323.95	0.47%
5/30/2025	320.91	-0.94%
6/2/2025	319.66	-0.39%
6/3/2025	320.96	0.41%
6/4/2025	324.07	0.97%
6/5/2025	326.11	0.63%
6/6/2025	326.25	0.04%
6/9/2025	328.81	0.78%
6/10/2025	330.94	0.65%
6/11/2025	333.08	0.65%
6/12/2025	331.02	-0.62%
6/13/2025	327.57	-1.04%
6/16/2025	329.36	0.55%
6/17/2025	328.52	-0.26%
6/18/2025	327.28	-0.38%
6/19/2025	322.56	-1.44%
6/20/2025	324.69	0.66%
6/23/2025	323.44	-0.38%
6/24/2025	330.54	2.20%
6/25/2025	333.11	0.78%
6/26/2025	335.25	0.64%
6/27/2025	336.18	0.28%

6/30/2025	334.75	-0.43%
7/1/2025	336.52	0.53%
7/2/2025	336.99	0.14%
7/3/2025	338.7	0.51%
7/4/2025	337.89	-0.24%
7/7/2025	336.26	-0.48%
7/8/2025	337.13	0.26%
7/9/2025	336.61	-0.15%
7/10/2025	336.96	0.10%
7/11/2025	336.57	-0.12%
7/14/2025	336.32	-0.07%
7/15/2025	339.19	0.85%
7/16/2025	339.25	0.02%
7/17/2025	340.28	0.30%
7/18/2025	342.4	0.62%
7/21/2025	343.12	0.21%
7/22/2025	341.96	-0.34%
7/23/2025	346.19	1.24%
7/24/2025	346.65	0.13%
7/25/2025	343.93	-0.78%
7/28/2025	342.88	-0.31%
7/29/2025	342.29	-0.17%
7/30/2025	341.55	-0.22%
7/31/2025	338.87	-0.78%
8/1/2025	335.99	-0.85%
8/4/2025	338.36	0.71%
8/5/2025	340.49	0.63%
8/6/2025	340.28	-0.06%
8/7/2025	343.87	1.06%
8/8/2025	342.46	-0.41%
8/11/2025	343.1	0.19%
8/12/2025	343.84	0.22%
8/13/2025	348.48	1.35%
8/14/2025	347.2	-0.37%
8/15/2025	347.79	0.17%
8/18/2025	349.38	0.46%
8/19/2025	349.18	-0.06%

8/20/2025	346.7	-0.71%
8/21/2025	347.3	0.17%
8/22/2025	348.26	0.28%
8/25/2025	352.98	1.36%
8/26/2025	350.5	-0.70%
8/27/2025	348.94	-0.45%
8/28/2025	347.04	-0.54%
8/29/2025	346.49	-0.16%
9/1/2025	348.88	0.69%
9/2/2025	347.55	-0.38%
9/3/2025	347.88	0.09%
9/4/2025	346.05	-0.53%
9/5/2025	350.41	1.26%
9/8/2025	352.03	0.46%
9/9/2025	354.69	0.76%
9/10/2025	357.03	0.66%
9/11/2025	357.66	0.18%
9/12/2025	360.76	0.87%
9/15/2025	361.51	0.21%
9/16/2025	363.77	0.63%
9/17/2025	366.53	0.76%
9/18/2025	365.54	-0.27%
9/19/2025	364.57	-0.27%
9/22/2025	364.61	0.01%
9/23/2025	364.54	-0.02%
9/24/2025	366.03	0.41%
9/25/2025	363.71	-0.63%
9/26/2025	359.51	-1.15%
9/29/2025	362.55	0.85%
9/30/2025	364.78	0.62%
10/1/2025	366.18	0.38%
10/2/2025	368.66	0.68%
10/3/2025	370.29	0.44%
10/6/2025	369.8	-0.13%
10/7/2025	370.48	0.18%
10/8/2025	369.62	-0.23%
10/9/2025	370.81	0.32%

10/10/2025	367.22	-0.97%
10/13/2025	364.46	-0.75%
10/14/2025	360.36	-1.12%
10/15/2025	366.8	1.79%
10/16/2025	368.91	0.58%
10/17/2025	363.53	-1.46%
10/20/2025	368.52	1.37%
10/21/2025	369.35	0.23%
10/22/2025	367.94	-0.38%
10/23/2025	368.92	0.27%
10/24/2025	370.04	0.30%
10/27/2025	373.91	1.05%
10/28/2025	373.25	-0.18%
10/29/2025	375.51	0.61%
10/30/2025	373.69	-0.48%
10/31/2025	370.73	-0.79%
11/3/2025	371.77	0.28%
11/4/2025	368.33	-0.93%
11/5/2025	367.59	-0.20%
11/6/2025	370.25	0.72%
11/7/2025	367.93	-0.63%
11/10/2025	371.25	0.90%
11/11/2025	371.58	0.09%
11/12/2025	372.68	0.30%
11/13/2025	373.46	0.21%
11/14/2025	368.52	-1.32%
11/17/2025	368.05	-0.13%
11/18/2025	362.38	-1.54%
11/19/2025	362.15	-0.06%
11/20/2025	364.11	0.54%
11/21/2025	355.42	-2.39%
11/24/2025	357.87	0.69%
11/25/2025	360.7	0.79%
11/26/2025	364.18	0.96%
11/27/2025	363.81	-0.10%
11/28/2025	364.17	0.10%
12/1/2025	364.42	0.07%
12/2/2025	364.7	0.08%
12/3/2025	364.08	-0.17%
12/4/2025	365.33	0.34%

12/5/2025	366.4	0.29%
12/8/2025	365.82	-0.16%
12/9/2025	364.47	-0.37%
12/10/2025	365.16	0.19%
12/11/2025	364.12	-0.28%
12/12/2025	367.39	0.90%
12/15/2025	364.23	-0.86%
12/16/2025	359.32	-1.35%
12/17/2025	360.28	0.27%
12/18/2025	360.35	0.02%
12/19/2025	363.45	0.86%
12/22/2025	366.63	0.87%
12/23/2025	367.5	0.24%
12/24/2025	368.01	0.14%
12/25/2025	367.91	-0.03%
12/26/2025	368.38	0.13%
12/29/2025	367.86	-0.14%
12/30/2025	368.49	0.17%
12/31/2025	369.16	0.18%

Following my attempts to estimate the relative equity market volatility, I know that I have struggled with estimating market sovereign bonds, and have used the coefficient of variation in yields to make that estimate with a return on an emerging market government bond ETF from iShares. The standard deviation for emerging equities and emerging government bonds. (Using the old yield coefficient of variation of 49 in this computation.)

[/overview](#)

Row Labels	Data	
	Average of Adj. Default Sprea	Average of Country Risk Premi
Africa	5.42%	8.26%
Asia	3.17%	4.83%
Australia & New Zealand	1.28%	1.94%
Caribbean	3.08%	4.69%
Central and South America	4.57%	6.96%
Eastern Europe & Russia	3.29%	5.02%
Middle East	3.10%	4.73%
North America	0.12%	0.12%
Western Europe	0.63%	0.96%
Total Result	3.28%	5.00%

Average of Equity Risk Premi		Average of Corporate Tax Rate	
	12.49%		27.59%
	9.06%		24.05%
	6.17%		26.00%
	8.92%		17.71%
	11.19%		27.13%
	9.25%		16.42%
	8.96%		14.68%
	4.35%		25.57%
	5.19%		19.94%
	9.23%		21.87%

<i>Region</i>	<i>Weighted Average: ERP</i>	<i>Weighted Average: CRP</i>	<i>Weighted Average: Default Spread</i>	<i>Tax Rate</i>	<i>Total GDP</i>	<i>Weight</i>
Africa	11.95%	7.72%	5.07%	27.39%	2340875.22920722	2.18%
Asia	5.72%	1.49%	0.98%	25.65%	34731689.3748114	32.28%
Australia & New Zealand	4.23%	0.00%	0.00%	29.74%	2018608.83675045	1.88%
Caribbean	11.72%	7.49%	4.92%	26.53%	347814.13246985	0.32%
Central and South America	8.46%	4.23%	2.78%	31.55%	6538486.8558457	6.08%
Eastern Europe	7.58%	3.35%	2.20%	17.64%	3491477.19313994	3.25%
Middle East	6.24%	2.01%	1.32%	16.12%	3565018.31247331	3.31%
North America	4.45%	0.22%	0.22%	25.08%	30994592.957365	28.81%
Western Europe	5.27%	1.04%	0.68%	25.56%	23561725.6351465	21.90%
Global	5.63%	1.40%	0.94%	25.37%	107590288.527209	100.00%

For updating industry average spreadsheets

	<i>ERP</i>	<i>Default Spread</i>	<i>Tax rate</i>
Africa & Mid East	8.50%	2.80%	20.59%
Australia, NZ & Canada	4.23%	0.00%	27.84%
Latin America & Caribbean	8.63%	2.89%	31.29%
Japan	5.14%	0.60%	29.74%
US	4.46%	0.23%	25.00%
Europe	5.27%	0.68%	25.56%
Emerging Markets	6.68%	1.57%	25.11%
Small Asia (No India, China & J)	6.70%	1.62%	25.27%
India	7.08%	1.87%	30.00%
China	5.14%	0.60%	25.00%
Global	5.63%	0.94%	25.37%

Country	GDP (in millions) in 2024	Moody's rating	Sovereign CDS
Abu Dhabi	310000.00	Aa2	0.60%
Albania	27046.43	Ba3	NA
Andorra (Principality of)	4039.84	Baa1	NA
Angola	100998.92	B3	6.24%
Argentina	638365.46	Caa1	NA
Armenia	25955.28	Ba3	NA
Aruba	4265.65	Baa3	NA
Australia	1757022.45	Aaa	0.19%
Austria	534790.72	Aa1	0.28%
Azerbaijan	74315.88	Baa3	NA
Bahamas	15832.80	B1	NA
Bahrain	47109.73	B2	2.47%
Bangladesh	450119.43	B2	NA
Barbados	7498.05	B2	NA
Belarus	75961.87	C	NA
Belgium	671370.08	Aa3	0.40%
Belize	3203.63	Caa1	NA
Benin	21482.64	B1	NA
Bermuda	9233.60	A2	NA
Bolivia	54881.33	Ca	NA
Bosnia and Herzegovina	29613.57	B3	NA
Botswana	19402.06	Baa1	NA
Brazil	2185821.65	Ba1	2.35%
Bulgaria	113343.36	Baa1	0.77%
Burkina Faso	23124.73	Caa1	NA
Cambodia	46352.65	B2	NA
Cameroon	53296.69	Caa1	6.87%
Canada	2243636.83	Aaa	0.33%
Cape Verde	2725.41	B2	NA
Cayman Islands	7241.24	Aa3	NA
Chile	330267.14	A2	0.87%
China	18743803.17	A1	0.64%
Colombia	418818.15	Baa3	3.34%
Congo (Democratic Republic of)	70962.19	B3	NA
Congo (Republic of)	15719.99	Caa2	NA
Cook Islands	1414.00	B1	NA
Costa Rica	95350.42	Ba2	1.75%
Côte d'Ivoire	87113.18	Ba2	NA
Croatia	92983.81	A3	1.02%
Cuba	107352.00	Ca	NA
Curacao	3561.18	Baa3	NA
Cyprus	37634.53	A3	0.75%

Czech Republic	347034.06	Aa3	0.48%
Denmark	424524.72	Aaa	0.18%
Dominican Republic	124282.25	Ba2	NA
Ecuador	124676.07	Caa3	5.67%
Egypt	389059.91	Caa1	3.55%
El Salvador	35364.96	B3	3.40%
Estonia	43130.42	A1	1.00%
Ethiopia	149740.30	Caa2	NA
Fiji	5968.13	B1	NA
Finland	298696.96	Aa1	0.26%
France	3160442.62	Aa3	0.64%
Gabon	20895.68	Caa2	NA
Georgia	34189.42	Ba2	NA
Germany	4685592.58	Aaa	0.21%
Ghana	82308.11	Caa1	NA
Greece	256238.37	Baa3	0.75%
Guatemala	113199.58	Ba1	NA
Guernsey (States of)	3446.00	A1	NA
Honduras	37093.57	B1	NA
Hong Kong	406863.40	Aa3	0.44%
Hungary	222722.74	Baa2	1.60%
Iceland	33255.18	A1	0.45%
India	3909891.53	Baa3	0.80%
Indonesia	1396300.10	Baa2	1.19%
Iraq	279641.26	Caa1	2.89%
Ireland	609157.46	Aa3	0.34%
Isle of Man	7431.26	Aa3	NA
Israel	540379.92	Baa1	1.13%
Italy	2380825.08	Baa2	0.61%
Jamaica	22014.40	Ba3	NA
Japan	4027597.52	A1	0.44%
Jersey (States of)	4890.00	Aa3	NA
Jordan	53352.29	Ba3	NA
Kazakhstan	291480.27	Baa1	1.31%
Kenya	120339.56	B3	4.51%
Korea	1875388.21	Aa2	0.34%
Kuwait	160227.27	A1	0.87%
Kyrgyzstan	17478.26	B3	NA
Laos	16502.93	Caa2	NA
Latvia	43684.25	A3	0.82%
Lebanon	20078.62	C	NA
Liechtenstein	8239.38	Aaa	NA
Lithuania	84869.22	A2	0.82%

Luxembourg	93279.85	Aaa	NA
Macao	49467.26	Aa3	NA
Macedonia	16951.68	Ba3	NA
Malaysia	422227.01	A3	0.67%
Maldives	7061.61	Caa2	NA
Mali	26794.75	Caa2	NA
Malta	24971.57	A2	NA
Mauritius	14937.86	Baa3	NA
Mexico	1856365.62	Baa2	1.66%
Moldova	18200.34	B3	NA
Mongolia	23794.54	B1	NA
Montenegro	8270.02	B1	NA
Montserrat	16199.00	Baa3	NA
Morocco	160610.99	Ba1	1.30%
Mozambique	22745.34	Caa3	NA
Namibia	13372.35	B1	NA
Nepal	42914.27	Ba3	NA
Netherlands	1214927.70	Aaa	0.20%
New Zealand	260172.39	Aaa	0.23%
Nicaragua	19693.98	B2	5.29%
Niger	19876.13	Caa3	NA
Nigeria	252261.88	B3	3.91%
Norway	483592.65	Aaa	0.18%
Oman	107137.20	Baa3	1.30%
Pakistan	371570.00	Caa1	5.23%
Panama	86523.96	Baa3	2.22%
Papua New Guinea	31800.43	B2	NA
Paraguay	44458.12	Baa3	NA
Peru	289221.97	Baa1	1.26%
Philippines	461617.51	Baa2	1.02%
Poland	917767.11	A2	1.00%
Portugal	313271.19	A3	0.45%
Qatar	219162.64	Aa2	0.61%
Ras Al Khaimah (Emirate of)	11000.00	A3	NA
Romania	382564.22	Baa3	2.11%
Rwanda	14251.64	B2	4.03%
Saudi Arabia	1239804.53	Aa3	1.12%
Senegal	32808.06	Caa1	9.93%
Serbia	90097.77	Ba2	1.96%
Sharjah	24800.00	Ba1	#N/A
Singapore	547386.65	Aaa	#N/A
Slovakia	140934.08	A3	0.57%
Slovenia	72972.02	A3	0.67%

Solomon Islands	1583.96	Caa1	#N/A
South Africa	401145.00	Ba2	2.40%
Spain	1725671.65	A3	0.45%
Sri Lanka	98963.19	Ca	NA
St. Maarten	1797.84	Ba2	#N/A
St. Vincent & the Grenadin	1157.21	B3	#N/A
Suriname	4416.78	Caa1	#N/A
Swaziland	4858.89	B2	#N/A
Sweden	603715.22	Aaa	0.20%
Switzerland	936564.20	Aaa	0.14%
Taiwan	791610.00	Aa3	#N/A
Tajikistan	14204.58	B3	#N/A
Tanzania	78844.41	B1	#N/A
Thailand	526517.66	Baa1	0.66%
Togo	10651.18	B3	#N/A
Trinidad and Tobago	25633.54	Ba2	#N/A
Tunisia	51332.29	Caa1	7.01%
Turkey	1359123.77	Ba3	2.99%
Turks and Caicos Islands	1745.38	Baa1	#N/A
Uganda	53911.91	B3	#N/A
Ukraine	190741.26	Ca	NA
United Arab Emirates	552324.85	Aa2	#N/A
United Kingdom	3686033.04	Aa3	0.37%
United States	28750956.13	Aa1	0.44%
Uruguay	80961.51	Baa1	0.77%
Uzbekistan	114965.29	Ba3	#N/A
Venezuela	119802.96	C	9.29%
Vietnam	476388.23	Ba2	1.51%
Zambia	25303.19	Caa2	4.08%

<i>Adj. Default Spread</i>	<i>Equity Risk Premium</i>	<i>Country Risk Premium</i>	<i>Corporate Tax Rate</i>
0.42%	4.87%	0.64%	9.00%
3.06%	8.89%	4.66%	15.00%
1.36%	6.30%	2.07%	10.00%
5.52%	12.64%	8.41%	25.00%
6.37%	13.94%	9.71%	35.00%
3.06%	8.89%	4.66%	18.00%
1.87%	7.08%	2.85%	22.00%
0.00%	4.23%	0.00%	30.00%
0.23%	4.59%	0.36%	23.00%
1.87%	7.08%	2.85%	20.00%
3.83%	10.06%	5.83%	0.00%
4.67%	11.35%	7.12%	0.00%
4.67%	11.35%	7.12%	27.50%
4.67%	11.35%	7.12%	9.00%
17.50%	30.89%	26.66%	25.00%
0.51%	5.01%	0.78%	25.00%
6.37%	13.94%	9.71%	0.00%
3.83%	10.06%	5.83%	30.00%
0.72%	5.33%	1.10%	0.00%
10.20%	19.77%	15.54%	25.00%
5.52%	12.64%	8.41%	10.00%
1.36%	6.30%	2.07%	22.00%
2.13%	7.47%	3.24%	34.00%
1.36%	6.30%	2.07%	10.00%
6.37%	13.94%	9.71%	27.50%
4.67%	11.35%	7.12%	20.00%
6.37%	13.94%	9.71%	33.00%
0.00%	4.23%	0.00%	26.14%
4.67%	11.35%	7.12%	26.86%
0.51%	5.01%	0.78%	0.00%
0.72%	5.33%	1.10%	27.00%
0.60%	5.14%	0.91%	25.00%
1.87%	7.08%	2.85%	35.00%
5.52%	12.64%	8.41%	30.00%
7.65%	15.89%	11.66%	26.86%
3.83%	10.06%	5.83%	20.00%
2.56%	8.13%	3.90%	30.00%
2.56%	8.13%	3.90%	25.00%
1.02%	5.78%	1.55%	18.00%
10.20%	19.77%	15.54%	35.00%
1.87%	7.08%	2.85%	22.00%
1.02%	5.78%	1.55%	12.50%

0.51%	5.01%	0.78%	21.00%
0.00%	4.23%	0.00%	22.00%
2.56%	8.13%	3.90%	27.00%
8.50%	17.18%	12.95%	25.00%
6.37%	13.94%	9.71%	22.50%
5.52%	12.64%	8.41%	30.00%
0.60%	5.14%	0.91%	20.00%
7.65%	15.89%	11.66%	30.00%
3.83%	10.06%	5.83%	25.00%
0.23%	4.59%	0.36%	20.00%
0.51%	5.01%	0.78%	25.83%
7.65%	15.89%	11.66%	30.00%
2.56%	8.13%	3.90%	15.00%
0.00%	4.23%	0.00%	29.93%
6.37%	13.94%	9.71%	25.00%
1.87%	7.08%	2.85%	22.00%
2.13%	7.47%	3.24%	25.00%
0.60%	5.14%	0.91%	0.00%
3.83%	10.06%	5.83%	30.00%
0.51%	5.01%	0.78%	16.50%
1.62%	6.69%	2.46%	9.00%
0.60%	5.14%	0.91%	21.00%
1.87%	7.08%	2.85%	30.00%
1.62%	6.69%	2.46%	22.00%
6.37%	13.94%	9.71%	15.00%
0.51%	5.01%	0.78%	12.50%
0.51%	5.01%	0.78%	0.00%
1.36%	6.30%	2.07%	23.00%
1.62%	6.69%	2.46%	27.81%
3.06%	8.89%	4.66%	25.00%
0.60%	5.14%	0.91%	29.74%
0.51%	5.01%	0.78%	0.00%
3.06%	8.89%	4.66%	20.00%
1.36%	6.30%	2.07%	20.00%
5.52%	12.64%	8.41%	30.00%
0.42%	4.87%	0.64%	26.40%
0.60%	5.14%	0.91%	15.00%
5.52%	12.64%	8.41%	10.00%
7.65%	15.89%	11.66%	20.00%
1.02%	5.78%	1.55%	20.00%
17.50%	30.89%	26.66%	17.00%
0.00%	4.23%	0.00%	12.50%
0.72%	5.33%	1.10%	15.00%

0.00%	4.23%	0.00%	24.94%
0.51%	5.01%	0.78%	25.00%
3.06%	8.89%	4.66%	10.00%
1.02%	5.78%	1.55%	24.00%
7.65%	15.89%	11.66%	15.00%
7.65%	15.89%	11.66%	30.00%
0.72%	5.33%	1.10%	35.00%
1.87%	7.08%	2.85%	15.00%
1.62%	6.69%	2.46%	30.00%
5.52%	12.64%	8.41%	12.00%
3.83%	10.06%	5.83%	25.00%
3.83%	10.06%	5.83%	15.00%
1.87%	7.08%	2.85%	30.00%
2.13%	7.47%	3.24%	33.00%
8.50%	17.18%	12.95%	32.00%
3.83%	10.06%	5.83%	32.00%
3.06%	8.89%	4.66%	25.00%
0.00%	4.23%	0.00%	25.80%
0.00%	4.23%	0.00%	28.00%
4.67%	11.35%	7.12%	30.00%
8.50%	17.18%	12.95%	30.00%
5.52%	12.64%	8.41%	30.00%
0.00%	4.23%	0.00%	22.00%
1.87%	7.08%	2.85%	15.00%
6.37%	13.94%	9.71%	29.00%
1.87%	7.08%	2.85%	25.00%
4.67%	11.35%	7.12%	30.00%
1.87%	7.08%	2.85%	10.00%
1.36%	6.30%	2.07%	29.50%
1.62%	6.69%	2.46%	25.00%
0.72%	5.33%	1.10%	19.00%
1.02%	5.78%	1.55%	31.50%
0.42%	4.87%	0.64%	10.00%
1.02%	5.78%	1.55%	18.94%
1.87%	7.08%	2.85%	16.00%
4.67%	11.35%	7.12%	28.00%
0.51%	5.01%	0.78%	20.00%
6.37%	13.94%	9.71%	30.00%
2.56%	8.13%	3.90%	15.00%
2.13%	7.47%	3.24%	18.94%
0.00%	4.23%	0.00%	17.00%
1.02%	5.78%	1.55%	21.00%
1.02%	5.78%	1.55%	22.00%

6.37%	13.94%	9.71%	30.00%
2.56%	8.13%	3.90%	27.00%
1.02%	5.78%	1.55%	25.00%
10.20%	19.77%	15.54%	30.00%
2.56%	8.13%	3.90%	20.00%
5.52%	12.64%	8.41%	28.00%
6.37%	13.94%	9.71%	36.00%
4.67%	11.35%	7.12%	25.00%
0.00%	4.23%	0.00%	20.60%
0.00%	4.23%	0.00%	19.61%
0.51%	5.01%	0.78%	20.00%
5.52%	12.64%	8.41%	18.00%
3.83%	10.06%	5.83%	30.00%
1.36%	6.30%	2.07%	20.00%
5.52%	12.64%	8.41%	27.00%
2.56%	8.13%	3.90%	30.00%
6.37%	13.94%	9.71%	15.00%
3.06%	8.89%	4.66%	25.00%
1.36%	6.30%	2.07%	0.00%
5.52%	12.64%	8.41%	30.00%
10.20%	19.77%	15.54%	18.00%
0.42%	4.87%	0.64%	9.00%
0.51%	5.01%	0.78%	25.00%
0.23%	4.46%	0.23%	25.00%
1.36%	6.30%	2.07%	25.00%
3.06%	8.89%	4.66%	15.00%
17.50%	30.89%	26.66%	34.00%
2.56%	8.13%	3.90%	20.00%
7.65%	15.89%	11.66%	30.00%

<i>Region</i>
Middle East
Eastern Europe & Russia
Western Europe
Africa
Central and South America
Eastern Europe & Russia
Caribbean
Australia & New Zealand
Western Europe
Eastern Europe & Russia
Caribbean
Middle East
Asia
Caribbean
Eastern Europe & Russia
Western Europe
Central and South America
Africa
Caribbean
Central and South America
Eastern Europe & Russia
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Eastern Europe & Russia
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Asia
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North America
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Caribbean
Central and South America
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Australia & New Zealand
Central and South America
Africa
Eastern Europe & Russia
Caribbean
Caribbean
Western Europe

Eastern Europe & Russia
Western Europe
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Central and South America
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Eastern Europe & Russia
Western Europe
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Middle East
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Eastern Europe & Russia
Eastern Europe & Russia

Asia
Africa
Western Europe
Asia
Caribbean
Caribbean
Central and South America
Africa
Western Europe
Western Europe
Asia
Eastern Europe & Russia
Africa
Asia
Africa
Caribbean
Africa
Western Europe
Caribbean
Africa
Eastern Europe & Russia
Middle East
Western Europe
North America
Central and South America
Eastern Europe & Russia
Central and South America
Asia
Africa

Country	S&P Rating	Fitch rating	Moody's rating
Abu Dhabi	#N/A	#N/A	Aa2
Albania	BB	NR	Ba3
Andorra (Principality of)	A-	#N/A	Baa1
Angola	B-	B-	B3
Argentina	CCC+	CCC+	Caa1
Armenia	BB-	BB-	Ba3
Aruba	BBB+	BBB-	Baa3
Australia	AAA	AAA	Aaa
Austria	AA+	AA	Aa1
Azerbaijan	NR	BBB-	Baa3
Bahamas	BB-	BB-	B1
Bahrain	B	B+	B2
Bangladesh	B+	B+	B2
Barbados	B+	B+	B2
Belarus	NR	CCC	C
Belgium	AA	A+	Aa3
Belize	B-	NR	Caa1
Benin	BB-	B+	B1
Bermuda	A+	NR	A2
Bolivia	CCC-	CCC	Ca
Bosnia and Herzegovina	B+	NR	B3
Botswana	BBB	NR	Baa1
Brazil	BB	BB	Ba1
Bulgaria	BBB+	BBB+	Baa1
Burkina Faso	CCC+	NR	Caa1
Cambodia	N/A	NR	B2
Cameroon	B-	B	Caa1
Canada	AAA	AA+	Aaa
Cape Verde	B	B	B2
Cayman Islands		#N/A	Aa3
Chile	A	A-	A2
China	A+	A	A1
Colombia	BB	BB	Baa3
Congo (Democratic Rep)	#N/A	#N/A	B3
Congo (Republic of)	CCC+	#N/A	Caa2
Cook Islands	#N/A	#N/A	B1
Costa Rica	BB	BB	Ba2
Côte d'Ivoire	#N/A	#N/A	Ba2
Croatia	A-	A-	A3
Cuba	#N/A	NR	Ca

Curacao	BBB-	#N/A	Baa3
Cyprus	A-	A-	A3
Czech Republic	AA-	#N/A	Aa3
Denmark	AAA	AAA	Aaa
Dominican Republic	BB	BB-	Ba2
Ecuador	B-	CCC+	Caa3
Egypt	B	B	Caa1
El Salvador	B-	B-	B3
Estonia	NR	A+	A1
Ethiopia	SD	RD	Caa2
Fiji	B+	NR	B1
Finland	AA+	AA	Aa1
France	A+	A+	Aa3
Gabon	N/A	CCC-	Caa2
Georgia	BB	BB	Ba2
Germany	AAA	AAA	Aaa
Ghana	B-	B-	Caa1
Greece	BBB	BBB	Baa3
Guatemala	BB+	BB+	Ba1
Guernsey (States of)	A+	#N/A	A1
Honduras	BB-	B	B1
Hong Kong	AA+	AA-	Aa3
Hungary	BBB-	BBB	Baa2
Iceland	A+	A	A1
India	BBB	BBB-	Baa3
Indonesia	BBB	BBB	Baa2
Iraq	B-	B-	Caa1
Ireland	AA	AA	Aa3
Isle of Man	NR	#N/A	Aa3
Israel	A	A	Baa1
Italy	BBB+	BBB+	Baa2
Jamaica	BB	BB-	Ba3
Japan	A+	A	A1
Jersey (States of)	AA-	#N/A	Aa3
Jordan	BB-	BB-	Ba3
Kazakhstan	BBB-	BBB	Baa1
Kenya	B	B-	B3
Korea	AA	#N/A	Aa2
Kuwait	AA-	AA-	A1
Kyrgyzstan	#N/A	#N/A	B3
Laos	CCC+	CCC+	Caa2

Latvia	A	A-	A3
Lebanon	SD	NR	C
Liechtenstein	AAA	NR	Aaa
Lithuania	A	A	A2
Luxembourg	AAA	AAA	Aaa
Macao	NR	AA	Aa3
Macedonia	BB-	#N/A	Ba3
Malaysia	A-	BBB+	A3
Maldives	NR	CC	Caa2
Mali	NR	B-	Caa2
Malta	A-	A+	A2
Mauritius	BBB-	NR	Baa3
Mexico	BBB	BBB-	Baa2
Moldova	BB-	B+	B3
Mongolia	BB-	B+	B1
Montenegro	B+	NR	B1
Montserrat	BBB-	NR	Baa3
Morocco	BBB-	BB+	Ba1
Mozambique	CCC+	CCC	Caa3
Namibia		BB-	B1
Nepal	NR	BB-	Ba3
Netherlands	AAA	AAA	Aaa
New Zealand	AA+	AA+	Aaa
Nicaragua	B+	B	B2
Niger		NR	Caa3
Nigeria	B-	B	B3
Norway	AAA	AAA	Aaa
Oman	BBB-	BBB-	Baa3
Pakistan	B-	B-	Caa1
Panama	BBB-	BB+	Baa3
Papua New Guinea	B-	B+	B2
Paraguay	BBB-	BB+	Baa3
Peru	BBB-	BBB	Baa1
Philippines	BBB+	BBB	Baa2
Poland	A-	A-	A2
Portugal	A+	A	A3
Qatar	AA	AA	Aa2
Ras Al Khaimah (Emirat	A-	#N/A	A3
Romania	BBB-	BBB-	Baa3
Rwanda	B+	B+	B2
Saudi Arabia	A+	A+	Aa3

Senegal	CCC+	NR	Caa1
Serbia	BBB-	BB+	Ba2
Sharjah	NR	#N/A	Ba1
Singapore	AAA	AAA	Aaa
Slovakia	A+	A-	A3
Slovenia	AA	A+	A3
Solomon Islands		NR	Caa1
South Africa	BB	BB-	Ba2
Spain	A+	A	A3
Sri Lanka	CCC+	CCC+	Ca
St. Maarten	NA	#N/A	Ba2
St. Vincent & the Grenadines	NR	#N/A	B3
Suriname	CCC+	C	Caa1
Swaziland	#N/A	#N/A	B2
Sweden	AAA	AAA	Aaa
Switzerland	AAA	AAA	Aaa
Taiwan	AA+	AA	Aa3
Tajikistan	B	NR	B3
Tanzania	NR	B+	B1
Thailand	BBB+	BBB+	Baa1
Togo	B+	NR	B3
Trinidad and Tobago	BBB-	NR	Ba2
Tunisia	N/A	B-	Caa1
Turkey	BB-	BB-	Ba3
Turks and Caicos Islands	BBB+	BBB+	Baa1
Uganda	B-	B	B3
Ukraine	SD	CCC	Ca
United Arab Emirates	AA	AA-	Aa2
United Kingdom	AA	AA-	Aa3
United States	AA+	AA+	Aa1
Uruguay	BBB+	BBB	Baa1
Uzbekistan	BB	BB	Ba3
Venezuela	N/A	NR	C
Vietnam	BB+	BB+	Ba2
Zambia	CCC+	B-	Caa2

Moody's Rating (unadj)

Aa2
Ba3
Baa1
B3
Caa1
Ba3
Baa3
Aaa
Aa1
Baa3
B1
B2
B2
B2
C
Aa3
Caa1
B1
A2
Ca
B3
Baa1
Ba1
Baa1
NR
B2
Caa1
Aaa
NR
Aa3
A2
A1
Baa3
B3
Caa2
B1
Ba2
Ba2
A3
Ca

GOVERNMENT

In local c

S&P: These are the S&P sovereign local currency ratings

Moody's: These are the Moody's local currency rating

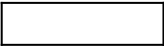
In red: These are countries where S&P has a rating and Moody's do
Moody's equivalent rating for these countries.

<i>S&P/Fitch</i>	<i>Moody's</i>
A	A2
A-	A3
A+	A1
AA	Aa2
AA-	Aa3
AA+	Aa1
AAA	Aaa
B	B2
B-	B3
B+	B1
BB	Ba2
BB-	Ba3
BB+	Ba1
BBB	Baa2
BBB-	Baa3
BBB+	Baa1
C	C2
C-	C3
C+	C1
CC	Ca2
CC-	Ca3
CC+	Ca1
CCC	Caa2
CCC-	Caa3
CCC+	Caa1
NR	NA

NR
A3
Aa3
Aaa
Ba2
Caa3
Caa1
NR
A1
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B1
Aa1
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Caa1
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Ba2
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Ca
Ba2
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Ba2
Caa1
Ba3
Baa1
B3
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Ba2
Caa2



BOND RATINGS

currency

es not. The following lookup table is used to estimate the

<i>Country</i>	<i>Region</i>
Abu Dhabi	Middle East
Albania	Eastern Europe & Russia
Andorra (Principality of)	Western Europe
Angola	Africa
Argentina	Central and South America
Armenia	Eastern Europe & Russia
Aruba	Caribbean
Australia	Australia & New Zealand
Austria	Western Europe
Azerbaijan	Eastern Europe & Russia
Bahamas	Caribbean
Bahrain	Middle East
Bangladesh	Asia
Barbados	Caribbean
Belarus	Eastern Europe & Russia
Belgium	Western Europe
Belize	Central and South America
Benin	Africa
Bermuda	Caribbean
Bolivia	Central and South America
Bosnia and Herzegovina	Eastern Europe & Russia
Botswana	Africa
Brazil	Central and South America
Bulgaria	Eastern Europe & Russia
Burkina Faso	Africa
Cambodia	Asia
Cameroon	Africa
Canada	North America
Cape Verde	Africa
Cayman Islands	Caribbean
Chile	Central and South America
China	Asia
Colombia	Central and South America
Congo (Democratic Republic of)	Africa
Congo (Republic of)	Africa
Cook Islands	Australia & New Zealand
Costa Rica	Central and South America
Côte d'Ivoire	Africa
Croatia	Eastern Europe & Russia
Cuba	Caribbean
Curacao	Caribbean
Cyprus	Western Europe

Czech Republic	Eastern Europe & Russia
Democratic Republic of Congo	Africa
Denmark	Western Europe
Dominican Republic	Caribbean
Ecuador	Central and South America
Egypt	Africa
El Salvador	Central and South America
Estonia	Eastern Europe & Russia
Ethiopia	Africa
Fiji	Asia
Finland	Western Europe
France	Western Europe
Gabon	Africa
Georgia	Eastern Europe & Russia
Germany	Western Europe
Ghana	Africa
Greece	Western Europe
Guatemala	Central and South America
Guernsey (States of)	Western Europe
Honduras	Central and South America
Hong Kong	Asia
Hungary	Eastern Europe & Russia
Iceland	Western Europe
India	Asia
Indonesia	Asia
Iraq	Middle East
Ireland	Western Europe
Isle of Man	Western Europe
Israel	Middle East
Italy	Western Europe
Jamaica	Caribbean
Japan	Asia
Jersey (States of)	Western Europe
Jordan	Middle East
Kazakhstan	Eastern Europe & Russia
Kenya	Africa
Korea	Asia
Kuwait	Middle East
Kyrgyzstan	Eastern Europe & Russia
Laos	Asia
Latvia	Eastern Europe & Russia
Lebanon	Middle East
Liechtenstein	Western Europe

Lithuania	Eastern Europe & Russia
Luxembourg	Western Europe
Macao	Asia
Macedonia	Eastern Europe & Russia
Malaysia	Asia
Maldives	Asia
Mali	Africa
Malta	Western Europe
Mauritius	Africa
Mexico	Central and South America
Moldova	Eastern Europe & Russia
Mongolia	Asia
Montenegro	Eastern Europe & Russia
Montserrat	Caribbean
Morocco	Africa
Mozambique	Africa
Namibia	Africa
Nepal	Asia
Netherlands	Western Europe
New Zealand	Australia & New Zealand
Nicaragua	Central and South America
Niger	Africa
Nigeria	Africa
Norway	Western Europe
Oman	Middle East
Pakistan	Asia
Panama	Central and South America
Papua New Guinea	Asia
Paraguay	Central and South America
Peru	Central and South America
Philippines	Asia
Poland	Eastern Europe & Russia
Portugal	Western Europe
Qatar	Middle East
Ras Al Khaimah (Emirate of)	Middle East
Republic of the Congo	Africa
Romania	Eastern Europe & Russia
Russia	Eastern Europe & Russia
Rwanda	Africa
Saudi Arabia	Middle East
Senegal	Africa
Serbia	Eastern Europe & Russia
Sharjah	Middle East

Singapore	Asia
Slovakia	Eastern Europe & Russia
Slovenia	Eastern Europe & Russia
Solomon Islands	Asia
South Africa	Africa
Spain	Western Europe
Sri Lanka	Asia
St. Maarten	Caribbean
St. Vincent & the Grenadines	Caribbean
Suriname	Central and South America
Swaziland	Africa
Sweden	Western Europe
Switzerland	Western Europe
Taiwan	Asia
Tajikistan	Eastern Europe & Russia
Tanzania	Africa
Thailand	Asia
Togo	Africa
Trinidad and Tobago	Caribbean
Tunisia	Africa
Turkey	Western Europe
Turkmenistan	Eastern Europe & Russia
Turks and Caicos Islands	Caribbean
Uganda	Africa
Ukraine	Eastern Europe & Russia
United Arab Emirates	Middle East
United Kingdom	Western Europe
United States	North America
Uruguay	Central and South America
Uzbekistan	Eastern Europe & Russia
Venezuela	Central and South America
Vietnam	Asia
Zambia	Africa

<i>Rating</i>	Default Spread (7/1/25)	Updated Default Spread (12/31/25)
A1	69	60
A2	83	72
A3	118	102
Aa1	27	23
Aa2	49	42
Aa3	59	51
Aaa	0	0
B1	443	383
B2	541	467
B3	639	552
Ba1	246	213
Ba2	296	256
Ba3	354	306
Baa1	157	136
Baa2	187	162
Baa3	216	187
Ca	1180	1020
Caa1	737	637
Caa2	885	765
Caa3	983	850
NR	NA	NA

* Adjusted to reflect US addition to Aa1 countries

Country
Abu Dhabi
Algeria
Angola
Australia
Austria
Bahrain
Belgium
Brazil
Bulgaria
Cameroon
Canada
Chile
China
Colombia
Costa Rica
Croatia
Cyprus
Czech Republic
Denmark
Dubai
Ecuador
Egypt
El Salvador
Estonia
Finland
France
Germany
Greece
Guatemala

Hong Kong
Hungary
Iceland
India
Indonesia
Iraq
Ireland
Israel
Italy
Japan
Kazakhstan
Kenya
Korea
Kuwait
Latvia
Lithuania
Malaysia
Mexico
Morocco
Namibia
Netherlands
New Zealand
Nicaragua
Nigeria
Norway
Oman
Pakistan
Panama
Peru
Philippines
Poland

Portugal
Qatar
Romania
Russia
Rwanda
Saudi Arabia
Senegal
Serbia
Slovakia
Slovenia
South Africa
Spain
Sweden
Switzerland
Thailand
Tunisia
Turkey
United Kingdom
United States
Uruguay
Vietnam
Average
Median

Moody's rating	7/1/2025	###	CDS % Change
Aa2	0.68%	0.60%	-11.76%
#N/A	1.40%	1.34%	-4.29%
B3	7.16%	6.24%	-12.85%
Aaa	0.18%	0.19%	5.56%
Aa1	0.32%	0.28%	-12.50%
B2	2.79%	2.47%	-11.47%
Aa3	0.50%	0.40%	-20.00%
Ba1	2.55%	2.35%	-7.84%
Baa1	1.12%	0.77%	-31.25%
Caa1	8.03%	6.87%	-14.45%
Aaa	0.37%	0.33%	-10.81%
A2	1.04%	0.87%	-16.35%
A1	0.75%	0.64%	-14.67%
Baa3	3.48%	3.34%	-4.02%
Ba2	2.50%	1.75%	-30.00%
A3	1.24%	1.02%	-17.74%
A3	0.82%	0.75%	-8.54%
Aa3	0.54%	0.48%	-11.11%
Aaa	0.21%	0.18%	-14.29%
#N/A	0.98%	0.82%	-16.33%
Caa3	12.42%	5.67%	-54.35%
Caa1	6.61%	3.55%	-46.29%
B3	4.47%	3.40%	-23.94%
A1	0.93%	1.00%	7.53%
Aa1	0.32%	0.26%	-18.75%
Aa3	0.68%	0.64%	-5.88%
Aaa	0.24%	0.21%	-12.50%
Baa3	1.04%	0.75%	-27.88%
Ba1	2.21%	1.71%	-22.62%

Country	###		
Abu Dhabi	0.60%		
Algeria	1.34%		
Angola	6.24%		
Argentina	6.94%		
Australia	0.19%		
Austria	0.28%		
Bahrain	2.47%		
Belgium	0.40%		
Brazil	2.35%		
Bulgaria	0.77%		
Cameroon	6.87%		
Canada	0.33%		
Chile	0.87%		
China	0.64%		
Colombia	3.34%		
Costa Rica	1.75%		
Croatia	1.02%		
Cyprus	0.75%		
Czech Republic	0.48%		
Denmark	0.18%		
Dubai	0.82%		
Ecuador	5.67%		
Egypt	3.55%		
El Salvador	3.40%		
Estonia	1.00%		
Ethiopia	NA		
Finland	0.26%		
France	0.64%		
Gabon	8.54%		

Aa3	0.59%	0.44%	-25.42%
Baa2	1.83%	1.60%	-12.57%
A1	0.47%	0.45%	-4.26%
Baa3	0.84%	0.80%	-4.76%
Baa2	1.34%	1.19%	-11.19%
Caa1	4.05%	2.89%	-28.64%
Aa3	0.38%	0.34%	-10.53%
Baa1	1.39%	1.13%	-18.71%
Baa2	0.98%	0.61%	-37.76%
A1	0.36%	0.44%	22.22%
Baa1	1.46%	1.31%	-10.27%
B3	6.34%	4.51%	-28.86%
Aa2	0.37%	0.34%	-8.11%
A1	0.87%	0.87%	0.00%
A3	0.98%	0.82%	-16.33%
A2	0.90%	0.82%	-8.89%
A3	0.81%	0.67%	-17.28%
Baa2	1.86%	1.66%	-10.75%
Ba1	1.60%	1.30%	-18.75%
B1	3.65%	3.41%	-6.58%
Aaa	0.25%	0.20%	-20.00%
Aaa	0.23%	0.23%	0.00%
B2	6.37%	5.29%	-16.95%
B3	6.05%	3.91%	-35.37%
Aaa	0.22%	0.18%	-18.18%
Baa3	1.66%	1.30%	-21.69%
Caa1	7.64%	5.23%	-31.54%
Baa3	2.96%	2.22%	-25.00%
Baa1	1.39%	1.26%	-9.35%
Baa2	1.15%	1.02%	-11.30%
A2	1.10%	1.00%	-9.09%

Germany	0.21%	
Greece	0.75%	
Guatemala	1.71%	
Hong Kong	0.44%	
Hungary	1.60%	
Iceland	0.45%	
India	0.80%	
Indonesia	1.19%	
Iraq	2.89%	
Ireland	0.34%	
Israel	1.13%	
Italy	0.61%	
Japan	0.44%	
Kazakhstan	1.31%	
Kenya	4.51%	
Korea	0.34%	
Kuwait	0.87%	
Latvia	0.82%	
Lebanon	NA	
Lithuania	0.82%	
Malaysia	0.67%	
Mexico	1.66%	
Mongolia	3.26%	
Morocco	1.30%	
Namibia	3.41%	
Netherlands	0.20%	
New Zealand	0.23%	
Nicaragua	5.29%	
Nigeria	3.91%	
Norway	0.18%	
Oman	1.30%	

A3	0.55%	0.45%	-18.18%
Aa2	0.68%	0.61%	-10.29%
Baa3	2.67%	2.11%	-20.97%
MR	NA	NA	NA
B2	5.43%	4.03%	-25.78%
Aa3	1.12%	1.12%	0.00%
Caa1	7.83%	9.93%	26.82%
Ba2	2.30%	1.96%	-14.78%
A3	0.61%	0.57%	-6.56%
A3	0.70%	0.67%	-4.29%
Ba2	3.01%	2.40%	-20.27%
A3	0.64%	0.45%	-29.69%
Aaa	0.25%	0.20%	-20.00%
Aaa	0.17%	0.14%	-17.65%
Baa1	0.72%	0.66%	-8.33%
Caa1	7.42%	7.01%	-5.53%
Ba3	3.80%	2.99%	-21.32%
Aa3	0.37%	0.37%	0.00%
Aa1	0.50%	0.44%	-12.00%
Baa1	1.19%	0.77%	-35.29%
Ba2	1.57%	1.51%	-3.82%
			-14.67%
			-13.57%

Pakistan	5.23%	
Panama	2.22%	
Peru	1.26%	
Philippines	1.02%	
Poland	1.00%	
Portugal	0.45%	
Qatar	0.61%	
Romania	2.11%	
Russia	NA	
Rwanda	4.03%	
Saudi Arabia	1.12%	
Senegal	9.93%	
Serbia	1.96%	
Slovakia	0.57%	
Slovenia	0.67%	
South Africa	2.40%	
Spain	0.45%	
Sri Lanka	NA	
Sweden	0.20%	
Switzerland	0.14%	
Thailand	0.66%	
Tunisia	7.01%	
Turkey	2.99%	
Ukraine	NA	
United Kingdom	0.37%	
United States	0.44%	
Uruguay	0.77%	
Venezuela	9.29%	
Vietnam	1.51%	
Zambia	4.08%	

Country	Moody's rating	CDS Spread (12/31/25)
Abu Dhabi	Aa2	0.60%
Albania	Ba3	NA
Andorra (Principality of)	Baa1	NA
Angola	B3	6.24%
Argentina	Caa1	NA
Armenia	Ba3	NA
Aruba	Baa3	NA
Australia	Aaa	0.19%
Austria	Aa1	0.28%
Azerbaijan	Baa3	NA
Bahamas	B1	NA
Bahrain	B2	2.47%
Bangladesh	B2	NA
Barbados	B2	NA
Belarus	C	NA
Belgium	Aa3	0.40%
Belize	Caa1	NA
Benin	B1	NA
Bermuda	A2	NA
Bolivia	Ca	NA
Bosnia and Herzegovina	B3	NA
Botswana	Baa1	NA
Brazil	Ba1	2.35%
Bulgaria	Baa1	0.77%
Burkina Faso	Caa1	NA
Cambodia	B2	NA
Cameroon	Caa1	6.87%
Canada	Aaa	0.33%
Cape Verde	B2	NA
Cayman Islands	Aa3	NA
Chile	A2	0.87%
China	A1	0.64%
Colombia	Baa3	3.34%
Congo (Democratic Rep)	B3	NA
Congo (Republic of)	Caa2	NA
Cook Islands	B1	NA
Costa Rica	Ba2	1.75%
Côte d'Ivoire	Ba2	NA
Croatia	A3	1.02%
Cuba	Ca	NA

Curacao	Baa3	NA
Cyprus	A3	0.75%
Czech Republic	Aa3	0.48%
Denmark	Aaa	0.18%
Dominican Republic	Ba2	NA
Ecuador	Caa3	5.67%
Egypt	Caa1	3.55%
El Salvador	B3	3.40%
Estonia	A1	1.00%
Ethiopia	Caa2	NA
Fiji	B1	NA
Finland	Aa1	0.26%
France	Aa3	0.64%
Gabon	Caa2	NA
Georgia	Ba2	NA
Germany	Aaa	0.21%
Ghana	Caa1	NA
Greece	Baa3	0.75%
Guatemala	Ba1	NA
Guernsey (States of)	A1	NA
Honduras	B1	NA
Hong Kong	Aa3	0.44%
Hungary	Baa2	1.60%
Iceland	A1	0.45%
India	Baa3	0.80%
Indonesia	Baa2	1.19%
Iraq	Caa1	2.89%
Ireland	Aa3	0.34%
Isle of Man	Aa3	NA
Israel	Baa1	1.13%
Italy	Baa2	0.61%
Jamaica	Ba3	NA
Japan	A1	0.44%
Jersey (States of)	Aa3	NA
Jordan	Ba3	NA
Kazakhstan	Baa1	1.31%
Kenya	B3	4.51%
Korea	Aa2	0.34%
Kuwait	A1	0.87%
Kyrgyzstan	B3	NA
Laos	Caa2	NA
Latvia	A3	0.82%
Lebanon	C	NA

Liechtenstein	Aaa	NA
Lithuania	A2	0.82%
Luxembourg	Aaa	NA
Macao	Aa3	NA
Macedonia	Ba3	NA
Malaysia	A3	0.67%
Maldives	Caa2	NA
Mali	Caa2	NA
Malta	A2	NA
Mauritius	Baa3	NA
Mexico	Baa2	1.66%
Moldova	B3	NA
Mongolia	B1	NA
Montenegro	B1	NA
Montserrat	Baa3	NA
Morocco	Ba1	1.30%
Mozambique	Caa3	NA
Namibia	B1	NA
Nepal	Ba3	NA
Netherlands	Aaa	0.20%
New Zealand	Aaa	0.23%
Nicaragua	B2	5.29%
Niger	Caa3	NA
Nigeria	B3	3.91%
Norway	Aaa	0.18%
Oman	Baa3	1.30%
Pakistan	Caa1	5.23%
Panama	Baa3	2.22%
Papua New Guinea	B2	NA
Paraguay	Baa3	NA
Peru	Baa1	1.26%
Philippines	Baa2	1.02%
Poland	A2	1.00%
Portugal	A3	0.45%
Qatar	Aa2	0.61%
Ras Al Khaimah (Emirat	A3	NA
Romania	Baa3	2.11%
Rwanda	B2	4.03%
Saudi Arabia	Aa3	1.12%
Senegal	Caa1	9.93%
Serbia	Ba2	1.96%
Sharjah	Ba1	#N/A
Singapore	Aaa	#N/A

Slovakia	A3	0.57%
Slovenia	A3	0.67%
Solomon Islands	Caa1	#N/A
South Africa	Ba2	2.40%
Spain	A3	0.45%
Sri Lanka	Ca	NA
St. Maarten	Ba2	#N/A
St. Vincent & the Grenadines	B3	#N/A
Suriname	Caa1	#N/A
Swaziland	B2	#N/A
Sweden	Aaa	0.20%
Switzerland	Aaa	0.14%
Taiwan	Aa3	#N/A
Tajikistan	B3	#N/A
Tanzania	B1	#N/A
Thailand	Baa1	0.66%
Togo	B3	#N/A
Trinidad and Tobago	Ba2	#N/A
Tunisia	Caa1	7.01%
Turkey	Ba3	2.99%
Turks and Caicos Islands	Baa1	#N/A
Uganda	B3	#N/A
Ukraine	Ca	NA
United Arab Emirates	Aa2	#N/A
United Kingdom	Aa3	0.37%
United States	Aa1	0.44%
Uruguay	Baa1	0.77%
Uzbekistan	Ba3	#N/A
Venezuela	C	9.29%
Vietnam	Ba2	1.51%
Zambia	Caa2	4.08%

CDS Spread adj for Switzerland
0.46%
NA
NA
6.10%
NA
NA
NA
0.05%
0.14%
NA
NA
2.33%
NA
NA
NA
0.26%
NA
NA
NA
NA
NA
NA
2.21%
0.63%
NA
NA
6.73%
0.19%
NA
NA
0.73%
0.50%
3.20%
NA
NA
NA
1.61%
NA
0.88%
NA

Country
Abu Dhabi
Algeria
Angola
Argentina
Australia
Austria
Bahrain
Belgium
Brazil
Bulgaria
Cameroon
Canada
Chile
China
Colombia
Costa Rica
Croatia
Cyprus

NA
0.61%
0.34%
0.04%
NA
5.53%
3.41%
3.26%
0.86%
NA
NA
0.12%
0.50%
NA
NA
0.07%
NA
0.61%
NA
NA
NA
0.30%
1.46%
0.31%
0.66%
1.05%
2.75%
0.20%
NA
0.99%
0.47%
NA
0.30%
NA
NA
1.17%
4.37%
0.20%
0.73%
NA
NA
0.68%
NA

Czech Republic
Denmark
Dubai
Ecuador
Egypt
El Salvador
Estonia
Ethiopia
Finland
France
Gabon
Germany
Greece
Guatemala
Hong Kong
Hungary
Iceland
India
Indonesia
Iraq
Ireland
Israel
Italy
Japan
Kazakhstan
Kenya
Korea
Kuwait
Latvia
Lebanon
Lithuania
Malaysia
Mexico
Mongolia
Morocco
Namibia
Netherlands
New Zealand
Nicaragua
Nigeria
Norway
Oman
Pakistan

NA
0.68%
NA
NA
NA
0.53%
NA
NA
NA
NA
1.52%
NA
NA
NA
NA
1.16%
NA
NA
NA
0.06%
0.09%
5.15%
NA
3.77%
0.04%
1.16%
5.09%
2.08%
NA
NA
1.12%
0.88%
0.86%
0.31%
0.47%
NA
1.97%
3.89%
0.98%
9.79%
1.82%
#N/A
#N/A

Panama
Peru
Philippines
Poland
Portugal
Qatar
Romania
Russia
Rwanda
Saudi Arabia
Senegal
Serbia
Slovakia
Slovenia
South Africa
Spain
Sri Lanka
Sweden
Switzerland
Thailand
Tunisia
Turkey
Ukraine
United Kingdom
United States
Uruguay
Venezuela

Vietnam

Zambia

0.43%
0.53%
#N/A
2.26%
0.31%
NA
#N/A
#N/A
#N/A
#N/A
0.06%
0.00%
#N/A
#N/A
#N/A
0.52%
#N/A
#N/A
6.87%
2.85%
#N/A
#N/A
NA
#N/A
0.23%
0.30%
0.63%
#N/A
9.15%
1.37%
3.94%

###	S Spread net of Switzerland
0.60%	0.46%
1.34%	1.20%
6.24%	6.10%
6.94%	6.80%
0.19%	0.05%
0.28%	0.14%
2.47%	2.33%
0.40%	0.26%
2.35%	2.21%
0.77%	0.63%
6.87%	6.73%
0.33%	0.19%
0.87%	0.73%
0.64%	0.50%
3.34%	3.20%
1.75%	1.61%
1.02%	0.88%
0.75%	0.61%

0.48%	0.34%
0.18%	0.04%
0.82%	0.68%
5.67%	5.53%
3.55%	3.41%
3.40%	3.26%
1.00%	0.86%
NA	NA
0.26%	0.12%
0.64%	0.50%
8.54%	8.40%
0.21%	0.07%
0.75%	0.61%
1.71%	1.57%
0.44%	0.30%
1.60%	1.46%
0.45%	0.31%
0.80%	0.66%
1.19%	1.05%
2.89%	2.75%
0.34%	0.20%
1.13%	0.99%
0.61%	0.47%
0.44%	0.30%
1.31%	1.17%
4.51%	4.37%
0.34%	0.20%
0.87%	0.73%
0.82%	0.68%
NA	NA
0.82%	0.68%
0.67%	0.53%
1.66%	1.52%
3.26%	3.12%
1.30%	1.16%
3.41%	3.27%
0.20%	0.06%
0.23%	0.09%
5.29%	5.15%
3.91%	3.77%
0.18%	0.04%
1.30%	1.16%
5.23%	5.09%

2.22%	2.08%
1.26%	1.12%
1.02%	0.88%
1.00%	0.86%
0.45%	0.31%
0.61%	0.47%
2.11%	#REF!
NA	1.97%
4.03%	3.89%
1.12%	0.98%
9.93%	9.79%
1.96%	1.82%
0.57%	0.43%
0.67%	0.53%
2.40%	2.26%
0.45%	0.31%
NA	NA
0.20%	0.06%
0.14%	0.00%
0.66%	0.52%
7.01%	6.87%
2.99%	2.85%
NA	NA
0.37%	0.23%
0.44%	0.30%
0.77%	0.63%
9.29%	9.15%
1.51%	1.37%
4.08%	3.94%

Date:	Feb-24	
<i>Country</i>	<i>σ_{Equity}</i>	<i>σ_{Bond}</i>
Argentina	56.52%	48.06%
Bahrain	5.11%	NA
Brazil	16.21%	13.32%
Colombia	17.44%	NA
Costa Rica	7.14%	4.92%
Egypt	27.33%	NA
Hungary	14.41%	12.26%
Jamaica	11.24%	9.08%
Kenya	17.67%	NA
Lebanon	31.00%	16.46%
Mexico	14.14%	8.01%
Mongolia	15.45%	NA
Morocco	9.90%	NA
Nigeria	16.74%	NA
Oman	7.33%	NA
Panama	7.94%	8.90%
Romania	11.09%	8.03%
Serbia	7.12%	NA
Singapore	11.62%	5.99%
South Africa	17.29%	NA
Taiwan	13.98%	13.73%
Tunisia	5.39%	NA
Turkey	31.89%	14.80%
Venezuela	30.81%	69.73%
Vietnam	15.96%	NA
Average		
Median		
High		
Low		

Just Risky markets

$\sigma_{\text{Equity}} / \sigma_{\text{Bond}}$	σ (CDS)	CDS	CV(CDS)	$\sigma_{\text{Equity}} / \sigma_{\text{CDS}}$
1.18	NA	NA	NA	NA
NA	0.29%	2.69%	10.78%	0.47
1.22	0.21%	2.31%	9.09%	1.78
NA	0.23%	2.75%	8.36%	2.09
1.45	0.45%	2.42%	18.60%	0.38
NA	0.65%	6.33%	10.27%	2.66
1.18	0.27%	1.75%	15.43%	0.93
1.24	0.38%	1.56%	24.36%	0.46
NA	0.72%	4.57%	15.75%	1.12
1.88	NA	NA	NA	NA
1.77	0.25%	1.58%	15.82%	0.89
NA	0.50%	3.65%	13.70%	1.13
NA	0.45%	1.53%	29.41%	0.34
NA	0.49%	6.94%	7.06%	2.37
NA	0.41%	1.68%	24.40%	0.30
0.89	0.26%	2.68%	9.70%	0.82
1.38	NA	NA	NA	NA
NA	0.32%	2.63%	12.17%	0.59
1.94	NA	NA	NA	NA
NA	0.27%	3.47%	7.78%	2.22
1.02	NA	NA	NA	NA
NA	0.40%	8.16%	4.90%	1.10
2.15	0.36%	4.14%	8.70%	3.67
0.44	1.23%	10.65%	11.55%	2.67
NA	0.24%	1.80%	13.33%	1.20
1.36				1.36
1.24				1.11
2.15				3.67
0.44				0.30

Country	GDP (in millions) in 2024
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Abu Dhabi	\$310,000
Albania	\$27,046
Andorra (Principality of)	\$4,040
Angola	\$100,999
Argentina	\$638,365
Armenia	\$25,955
Aruba	\$4,266
Australia	\$1,757,022
Austria	\$534,791
Azerbaijan	\$74,316
Bahamas	\$15,833
Bahrain	\$47,110
Bangladesh	\$450,119
Barbados	\$7,498
Belarus	\$75,962
Belgium	\$671,370
Belize	\$3,204
Benin	\$21,483
Bermuda	\$9,234
Bolivia	\$54,881
Bosnia and Herzegovina	\$29,614
Botswana	\$19,402
Brazil	\$2,185,822
Bulgaria	\$113,343
Burkina Faso	\$23,125
Cambodia	\$46,353
Cameroon	\$53,297
Canada	\$2,243,637
Cape Verde	\$2,725
Cayman Islands	\$7,241
Chile	\$330,267
China	\$18,743,803
Colombia	\$418,818
Congo (Democratic Repu	\$70,962
Congo (Republic of)	\$15,720
Cook Islands	\$1,414
Costa Rica	\$95,350
Côte d'Ivoire	\$87,113
Croatia	\$92,984

Source: [World Bank](#)
Year: 2024

Cuba	\$107,352
Curacao	\$3,561
Cyprus	\$37,635
Czech Republic	\$347,034
Denmark	\$424,525
Dominican Republic	\$124,282
Ecuador	\$124,676
Egypt	\$389,060
El Salvador	\$35,365
Estonia	\$43,130
Ethiopia	\$149,740
Fiji	\$5,968
Finland	\$298,697
France	\$3,160,443
Gabon	\$20,896
Georgia	\$34,189
Germany	\$4,685,593
Ghana	\$82,308
Greece	\$256,238
Guatemala	\$113,200
Guernsey (States of)	\$3,446
Honduras	\$37,094
Hong Kong	\$406,863
Hungary	\$222,723
Iceland	\$33,255
India	\$3,909,892
Indonesia	\$1,396,300
Iraq	\$279,641
Ireland	\$609,157
Isle of Man	\$7,431
Israel	\$540,380
Italy	\$2,380,825
Jamaica	\$22,014
Japan	\$4,027,598
Jersey (States of)	\$4,890
Jordan	\$53,352
Kazakhstan	\$291,480
Kenya	\$120,340
Korea	\$1,875,388
Kuwait	\$160,227
Kyrgyzstan	\$17,478

Laos	\$16,503
Latvia	\$43,684
Lebanon	\$20,079
Liechtenstein	\$8,239
Lithuania	\$84,869
Luxembourg	\$93,280
Macao	\$49,467
Macedonia	\$16,952
Malaysia	\$422,227
Maldives	\$7,062
Mali	\$26,795
Malta	\$24,972
Mauritius	\$14,938
Mexico	\$1,856,366
Moldova	\$18,200
Mongolia	\$23,795
Montenegro	\$8,270
Montserrat	\$16,199
Morocco	\$160,611
Mozambique	\$22,745
Namibia	\$13,372
Nepal	\$42,914
Netherlands	\$1,214,928
New Zealand	\$260,172
Nicaragua	\$19,694
Niger	\$19,876
Nigeria	\$252,262
Norway	\$483,593
Oman	\$107,137
Pakistan	\$371,570
Panama	\$86,524
Papua New Guinea	\$31,800
Paraguay	\$44,458
Peru	\$289,222
Philippines	\$461,618
Poland	\$917,767
Portugal	\$313,271
Qatar	\$219,163
Ras Al Khaimah (Emirate)	\$11,000
Romania	\$382,564
Russia	\$2,173,836

Rwanda	\$14,252
Saudi Arabia	\$1,239,805
Senegal	\$32,808
Serbia	\$90,098
Sharjah	\$24,800
Singapore	\$547,387
Slovakia	\$140,934
Slovenia	\$72,972
Solomon Islands	\$1,584
South Africa	\$401,145
Spain	\$1,725,672
Sri Lanka	\$98,963
St. Maarten	\$1,798
St. Vincent & the Grenad	\$1,157
Suriname	\$4,417
Swaziland	\$4,859
Sweden	\$603,715
Switzerland	\$936,564
Taiwan	\$791,610
Tajikistan	\$14,205
Tanzania	\$78,844
Thailand	\$526,518
Togo	\$10,651
Trinidad and Tobago	\$25,634
Tunisia	\$51,332
Turkey	\$1,359,124
Turks and Caicos Islands	\$1,745
Uganda	\$53,912
Ukraine	\$190,741
United Arab Emirates	\$552,325
United Kingdom	\$3,686,033
United States	\$28,750,956
Uruguay	\$80,962
Uzbekistan	\$114,965
Venezuela	\$119,803
Vietnam	\$476,388
Zambia	\$25,303

Algeria	\$269,322
Brunei	\$15,341

Gambia	\$2,405
Guinea	\$25,009
Guinea-Bissau	\$2,218
Guyana	\$24,663
Haiti	\$25,224
Iran	\$475,252
Korea, D.P.R.	\$0
Liberia	\$4,779
Libya	\$48,487
Madagascar	\$17,421
Malawi	\$11,316
Myanmar	\$74,068
Sierra Leone	\$6,971
Somalia	\$11,967
Sudan	\$49,672
Syria	\$23,623
Yemen, Republic	\$21,606
Zimbabwe	\$41,539

Country Name	GDP in 2024
Afghanistan	\$17,152
Africa Eastern and Southern	\$1,242,694
Africa Western and Central	\$736,385
Albania	\$27,046
Algeria	\$269,322
American Samoa	\$871
Andorra	\$4,040
Angola	\$100,999
Antigua and Barbuda	\$2,208
Arab World	\$3,735,881
Argentina	\$638,365
Armenia	\$25,955
Aruba	\$4,266
Australia	\$1,757,022
Austria	\$534,791
Azerbaijan	\$74,316
Bahamas, The	\$15,833
Bahrain	\$47,110
Bangladesh	\$450,119
Barbados	\$7,498
Belarus	\$75,962
Belgium	\$671,370
Belize	\$3,204
Benin	\$21,483
Bermuda	\$9,234
Bhutan	\$3,013
Bolivia	\$54,881
Bosnia and Herzegovina	\$29,614
Botswana	\$19,402
Brazil	\$2,185,822
British Virgin Islands	\$0
Brunei Darussalam	\$15,341
Bulgaria	\$113,343
Burkina Faso	\$23,125
Burundi	\$3,082
Cabo Verde	\$2,725
Cambodia	\$46,353
Cameroon	\$53,297
Canada	\$2,243,637

Caribbean small states	\$90,345
Cayman Islands	\$7,241
Central African Republic	\$2,751
Central Europe and the Baltic	\$2,462,005
Chad	\$19,519
Channel Islands	\$12,507
Chile	\$330,267
China	\$18,743,803
Colombia	\$418,818
Comoros	\$1,441
Congo, Dem. Rep.	\$70,962
Congo, Rep.	\$15,720
Costa Rica	\$95,350
Cote d'Ivoire	\$87,113
Croatia	\$92,984
Cuba	\$107,352
Curacao	\$3,561
Cyprus	\$37,635
Czechia	\$347,034
Denmark	\$424,525
Djibouti	\$4,152
Dominica	\$689
Dominican Republic	\$124,282
Early-demographic dividend	\$15,808,304
East Asia & Pacific	\$32,017,180
East Asia & Pacific (excludin	\$22,260,811
East Asia & Pacific (IDA & IBR	\$22,233,363
Ecuador	\$124,676
Egypt, Arab Rep.	\$389,060
El Salvador	\$35,365
Equatorial Guinea	\$12,766
Eritrea	\$0
Estonia	\$43,130
Eswatini	\$4,859
Ethiopia	\$149,740
Euro area	\$16,485,444
Europe & Central Asia	\$29,291,829
Europe & Central Asia (exclu	\$2,451,180
Europe & Central Asia (IDA &	\$6,131,674
European Union	\$19,497,116
Faroe Islands	\$4,053

Fiji	\$5,968
Finland	\$298,697
Fragile and conflict affected s	\$1,652,896
France	\$3,160,443
French Polynesia	\$6,324
Gabon	\$20,896
Gambia, The	\$2,405
Georgia	\$34,189
Germany	\$4,685,593
Ghana	\$82,308
Gibraltar	\$0
Greece	\$256,238
Greenland	\$3,327
Grenada	\$1,372
Guam	\$6,910
Guatemala	\$113,200
Guinea	\$25,009
Guinea-Bissau	\$2,218
Guyana	\$24,663
Haiti	\$25,224
Heavily indebted poor countr	\$1,145,111
High income	\$71,321,143
Honduras	\$37,094
Hong Kong SAR, China	\$406,863
Hungary	\$222,723
IBRD only	\$40,930,410
Iceland	\$33,255
IDA & IBRD total	\$43,823,205
IDA blend	\$1,117,412
IDA only	\$1,775,382
IDA total	\$2,892,794
India	\$3,909,892
Indonesia	\$1,396,300
Iran, Islamic Rep.	\$475,252
Iraq	\$279,641
Ireland	\$609,157
Isle of Man	\$7,431
Israel	\$540,380
Italy	\$2,380,825
Jamaica	\$22,014
Japan	\$4,027,598

Jordan	\$53,352
Kazakhstan	\$291,480
Kenya	\$120,340
Kiribati	\$308
Korea, Dem. People's Rep.	\$0
Korea, Rep.	\$1,875,388
Kosovo	\$11,197
Kuwait	\$160,227
Kyrgyz Republic	\$17,478
Lao PDR	\$16,503
Late-demographic dividend	\$29,979,152
Latin America & Caribbean	\$7,110,323
Latin America & Caribbean (e	\$6,168,787
Latin America & the Caribbea	\$6,776,899
Latvia	\$43,684
Least developed countries: U	\$1,536,273
Lebanon	\$0
Lesotho	\$2,272
Liberia	\$4,779
Libya	\$48,487
Liechtenstein	\$0
Lithuania	\$84,869
Low & middle income	\$39,397,099
Low income	\$469,282
Lower middle income	\$7,941,312
Luxembourg	\$93,280
Macao SAR, China	\$49,467
Madagascar	\$17,421
Malawi	\$11,316
Malaysia	\$422,227
Maldives	\$7,062
Mali	\$26,795
Malta	\$24,972
Marshall Islands	\$290
Mauritania	\$10,908
Mauritius	\$14,938
Mexico	\$1,856,366
Micronesia, Fed. Sts.	\$471
Middle East, North Africa, Af	\$5,094,722
Middle East, North Africa, Af	\$2,203,604
Middle East, North Africa, Af	\$2,189,893

Middle income	\$38,925,709
Moldova	\$18,200
Monaco	\$11,126
Mongolia	\$23,795
Montenegro	\$8,270
Morocco	\$160,611
Mozambique	\$22,745
Myanmar	\$74,068
Namibia	\$13,372
Nauru	\$163
Nepal	\$42,914
Netherlands	\$1,214,928
New Caledonia	\$8,549
New Zealand	\$260,172
Nicaragua	\$19,694
Niger	\$19,876
Nigeria	\$252,262
North America	\$31,003,827
North Macedonia	\$16,952
Northern Mariana Islands	\$0
Norway	\$483,593
Not classified	\$0
OECD members	\$67,500,142
Oman	\$107,137
Other small states	\$208,127
Pacific island small states	\$12,125
Pakistan	\$371,570
Palau	\$0
Panama	\$86,524
Papua New Guinea	\$31,800
Paraguay	\$44,458
Peru	\$289,222
Philippines	\$461,618
Poland	\$917,767
Portugal	\$313,271
Post-demographic dividend	\$62,787,811
Pre-demographic dividend	\$1,510,803
Puerto Rico (US)	\$126,030
Qatar	\$219,163
Romania	\$382,564
Russian Federation	\$2,173,836

Rwanda	\$14,252
Samoa	\$1,176
San Marino	\$0
Sao Tome and Principe	\$822
Saudi Arabia	\$1,239,805
Senegal	\$32,808
Serbia	\$90,098
Seychelles	\$2,167
Sierra Leone	\$6,971
Singapore	\$547,387
Sint Maarten (Dutch part)	\$1,798
Slovak Republic	\$140,934
Slovenia	\$72,972
Small states	\$310,642
Solomon Islands	\$1,584
Somalia, Fed. Rep.	\$11,967
South Africa	\$401,145
South Asia	\$4,512,297
South Asia (IDA & IBRD)	\$4,512,297
South Sudan	\$0
Spain	\$1,725,672
Sri Lanka	\$98,963
St. Kitts and Nevis	\$1,122
St. Lucia	\$2,549
St. Martin (French part)	\$0
St. Vincent and the Grenadin	\$1,157
Sub-Saharan Africa	\$1,979,079
Sub-Saharan Africa (excludin	\$1,827,171
Sub-Saharan Africa (IDA & IBI	\$1,979,079
Sudan	\$49,672
Suriname	\$4,417
Sweden	\$603,715
Switzerland	\$936,564
Syrian Arab Republic	\$0
Tajikistan	\$14,205
Tanzania	\$78,844
Thailand	\$526,518
Timor-Leste	\$1,866
Togo	\$10,651
Tonga	\$0
Trinidad and Tobago	\$25,634

Tunisia	\$51,332
Turkiye	\$1,359,124
Turkmenistan	\$51,387
Turks and Caicos Islands	\$1,745
Tuvalu	\$0
Uganda	\$53,912
Ukraine	\$190,741
United Arab Emirates	\$552,325
United Kingdom	\$3,686,033
United States	\$28,750,956
Uruguay	\$80,962
Uzbekistan	\$114,965
Vanuatu	\$1,118
Venezuela, RB	\$119,803
Viet Nam	\$476,388
Virgin Islands (U.S.)	\$0
West Bank and Gaza	\$13,711
World	\$110,982,661
Yemen, Rep.	\$0
Zambia	\$25,303
Zimbabwe	\$41,539

Country	GDP (in millions) in 2022
Abu Dhabi	299500
Albania	27046
Andorra (Principality of)	3330
Angola	100999
Argentina	638365
Armenia	25955
Aruba	4266
Australia	1757022
Austria	534791
Azerbaijan	74316
Bahamas	11210
Bahrain	47110
Bangladesh	450119
Barbados	7498
Belarus	75962
Belgium	671370
Belize	3204
Benin	21483
Bermuda	9234
Bolivia	54881
Bosnia and Herzegovina	29614
Botswana	19402
Brazil	2185822
Bulgaria	113343
Burkina Faso	23125
Cambodia	46353
Cameroon	53297
Canada	2243637
Cape Verde	1936
Cayman Islands	7241
Chile	330267
China	18743803
Colombia	418818
Congo (Democratic Repu	0
Congo (Republic of)	14616
Cook Islands	1414
Costa Rica	95350
Côte d'Ivoire	70019
Croatia	92984

Cuba	107352
Curacao	3561
Cyprus	37635
Czech Republic	0
Denmark	424525
Dominican Republic	124282
Ecuador	124676
Egypt	0
El Salvador	35365
Estonia	43130
Ethiopia	149740
Fiji	5968
Finland	298697
France	3160443
Gabon	20896
Georgia	34189
Germany	4685593
Ghana	82308
Greece	256238
Guatemala	113200
Guernsey (States of)	3446
Honduras	37094
Hong Kong	0
Hungary	222723
Iceland	33255
India	3909892
Indonesia	1396300
Iraq	279641
Ireland	609157
Isle of Man	7431
Israel	540380
Italy	2380825
Jamaica	22014
Japan	4027598
Jersey (States of)	4890
Jordan	53352
Kazakhstan	291480
Kenya	120340
Korea	0
Kuwait	160227
Kyrgyzstan	0

Laos	0
Latvia	43684
Lebanon	0
Liechtenstein	0
Lithuania	84869
Luxembourg	93280
Macao	0
Macedonia	0
Malaysia	422227
Maldives	7062
Mali	26795
Malta	24972
Mauritius	14938
Mexico	1856366
Moldova	18200
Mongolia	23795
Montenegro	8270
Montserrat	16199
Morocco	160611
Mozambique	22745
Namibia	13372
Netherlands	1214928
New Zealand	260172
Nicaragua	19694
Niger	19876
Nigeria	252262
Norway	483593
Oman	107137
Pakistan	371570
Panama	86524
Papua New Guinea	31800
Paraguay	44458
Peru	289222
Philippines	461618
Poland	917767
Portugal	313271
Qatar	219163
Ras Al Khaimah (Emirate	11000
Romania	382564
Russia	0
Rwanda	14252

Saudi Arabia	1239805
Senegal	32808
Serbia	90098
Sharjah	24800
Singapore	547387
Slovakia	0
Slovenia	72972
Solomon Islands	1584
South Africa	401145
Spain	1725672
Sri Lanka	98963
St. Maarten	11900
St. Vincent & the Grenad	8100
Suriname	4417
Swaziland	0
Sweden	603715
Switzerland	936564
Taiwan	761690
Tajikistan	14205
Tanzania	78844
Thailand	526518
Togo	10651
Trinidad and Tobago	25634
Tunisia	51332
Turkey	0
Turks and Caicos Islands	1745
Uganda	53912
Ukraine	190741
United Arab Emirates	552325
United Kingdom	3686033
United States of America	0
Uruguay	80962
Uzbekistan	114965
Venezuela	98400
Vietnam	408802
Zambia	25303
Algeria	269322
Brunei	14010
Gambia	2038

Guinea	25009
Guinea-Bissau	2218
Guyana	24663
Haiti	25224
Iran	0
Korea, D.P.R.	28500
Liberia	4779
Libya	48487
Madagascar	17421
Malawi	11316
Myanmar	74068
Sierra Leone	6971
Somalia	#N/A
Sudan	49672
Syria	0
Yemen, Republic	21610
Zimbabwe	41539

**GDP (in millions) in
2023**

310000
27046
3352
100999
638365
25955
4266
1757022
534791
74316
11210
47110
450119
7498
75962
671370
3204
21483
9234
54881
29614
19402
2185822
113343
23125
46353
53297
2243637
1936
7241
330267
18743803
418818
12507
330267
1414
95350
418818
92984

107351
3561
37635
87113
424525
124282
124676
424525
35365
43130
149740
5968
298697
3160443
20896
34189
4685593
82308
256238
113200
3446
37094
2405
222723
33255
3909892
1396300
279641
609157
7431
540380
2380825
22014
4027598
4890
53352
291480
120340
222723
11000
43823205

1117412
43684
0
0
84869
93280
540380
7941312
422227
7062
26795
24972
14938
1856366
18200
23795
8270
16199
160611
22745
13372
42914
1214928
260172
19694
19876
252262
483593
107137
371570
86524
31800
44458
289222
461618
917767
313271
219163
11000
382564
2189893

14252
1239805
32808
90098
24800
547387
1214928
72972
1584
#N/A
#N/A
#N/A
11900
8100
#N/A
32017180
#N/A
#N/A
791610
#N/A
#N/A
#N/A
#N/A
#N/A
#N/A
1510803
#N/A
#N/A
#N/A
#N/A
#N/A
822
#N/A
#N/A
98400
6971
#N/A
269322
14010

43130
25009
2218
24663
25224
256238
28500
4779
48487
17421
11316
74068
6971
#N/A
#N/A
8980
21610
#N/A

Country	S&P Rating	Moody's Rating
Abu Dhabi	#N/A	Aa2
Albania	BB	Ba3
Andorra (Principality of)	A-	Baa1
Angola	B-	B3
Argentina	CCC+	Caa1
Armenia	BB-	Ba3
Aruba	BBB+	Baa3
Australia	AAA	Aaa
Austria	AA+	Aa1
Azerbaijan	NR	Baa3
Bahamas	BB-	B1
Bahrain	B	B2
Bangladesh	B+	B2
Barbados	B+	B2
Belarus	NR	C
Belgium	AA	Aa3
Belize	B-	Caa1
Benin	BB-	B1
Bermuda	A+	A2
Bolivia	CCC-	Ca
Bosnia and Herzegovina	B+	B3
Botswana	BBB	Baa1
Brazil	BB	Ba1
Bulgaria	BBB+	Baa1
Burkina Faso	CCC+	NR
Cambodia	N/A	B2
Cameroon	B-	Caa1
Canada	AAA	Aaa
Cape Verde	B	NR
Cayman Islands	0	Aa3

Chile	A	A2
China	A+	A1
Colombia	BB	Baa3
Congo (Democratic Republic of)	#N/A	B3
Congo (Republic of)	CCC+	Caa2
Cook Islands	#N/A	B1
Costa Rica	BB	Ba2
Côte d'Ivoire	#N/A	Ba2
Croatia	A-	A3
Cuba	#N/A	Ca
Curacao	BBB-	NR
Cyprus	A-	A3
Czech Republic	AA-	Aa3
Denmark	AAA	Aaa
Dominican Republic	BB	Ba2
Ecuador	B-	Caa3
Egypt	B	Caa1
El Salvador	B-	NR
Estonia	NR	A1
Ethiopia	SD	Caa2
Fiji	B+	B1
Finland	AA+	Aa1
France	A+	Aa3
Gabon	N/A	Caa2
Georgia	BB	Ba2
Germany	AAA	Aaa
Ghana	B-	Caa1
Greece	BBB	Baa3
Guatemala	BB+	Ba1
Guernsey (States of)	A+	NR
Honduras	BB-	B1
Hong Kong	AA+	Aa3
Hungary	BBB-	Baa2

Iceland	A+	A1
India	BBB	Baa3
Indonesia	BBB	Baa2
Iraq	B-	Caa1
Ireland	AA	Aa3
Isle of Man	NR	Aa3
Israel	A	Baa1
Italy	BBB+	Baa2
Jamaica	BB	Ba3
Japan	A+	A1
Jersey (States of)	AA-	NR
Jordan	BB-	Ba3
Kazakhstan	BBB-	Baa1
Kenya	B	B3
Korea	AA	Aa2
Kuwait	AA-	A1
Kyrgyzstan	#N/A	B3
Laos	CCC+	Caa2
Latvia	A	A3
Lebanon	SD	C
Liechtenstein	AAA	NR
Lithuania	A	A2
Luxembourg	AAA	Aaa
Macao	NR	Aa3
Macedonia	BB-	NR
Malaysia	A-	A3
Maldives	NR	Caa2
Mali	NR	Caa2
Malta	A-	A2
Mauritius	BBB-	Baa3
Mexico	BBB	Baa2
Moldova	BB-	B3
Mongolia	BB-	B1

Montenegro	B+	B1
Montserrat	BBB-	NR
Morocco	BBB-	Ba1
Mozambique	CCC+	Caa3
Namibia	0	B1
Nepal	NR	NR
Netherlands	AAA	Aaa
New Zealand	AA+	Aaa
Nicaragua	B+	B2
Niger	0	Caa3
Nigeria	B-	B3
Norway	AAA	Aaa
Oman	BBB-	Baa3
Pakistan	B-	Caa1
Panama	BBB-	Baa3
Papua New Guinea	B-	B2
Paraguay	BBB-	Baa3
Peru	BBB-	Baa1
Philippines	BBB+	Baa2
Poland	A-	A2
Portugal	A+	A3
Qatar	AA	Aa2
Ras Al Khaimah (Emirate)	A-	A3
Romania	BBB-	Baa3
Russia	NR	NR
Rwanda	B+	B2
Saudi Arabia	A+	Aa3
Senegal	CCC+	Caa1
Serbia	BBB-	Ba2
Sharjah	NR	Ba1
Singapore	AAA	Aaa
Slovakia	A+	A3
Slovenia	AA	A3

Solomon Islands	0	Caa1
South Africa	BB	Ba2
Spain	A+	A3
Sri Lanka	CCC+	Ca
St. Maarten	NA	Ba2
St. Vincent & the Grenadines	NR	B3
Suriname	CCC+	Caa1
Swaziland	#N/A	B2
Sweden	AAA	Aaa
Switzerland	AAA	Aaa
Taiwan	AA+	Aa3
Tajikistan	B	B3
Tanzania	NR	B1
Thailand	BBB+	Baa1
Togo	B+	B3
Trinidad and Tobago	BBB-	Ba2
Tunisia	N/A	Caa1
Turkey	BB-	Ba3
Turks and Caicos Islands	BBB+	Baa1
Uganda	B-	B3
Ukraine	SD	Ca
United Arab Emirates	AA	Aa2
United Kingdom	AA	Aa3
United States	AA+	Aa1
Uruguay	BBB+	Baa1
Uzbekistan	BB	Ba3
Venezuela	N/A	C
Vietnam	BB+	Ba2
Zambia	CCC+	Caa2

Country	Moody's	S&P	FITCH
Abu Dhabi	Aa2	#N/A	#N/A
Albania	Ba3	BB	NR
Andorra	Baa1	A-	A-
Angola	B3	B-	B-
Argentina	Caa1	CCC+	CCC+
Armenia	Ba3	BB-	BB-
Aruba	Baa3	BBB+	BBB-
Australia	Aaa	AAA	AAA
Austria	Aa1	AA+	AA
Azerbaijan	Baa3	NR	BBB-
Bahamas	B1	BB-	BB-
Bahrain	B2	B	B+
Bangladesh	B2	B+	B+
Barbados	B2	B+	B+
Belarus	C	NR	CCC
Belgium	Aa3	AA	A+
Belize	Caa1	B-	NR
Benin	B1	BB-	B+
Bermuda	A2	A+	NR
Bolivia	Ca	CCC-	CCC
Bosnia and Herzegovina	B3	B+	NR
Botswana	Baa1	BBB	NR
Brazil	Ba1	BB	BB
Bulgaria	Baa1	BBB+	BBB+
Burkina Faso	NR	CCC+	NR
Cambodia	B2	N/A	NR
Cameroon	Caa1	B-	B
Canada	Aaa	AAA	AA+
Cape Verde	NR	B	B
Cayman Islands	Aa3		#N/A

Chile	A2	A	A-
China	A1	A+	A
Colombia	Baa3	BB	BB
Congo (Democratic Repub	B3	#N/A	#N/A
Congo (Republic of)	Caa2	CCC+	#N/A
Cook Islands	B1	#N/A	#N/A
Costa Rica	Ba2	BB	BB
Cote d'Ivoire	Ba2	#N/A	#N/A
Croatia	A3	A-	A-
Cuba	Ca	#N/A	NR
Curacao	NR	#N/A	#N/A
Cyprus	A3	A-	A-
Czech Republic	Aa3	AA-	#N/A
Denmark	Aaa	AAA	AAA
Dominican Republic	Ba2	BB	BB-
Ecuador	Caa3	B-	CCC+
Egypt	Caa1	B	B
El Salvador	-	B-	B-
Estonia	A1	NR	A+
Ethiopia	Caa2	SD	RD
Fiji	B1	B+	NR
Finland	Aa1	AA+	AA
France	Aa3	A+	A+
Gabon	Caa2	N/A	CCC-
Georgia	Ba2	BB	BB
Germany	Aaa	AAA	AAA
Ghana	Caa1	B-	B-
Greece	Baa3	BBB	BBB
Guatemala	Ba1	BB+	BB+
Guernsey (Channel Islands)	-	#N/A	#N/A
Honduras	B1	BB-	B
Hong Kong	Aa3	AA+	AA-
Hungary	Baa2	BBB-	BBB

Iceland	A1	A+	A
India	Baa3	BBB	BBB-
Indonesia	Baa2	BBB	BBB
Iraq	Caa1	B-	B-
Ireland	Aa3	AA	AA
Isle of Man	Aa3	NR	#N/A
Israel	Baa1	A	A
Italy	Baa2	BBB+	BBB+
Jamaica	Ba3	BB	BB-
Japan	A1	A+	A
Jersey (Channel Islands)	-	#N/A	#N/A
Jordan	Ba3	BB-	BB-
Kazakhstan	Baa1	BBB-	BBB
Kenya	B3	B	B-
Korea	Aa2	AA	#N/A
Kuwait	A1	AA-	AA-
Kyrgyz Republic	B3	#N/A	#N/A
Laos	Caa2	CCC+	CCC+
Latvia	A3	A	A-
Lebanon	C	SD	NR
Liechtenstein	NR	AAA	NR
Lithuania	A2	A	A
Luxembourg	Aaa	AAA	AAA
Macao	Aa3	NR	AA
Macedonia	NR	BB-	#N/A
Malaysia	A3	A-	BBB+
Maldives	Caa2	NR	CC
Mali	Caa2	NR	B-
Malta	A2	A-	A+
Mauritius	Baa3	BBB-	NR
Mexico	Baa2	BBB	BBB-
Moldova	B3	BB-	B+
Mongolia	B1	BB-	B+

Montenegro	B1	B+	NR
Montserrat	NR	BBB-	NR
Morocco	Ba1	BBB-	BB+
Mozambique	Caa3	CCC+	CCC
Namibia	B1		BB-
Nepal	NR	NR	BB-
Netherlands	Aaa	AAA	AAA
New Zealand	Aaa	AA+	AA+
Nicaragua	B2	B+	B
Niger	Caa3		NR
Nigeria	B3	B-	B
Norway	Aaa	AAA	AAA
Oman	Baa3	BBB-	BBB-
Pakistan	Caa1	B-	B-
Panama	Baa3	BBB-	BB+
Papua New Guinea	B2	B-	B+
Paraguay	Baa3	BBB-	BB+
Peru	Baa1	BBB-	BBB
Philippines	Baa2	BBB+	BBB
Poland	A2	A-	A-
Portugal	A3	A+	A
Qatar	Aa2	AA	AA
Ras Al Khaimah (Emirate of)	NR	NR	#N/A
Romania	Baa3	BBB-	BBB-
Russia	NR	NR	BBB-
Rwanda	B2	B+	B+
Saudi Arabia	Aa3	A+	A+
Senegal	Caa1	CCC+	NR
Serbia	Ba2	BBB-	BB+
Sharjah	Ba1	NR	#N/A
Singapore	Aaa	AAA	AAA
Slovakia	A3	A+	A-
Slovenia	A3	AA	A+

Solomon Islands	Caa1		NR
South Africa	Ba2	BB	BB-
Spain	A3	A+	A
Sri Lanka	Ca	CCC+	CCC+
Sint Maarten	WR	NR	#N/A
St. Vincent & the Grenadines	B3	NR	#N/A
Suriname	Caa1	CCC+	C
eSwatini	B2	#N/A	#N/A
Sweden	Aaa	AAA	AAA
Switzerland	Aaa	AAA	AAA
Taiwan	Aa3	AA+	AA
Tajikistan	B3	B	NR
Tanzania	B1	NR	B+
Thailand	Baa1	BBB+	BBB+
Togo	B3	B+	NR
Trinidad and Tobago	Ba2	BBB-	NR
Tunisia	Caa1	N/A	B-
Turkey	Ba3	BB-	BB-
Turks and Caicos Islands	NR	NR	BBB+
Uganda	B3	B-	B
Ukraine	Ca	SD	CCC
United Arab Emirates	Aa2	AA	AA-
United Kingdom	Aa3	AA	AA-
United States	Aa1	AA+	AA+
Uruguay	Baa1	BBB+	BBB
Uzbekistan	Ba3	BB	BB
Venezuela	C	N/A	NR
Vietnam	Ba2	BB+	BB+
Zambia	Caa2	CCC+	B-

	FC	LC	Country	S&P
Abu Dhabi	Aa2	Aa2	Albania	BB
Albania	Ba3	Ba3	Andorra	A-
Andorra	Baa1	Baa1	Angola	B-
Angola	B3	B3	Argentina	CCC+
Argentina	Caa1	Caa1	Armenia	BB-
Armenia	Ba3	Ba3	Aruba	BBB+
Aruba	Baa3	Baa3	Australia	AAA
Australia	Aaa	Aaa	Austria	AA+
Austria	Aa1	Aa1	Azerbaijan	NR
Azerbaijan	Baa3	Baa3	Bahamas	BB-
Bahamas	B1	B1	Bahrain	B
Bahamas-Offshore Banks	-	-	Bangladesh	B+
Bahrain	B2	B2	Barbados	B+
Bahrain-Offshore Banks [1]	-	-	Belarus	NR
Bangladesh	B2	B2	Belgium	AA
Barbados	B2	B2	Belize	B-
Belarus	C	C	Benin	BB-
Belgium	Aa3	Aa3	Bermuda	A+
Belize	Caa1	Caa1	Bolivia	CCC-
Benin	B1	B1	Bosnia and Herzegovina	B+
Bermuda	A2	A2	Botswana	BBB
Bolivia	Ca	Ca	Brazil	BB
Bosnia and Herzegovina	B3	B3	Bulgaria	BBB+
Botswana	Baa1	Baa1	Burkina Faso	CCC+
Brazil	Ba1	Ba1	Cambodia	N/A
Bulgaria	Baa1	Baa1	Cameroon	B-
Cambodia	B2	B2	Canada	AAA
Cameroon	Caa1	Caa1	Cape Verde	B
Canada	Aaa	Aaa	Cayman Islands	
Cayman Islands	Aa3	Aa3	Chad	B-

Cayman Islands-Offshore Banl	-	-	Chile	A
Chile	A2	A2	China	A+
China	A1	A1	Colombia	BB
Colombia	Baa3	Baa3	Congo	B-
Costa Rica	Ba2	Ba2	Costa Rica	BB
Cote d'Ivoire	Ba2	Ba2	Croatia	A-
Croatia	A3	A3	Cyprus	A-
Cuba	WR	WR	Czech Republic	AA-
Cyprus	A3	A3	Denmark	AAA
Czech Republic	Aa3	Aa3	Dominican Republic	BB
Democratic Republic of the Co	B3	B3	Ecuador	B-
Denmark	Aaa	Aaa	Egypt	B
Dominican Republic	Ba2	Ba2	El Salvador	B-
Ecuador	Caa1	-	Estonia	NR
Egypt	Caa1	Caa1	Ethiopia	SD
El Salvador	B3	-	European Union	AA+
Estonia	A1	A1	Fiji	B+
eSwatini	B2	B2	Finland	AA+
Ethiopia	Caa3	Caa2	France	A+
Fiji	B1	B1	Gabon	N/A
Finland	Aa1	Aa1	Georgia	BB
France	Aa3	Aa3	Germany	AAA
Gabon	Caa2	Caa2	Ghana	B-
Georgia	Ba2	Ba2	Greece	BBB
Germany	Aaa	Aaa	Guatemala	BB+
Ghana	Caa1	Caa1	Honduras	BB-
Greece	Baa3	Baa3	Hong Kong	AA+
Guatemala	Ba1	Ba1	Hungary	BBB-
Guernsey (Channel Islands)	-	-	Iceland	A+
Honduras	B1	B1	India	BBB
Hong Kong SAR, China	Aa3	Aa3	Indonesia	BBB
Hungary	Baa2	Baa2	Iraq	B-
Iceland	A1	A1	Ireland	AA

India	Baa3	Baa3	Isle of Man	N/A
Indonesia	Baa2	Baa2	Israel	A
Iraq	Caa1	Caa1	Italy	BBB+
Ireland	Aa3	Aa3	Ivory Coast	BB
Isle of Man	Aa3	Aa3	Jamaica	BB
Israel	Baa1	Baa1	Japan	A+
Italy	Baa2	Baa2	Jordan	BB-
Jamaica	Ba3	Ba3	Kazakhstan	BBB-
Japan	A1	A1	Kenya	B
Jersey (Channel Islands)	-	-	Kuwait	AA-
Jordan	Ba3	Ba3	Kyrgyzstan	B+
Kazakhstan	Baa1	Baa1	Laos	CCC+
Kenya	B3	B3	Latvia	A
Korea	Aa2	Aa2	Lebanon	SD
Kuwait	A1	A1	Lesotho	
Kyrgyz Republic	B3	B3	Liechtenstein	AAA
Laos	Caa2	Caa2	Lithuania	A
Latvia	A3	A3	Luxembourg	AAA
Lebanon	C	C	Macau	
Liechtenstein	-	-	Macedonia	BB-
Lithuania	A2	A2	Madagascar	B-
Luxembourg	Aaa	Aaa	Malaysia	A-
Macao SAR, China	Aa3	Aa3	Maldives	
Malaysia	A3	A3	Mali	
Maldives	Caa2	Caa2	Malta	A-
Mali	Caa2	Caa2	Mauritius	BBB-
Malta	A2	A2	Mexico	BBB
Mauritius	Baa3	Baa3	Moldova	BB-
Mexico	Baa2	Baa2	Mongolia	BB-
Moldova	B3	B3	Montenegro	B+
Mongolia	B1	B1	Montserrat	BBB-
Montenegro	Ba3	-	Morocco	BBB-
Morocco	Ba1	Ba1	Mozambique	CCC+

Mozambique	Caa2	Caa3	Namibia	
Namibia	B1	B1	Netherlands	AAA
Netherlands	Aaa	Aaa	New Zealand	AA+
New Zealand	Aaa	Aaa	Nicaragua	B+
Nicaragua	B2	B2	Niger	
Niger	Caa3	Caa3	Nigeria	B-
Nigeria	B3	B3	Norway	AAA
Norway	Aaa	Aaa	Oman	BBB-
Oman	Baa3	Baa3	Pakistan	B-
Pakistan	Caa1	Caa1	Panama	BBB-
Panama	Baa3	-	Papua New Guinea	B-
Panama-Offshore Banks	-	-	Paraguay	BBB-
Papua New Guinea	B2	B2	Peru	BBB-
Paraguay	Baa3	Baa3	Philippines	BBB+
Peru	Baa1	Baa1	Poland	A-
Philippines	Baa2	Baa2	Portugal	A+
Poland	A2	A2	Qatar	AA
Portugal	A3	A3	Republic of the Congo	CCC+
Qatar	Aa2	Aa2	Romania	BBB-
Republic of the Congo	Caa2	Caa2	Russia	NR
Romania	Baa3	Baa3	Rwanda	B+
Russia	WR	WR	San Marino	BBB+
Rwanda	B2	B2	Saudi Arabia	A+
Saudi Arabia	Aa3	Aa3	Senegal	CCC+
Senegal	Caa1	Caa1	Serbia	BBB-
Serbia	Ba2	Ba2	Seychelles	
Sharjah	Ba1	Ba1	Singapore	AAA
Singapore	Aaa	Aaa	Slovakia	A+
Slovakia	A3	A3	Slovenia	AA
Slovenia	A3	A3	Solomon Islands	
Solomon Islands	Caa1	Caa1	South Africa	BB
South Africa	Ba2	Ba2	South Korea	AA
Spain	A3	A3	Spain	A+

Sri Lanka	Caa1	-	Sri Lanka	CCC+
Sint Maarten	WR	WR	St Vincent and the Grenadines	
St. Vincent & the Grenadines	B3	B3	Suriname	CCC+
Suriname	Caa1	Caa1	Swaziland	
Sweden	Aaa	Aaa	Sweden	AAA
Switzerland	Aaa	Aaa	Switzerland	AAA
Taiwan, China	Aa3	Aa3	Taiwan	AA+
Tajikistan	B3	B3	Tajikistan	B
Tanzania	B1	B1	Tanzania	
Thailand	Baa1	Baa1	Thailand	BBB+
Togo	B3	B3	Togo	B+
Trinidad and Tobago	Ba2	Ba2	Trinidad and Tobago	BBB-
Tunisia	Caa1	Caa1	Tunisia	N/A
Turkey	Ba3	Ba3	Turkey	BB-
Uganda	B3	B3	Turkmenistan	
Ukraine	Ca	Ca	Uganda	B-
United Arab Emirates	Aa2	Aa2	Ukraine	SD
United Kingdom	Aa3	Aa3	United Arab Emirates	AA
United States of America	Aa1	Aa1	United Kingdom	AA
Uzbekistan	Ba3	Ba3	United States	AA+
Uruguay	Baa1	Baa1	Uruguay	BBB+
Venezuela [2]	C	WR	Uzbekistan	BB
Vietnam	Ba2	Ba2	Venezuela	N/A
Zambia	Caa2	Caa2	Vietnam	BB+
			Zambia	CCC+

	Fitch
Albania	NR
Andorra	A-
Angola	B-
Argentina	CCC+
Armenia	BB-
Aruba	BBB-
Australia	AAA
Austria	AA
Azerbaijan	BBB-
Bahamas	BB-
Bahrain	B+
Bangladesh	B+
Barbados	B+
Belarus	CCC
Belgium	A+
Belize	NR
Benin	B+
Bermuda	NR
Bolivia	CCC
Bosnia and H	NR
Botswana	NR
Brazil	BB
Bulgaria	BBB+
Burkina Faso	NR
Cambodia	NR
Cameroon	B
Canada	AA+
Cape Verde	B
Chad	B-
Chile	A-

China	A
Colombia	BB
Costa Rica	BB
Croatia	A-
Cuba	NR
Cyprus	A-
Czechia	AA-
Democratic R	NR
Denmark	AAA
Dominican R	BB-
Ecuador	CCC+
Egypt	B
El Salvador	B-
Estonia	A+
Ethiopia	RD
Fiji	NR
Finland	AA
France	A+
Gabon	CCC-
Gambia	CCC
Georgia	BB
Germany	AAA
Ghana	B-
Greece	BBB
Grenada	NR
Guatemala	BB+
Guinea	NR
Honduras	B
Hong Kong	AA-
Hungary	BBB
Iceland	A
India	BBB-
Indonesia	BBB

Iran	B+
Iraq	B-
Ireland	AA
Israel	A
Italy	BBB+
Ivory Coast	BB
Jamaica	BB-
Japan	A
Jordan	BB-
Kazakhstan	BBB
Kenya	B-
Kuwait	AA-
Kyrgyzstan	B
Laos	CCC+
Latvia	A-
Lebanon	NR
Lesotho	B
Libya	B
Liechtenstein	NR
Lithuania	A
Luxembourg	AAA
Macao	AA
Madagascar	NR
Malawi	B-
Malaysia	BBB+
Maldives	CC
Mali	B-
Malta	A+
Mauritius	NR
Mexico	BBB-
Moldova	B+
Mongolia	B+
Montenegro	NR

Montserrat	NR
Morocco	BB+
Mozambique	CCC
Namibia	BB-
Nepal	BB-
Netherlands	AAA
New Zealand	AA+
Nicaragua	B
Niger	NR
Nigeria	B
North Macedonia	BB+
Norway	AAA
Oman	BBB-
Pakistan	B-
Panama	BB+
Papua New Guinea	B+
Paraguay	BB+
Peru	BBB
Philippines	BBB
Poland	A-
Portugal	A
Puerto Rico	NR
Qatar	AA
Republic of the Czech Republic	CCC+
Romania	BBB-
Russia	BBB-
Rwanda	B+
Saint Vincent and the Grenadines	NR
San Marino	BBB-
Saudi Arabia	A+
Senegal	NR
Serbia	BB+
Seychelles	BB

Singapore	AAA
Slovakia	A-
Slovenia	A+
Solomon Islands	NR
South Africa	BB-
South Korea	AA-
Spain	A
Sri Lanka	CCC+
Suriname	C
Swaziland	NR
Sweden	AAA
Switzerland	AAA
Taiwan	AA
Tajikistan	NR
Tanzania	B+
Thailand	BBB+
Togo	NR
Trinidad and Tobago	NR
Tunisia	B-
Turkey	BB-
Turkmenistan	BB-
Uganda	B
Ukraine	CCC
United Arab Emirates	AA-
United Kingdom	AA-
Uruguay	BBB
United States	AA+
Uzbekistan	BB
Venezuela	NR
Vietnam	BB+
Zambia	B-

Country	Tax Rate	Looked up for 2025	Country
Abu Dhabi	9.00%	0.09	Afghanistan
Albania	15.00%	0.15	Albania
Andorra (Principality of)	10.00%	10.00%	Algeria
Angola	25.00%	0.25	Andorra
Argentina	35.00%	0.35	Angola
Armenia	18.00%	0.18	Anguilla
Aruba	22.00%	0.22	Antigua and Barbuda
Australia	30.00%	0.3	Argentina
Austria	23.00%	0.23	Armenia
Azerbaijan	20.00%	0.2	Aruba
Bahamas	0.00%	0	Australia
Bahrain	0.00%	0	Austria
Bangladesh	27.50%	0.275	Azerbaijan
Barbados	9.00%	0.09	Bahamas
Belarus	25.00%	0.25	Bahrain
Belgium	25.00%	0.25	Bangladesh
Belize	0.00%	0	Barbados
Benin	30.00%	0.3	Belarus
Bermuda	0.00%	0	Belgium
Bolivia	25.00%	0.25	Benin
Bosnia and Herzegovina	10.00%	0.1	Bermuda
Botswana	22.00%	0.22	Bolivia
Brazil	34.00%	0.34	Bonaire, Saint Eustatius a
Bulgaria	10.00%	0.1	Bosnia and Herzegovina
Burkina Faso	27.50%	0.275	Botswana
Cambodia	20.00%	0.2	Brazil
Cameroon	33.00%	0.33	Brunei Darussalam
Canada	26.14%	0.2614	Bulgaria
Cape Verde	26.86%	26.86%	Burkina Faso
Cayman Islands	0.00%	0	Burundi
Chile	27.00%	0.27	Cambodia
China	25.00%	0.25	Cameroon
Colombia	35.00%	0.35	Canada
Congo (Democratic Republic of)	30.00%	0.3	Cayman Islands
Congo (Republic of)	26.86%	26.86%	Chile
Cook Islands	20.00%	0.2	China
Costa Rica	30.00%	0.3	Colombia
Côte d'Ivoire	25.00%	0.25	Congo
Croatia	18.00%	0.18	Congo (Democratic Repub
Cuba	35.00%	0.35	Costa Rica
Curacao	22.00%	0.22	Croatia
Cyprus	12.50%	0.125	Curacao

Czech Republic	21.00%	0.21	Cyprus
Denmark	22.00%	0.22	Czech Republic
Dominican Republic	27.00%	0.27	Denmark
Ecuador	25.00%	0.25	Djibouti
Egypt	22.50%	0.225	Dominica
El Salvador	30.00%	0.3	Dominican Republic
Estonia	20.00%	0.2	Ecuador
Ethiopia	30.00%	0.3	Egypt
Fiji	25.00%	0.25	El Salvador
Finland	20.00%	0.2	Estonia
France	25.83%	0.2583	Ethiopia
Gabon	30.00%	0.3	Fiji
Georgia	15.00%	0.15	Finland
Germany	29.93%	0.2993	France
Ghana	25.00%	0.25	Gabon
Greece	22.00%	0.22	Gambia
Guatemala	25.00%	0.25	Georgia
Guernsey (States of)	0.00%	0	Germany
Honduras	30.00%	0.3	Ghana
Hong Kong	16.50%	0.165	Gibraltar
Hungary	9.00%	0.09	Greece
Iceland	21.00%	0.21	Grenada
India	30.00%	0.3	Guatemala
Indonesia	22.00%	0.22	Guernsey
Iraq	15.00%	0.15	Honduras
Ireland	12.50%	0.125	Hong Kong SAR
Isle of Man	0.00%	0	Hungary
Israel	23.00%	0.23	Iceland
Italy	27.81%	0.2781	India
Jamaica	25.00%	0.25	Indonesia
Japan	29.74%	0.2974	Iraq
Jersey (States of)	0.00%	0	Ireland
Jordan	20.00%	0.2	Isle of Man
Kazakhstan	20.00%	0.2	Israel
Kenya	30.00%	0.3	Italy
Korea	26.40%	0.264	Ivory Coast
Kuwait	15.00%	0.15	Jamaica
Kyrgyzstan	10.00%	0.1	Japan
Laos	20.00%	0.2	Jersey
Latvia	20.00%	0.2	Jordan
Lebanon	17.00%	0.17	Kazakhstan
Liechtenstein	12.50%	0.125	Kenya
Lithuania	15.00%	0.15	Korea, Republic of

Luxembourg	24.94%
Macao	25.00%
Macedonia	10.00%
Malaysia	24.00%
Maldives	15.00%
Mali	30.00%
Malta	35.00%
Mauritius	15.00%
Mexico	30.00%
Moldova	12.00%
Mongolia	25.00%
Montenegro	15.00%
Montserrat	30.00%
Morocco	33.00%
Mozambique	32.00%
Namibia	32.00%
Nepal	25.00%
Netherlands	25.80%
New Zealand	28.00%
Nicaragua	30.00%
Niger	30.00%
Nigeria	30.00%
Norway	22.00%
Oman	15.00%
Pakistan	29.00%
Panama	25.00%
Papua New Guinea	30.00%
Paraguay	10.00%
Peru	29.50%
Philippines	25.00%
Poland	19.00%
Portugal	31.50%
Qatar	10.00%
Ras Al Khaimah (Emirate of)	18.94%
Romania	16.00%
Russia	20.00%
Rwanda	28.00%
Saudi Arabia	20.00%
Senegal	30.00%
Serbia	15.00%
Sharjah	18.94%
Singapore	17.00%
Slovakia	21.00%

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Kuwait

Kyrgyzstan

Latvia

Lebanon

Libya

Liechtenstein

Lithuania

Luxembourg

Macao

Macedonia

Madagascar

Malawi

Malaysia

Malta

Mauritania

Mauritius

Mexico

Moldova

Monaco

Mongolia

Montenegro

Morocco

Mozambique

Myanmar

Namibia

Nepal

Netherlands

New Zealand

Nicaragua

Nigeria

Norway

Oman

Pakistan

Palestinian Territory

Panama

Papua New Guinea

Paraguay

Peru

Philippines

Poland

Portugal

Qatar

Romania

Slovenia	22.00%
Solomon Islands	30.00%
South Africa	27.00%
Spain	25.00%
Sri Lanka	30.00%
St. Maarten	20.00%
St. Vincent & the Grenadines	28.00%
Suriname	36.00%
Swaziland	25.00%
Sweden	20.60%
Switzerland	19.61%
Taiwan	20.00%
Tajikistan	18.00%
Tanzania	30.00%
Thailand	20.00%
Togo	27.00%
Trinidad and Tobago	30.00%
Tunisia	15.00%
Turkey	25.00%
Turks and Caicos Islands	0.00%
Uganda	30.00%
Ukraine	18.00%
United Arab Emirates	9.00%
United Kingdom	25.00%
United States	25.00%
Uruguay	25.00%
Uzbekistan	15.00%
Venezuela	34.00%
Vietnam	20.00%
Zambia	30.00%

0.22	Russia
0.3	Rwanda
0.27	Saint Kitts and Nevis
0.25	Saint Lucia
0.3	Saint Vincent and the Gre
0.2	Samoa
0.28	Saudi Arabia
0.36	Senegal
0.25	Serbia
0.206	Sierra Leone
0.1961	Singapore
0.2	Sint Maarten (Dutch part)
0.18	Slovakia
0.3	Slovenia
0.2	Solomon Islands
0.27	South Africa
0.3	Spain
0.15	Sri Lanka
0.25	St Maarten
0	Sudan
0.3	Suriname
0.18	Swaziland
0.09	Sweden
0.25	Switzerland
0.25	Syria
0.25	Taiwan
0.15	Tajikistan
0.34	Tanzania
0.2	Thailand
0.3	Trinidad and Tobago
	Tunisia
	Turkey
	Turkmenistan
	Turks and Caicos Islands
	Uganda
	Ukraine
	United Arab Emirates
	United Kingdom
	United States
	Uruguay
	Uzbekistan
	Vanuatu
	Venezuela
	Vietnam
	Yemen

Zambia
Zimbabwe

<i>2023</i>	Region	Average Tax Rate
20.00%	Europe average	19.52%
15.00%	Africa average	26.86%
26.00%	Latin America average	28.53%
10.00%	North America average	25.75%
25.00%	Oceania average	29.74%
0.00%	OECD average	22.81%
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Country	2025 tax rate %
Afghanistan	20.00%
Aland Islands	20.00%
Albania	15.00%
Algeria	26.00%
American Samoa	34.00%
Andorra	10.00%
Angola	25.00%
Anguilla	0.00%
Antigua and Barbuda	25.00%
Argentina	35.00%
Armenia	18.00%
Aruba	22.00%
Australia	30.00%
Austria	23.00%
Azerbaijan	20.00%
Bahamas	0.00%
Bahrain	0.00%
Bangladesh	27.50%
Barbados	9.00%
Belarus	25.00%
Belgium	25.00%
Belize	0.00%
Benin	30.00%
Bermuda	0.00%
Bhutan	25.00%
Bolivia	25.00%
Bonaire, Sint Eustatius and	25.80%
Bosnia and Herzegovina	10.00%
Botswana	22.00%
Brazil	34.00%
British Virgin Islands	0.00%
Brunei Darussalam	18.50%
Bulgaria	10.00%
Burkina Faso	27.50%
Burundi	30.00%
Cabo Verde	21.42%
Cambodia	20.00%
Cameroon	33.00%
Canada	26.14%
Cayman Islands	0.00%
Central African Republic	30.00%
Chad	35.00%

Chile	27.00%
China	25.00%
China, Hong Kong Special A	16.50%
China, Macao Special Admini	12.00%
Colombia	35.00%
Comoros	50.00%
Congo	28.00%
Cook Islands	20.00%
Costa Rica	30.00%
Cote d'Ivoire	25.00%
Croatia	18.00%
Cuba	35.00%
Curacao	22.00%
Cyprus	12.50%
Czechia	21.00%
Democratic Republic of the	30.00%
Denmark	22.00%
Djibouti	25.00%
Dominica	25.00%
Dominican Republic	27.00%
Ecuador	25.00%
Egypt	22.50%
El Salvador	30.00%
Equatorial Guinea	35.00%
Eritrea	30.00%
Estonia	20.00%
Ethiopia	30.00%
Falkland Islands (Malvinas)	26.00%
Faroe Islands	18.00%
Federated States of Microni	30.00%
Fiji	25.00%
Finland	20.00%
France	25.83%
French Polynesia	25.00%
Gabon	30.00%
Gambia	27.00%
Georgia	15.00%
Germany	29.93%
Ghana	25.00%
Gibraltar	15.00%
Greece	22.00%
Greenland	26.50%
Grenada	28.00%

Guam	21.00%
Guatemala	25.00%
Guernsey	0.00%
Guinea	25.00%
Guinea-Bissau	25.00%
Guyana	25.00%
Haiti	30.00%
Honduras	30.00%
Hungary	9.00%
Iceland	21.00%
India	30.00%
Indonesia	22.00%
Iran	25.00%
Iraq	15.00%
Ireland	12.50%
Isle of Man	0.00%
Israel	23.00%
Italy	27.81%
Jamaica	25.00%
Japan	29.74%
Jersey	0.00%
Jordan	20.00%
Kazakhstan	20.00%
Kenya	30.00%
Kiribati	30.00%
Korea	26.40%
Kosovo	10.00%
Kuwait	15.00%
Kyrgyzstan	10.00%
Laos	20.00%
Latvia	20.00%
Lebanon	17.00%
Lesotho	25.00%
Liberia	25.00%
Libya	20.00%
Liechtenstein	12.50%
Lithuania	15.00%
Luxembourg	24.94%
Macedonia	10.00%
Madagascar	20.00%
Malawi	30.00%
Malaysia	24.00%
Maldives	15.00%

Mali	30.00%
Malta	35.00%
Mauritania	25.00%
Mauritius	15.00%
Mexico	30.00%
Moldova	12.00%
Monaco	25.00%
Mongolia	25.00%
Montenegro	15.00%
Montserrat	30.00%
Morocco	33.00%
Mozambique	32.00%
Myanmar	22.00%
Namibia	32.00%
Nauru	25.00%
Nepal	25.00%
Netherlands	25.80%
New Caledonia	30.00%
New Zealand	28.00%
Nicaragua	30.00%
Niger	30.00%
Nigeria	30.00%
Niue	30.00%
Northern Mariana Islands	21.00%
Norway	22.00%
Oman	15.00%
Pakistan	29.00%
Palestine	15.00%
Panama	25.00%
Papua New Guinea	30.00%
Paraguay	10.00%
Peru	29.50%
Philippines	25.00%
Poland	19.00%
Portugal	31.50%
Puerto Rico	37.50%
Qatar	10.00%
Romania	16.00%
Russia	20.00%
Rwanda	28.00%
Saint Barthelemy	0.00%
Saint Helena	25.00%
Saint Kitts and Nevis	33.00%
Saint Lucia	30.00%
Saint Martin (French Part)	20.00%

Saint Vincent and the Gren	28.00%
Samoa	27.00%
San Marino	17.00%
Sao Tome and Principe	25.00%
Saudi Arabia	20.00%
Senegal	30.00%
Serbia	15.00%
Seychelles	25.00%
Sierra Leone	25.00%
Singapore	17.00%
Sint Maarten	34.50%
Slovakia	21.00%
Slovenia	22.00%
Solomon Islands	30.00%
South Africa	27.00%
South Sudan	30.00%
Spain	25.00%
Sri Lanka	30.00%
Sudan	35.00%
Suriname	36.00%
Swaziland	25.00%
Sweden	20.60%
Switzerland	19.61%
Syrian Arab Republic	25.00%
Taiwan	20.00%
Tajikistan	18.00%
Tanzania	30.00%
Thailand	20.00%
Timor-Leste	10.00%
Togo	27.00%
Tokelau	0.00%
Tonga	25.00%
Trinidad and Tobago	30.00%
Tunisia	15.00%
Turkey	25.00%
Turkmenistan	8.00%
Turks and Caicos Islands	0.00%
Uganda	30.00%
Ukraine	18.00%
United Arab Emirates	9.00%
United Kingdom	25.00%
United States	25.63%
United States Virgin Islands	23.10%
Uruguay	25.00%
Uzbekistan	15.00%
Vanuatu	0.00%
Venezuela	34.00%
Vietnam	20.00%
Wallis and Futuna Islands	0.00%
Yemen	20.00%

Zambia	30.00%
Zimbabwe	24.72%

Country	PRS Score	Global Equity Risk Prem	Rating-based Defa
Abu Dhabi	#N/A	4.87%	0.42%
Albania	74.00	8.89%	3.06%
Andorra (Principality of)	#N/A	6.30%	1.36%
Angola	63.50	12.64%	5.52%
Argentina	69.50	13.94%	6.37%
Armenia	66.25	8.89%	3.06%
Aruba	#N/A	7.08%	1.87%
Australia	79.50	4.23%	0.00%
Austria	77.25	4.59%	0.23%
Azerbaijan	74.25	7.08%	1.87%
Bahamas	77.00	10.06%	3.83%
Bahrain	70.50	11.35%	4.67%
Bangladesh	66.00	11.35%	4.67%
Barbados	#N/A	11.35%	4.67%
Belarus	64.75	30.89%	17.50%
Belgium	75.25	5.01%	0.51%
Belize	#N/A	13.94%	6.37%
Benin	#N/A	10.06%	3.83%
Bermuda	#N/A	5.33%	0.72%
Bolivia	60.50	19.77%	10.20%
Bosnia and Herzegovina	#N/A	12.64%	5.52%
Botswana	77.50	6.30%	1.36%
Brazil	69.50	7.47%	2.13%
Bulgaria	71.75	6.30%	1.36%
Burkina Faso	60.00	13.94%	6.37%
Cambodia	#N/A	11.35%	4.67%
Cameroon	61.00	13.94%	6.37%
Canada	82.50	4.23%	0.00%
Cape Verde	#N/A	11.35%	4.67%
Cayman Islands	#N/A	5.01%	0.51%
Chile	76.00	5.33%	0.72%
China	#N/A	5.14%	0.60%
Colombia	64.50	7.08%	1.87%
Congo (Democratic Repu	56.75	12.64%	5.52%
Congo (Republic of)	63.00	15.89%	7.65%
Cook Islands	#N/A	10.06%	3.83%
Costa Rica	75.50	8.13%	2.56%
Côte d'Ivoire	65.00	8.13%	2.56%
Croatia	72.75	5.78%	1.02%
Cuba	56.75	19.77%	10.20%

Curacao	#N/A	7.08%	1.87%
Cyprus	74.75	5.78%	1.02%
Czech Republic	80.50	5.01%	0.51%
Denmark	86.75	4.23%	0.00%
Dominican Republic	73.50	8.13%	2.56%
Ecuador	69.75	17.18%	8.50%
Egypt	62.75	13.94%	6.37%
El Salvador	72.25	12.64%	5.52%
Estonia	69.50	5.14%	0.60%
Ethiopia	57.50	15.89%	7.65%
Fiji	#N/A	10.06%	3.83%
Finland	78.50	4.59%	0.23%
France	73.50	5.01%	0.51%
Gabon	70.25	15.89%	7.65%
Georgia	#N/A	8.13%	2.56%
Germany	80.25	4.23%	0.00%
Ghana	69.00	13.94%	6.37%
Greece	70.00	7.08%	1.87%
Guatemala	71.75	7.47%	2.13%
Guernsey (States of)	#N/A	5.14%	0.60%
Honduras	67.25	10.06%	3.83%
Hong Kong	80.00	5.01%	0.51%
Hungary	72.00	6.69%	1.62%
Iceland	83.50	5.14%	0.60%
India	72.00	7.08%	1.87%
Indonesia	68.50	6.69%	1.62%
Iraq	66.25	13.94%	6.37%
Ireland	84.50	5.01%	0.51%
Isle of Man	#N/A	5.01%	0.51%
Israel	74.00	6.30%	1.36%
Italy	76.75	6.69%	1.62%
Jamaica	76.75	8.89%	3.06%
Japan	80.25	5.14%	0.60%
Jersey (States of)	#N/A	5.01%	0.51%
Jordan	65.00	8.89%	3.06%
Kazakhstan	74.75	6.30%	1.36%
Kenya	62.00	12.64%	5.52%
Korea	#N/A	4.87%	0.42%
Kuwait	80.25	5.14%	0.60%
Kyrgyzstan	#N/A	12.64%	5.52%
Laos	#N/A	15.89%	7.65%

Latvia	72.00	5.78%	1.02%
Lebanon	51.50	30.89%	17.50%
Liechtenstein	#N/A	4.23%	0.00%
Lithuania	74.00	5.33%	0.72%
Luxembourg	84.75	4.23%	0.00%
Macao	#N/A	5.01%	0.51%
Macedonia	#N/A	8.89%	3.06%
Malaysia	75.25	5.78%	1.02%
Maldives	#N/A	15.89%	7.65%
Mali	57.75	15.89%	7.65%
Malta	76.50	5.33%	0.72%
Mauritius	#N/A	7.08%	1.87%
Mexico	69.00	6.69%	1.62%
Moldova	64.50	12.64%	5.52%
Mongolia	71.25	10.06%	3.83%
Montenegro	#N/A	10.06%	3.83%
Montserrat	#N/A	7.08%	1.87%
Morocco	72.25	7.47%	2.13%
Mozambique	57.00	17.18%	8.50%
Namibia	70.75	10.06%	3.83%
Nepal	#N/A	8.89%	3.06%
Netherlands	81.50	4.23%	0.00%
New Zealand	78.50	4.23%	0.00%
Nicaragua	70.25	11.35%	4.67%
Niger	56.00	17.18%	8.50%
Nigeria	61.25	12.64%	5.52%
Norway	87.75	4.23%	0.00%
Oman	78.50	7.08%	1.87%
Pakistan	58.25	13.94%	6.37%
Panama	70.50	7.08%	1.87%
Papua New Guinea	70.00	11.35%	4.67%
Paraguay	70.75	7.08%	1.87%
Peru	69.75	6.30%	1.36%
Philippines	70.00	6.69%	1.62%
Poland	74.00	5.33%	0.72%
Portugal	78.25	5.78%	1.02%
Qatar	80.25	4.87%	0.42%
Ras Al Khaimah (Emirate)	#N/A	5.78%	1.02%
Romania	67.25	7.08%	1.87%
Rwanda	#N/A	11.35%	4.67%
Saudi Arabia	81.00	5.01%	0.51%

Senegal	60.00	13.94%	6.37%
Serbia	#N/A	8.13%	2.56%
Sharjah	#N/A	7.47%	2.13%
Singapore	86.75	4.23%	0.00%
Slovakia	67.00	5.78%	1.02%
Slovenia	76.00	5.78%	1.02%
Solomon Islands	#N/A	13.94%	6.37%
South Africa	68.50	8.13%	2.56%
Spain	73.50	5.78%	1.02%
Sri Lanka	69.00	19.77%	10.20%
St. Maarten	#N/A	8.13%	2.56%
St. Vincent & the Grenadines	#N/A	12.64%	5.52%
Suriname	67.50	13.94%	6.37%
Swaziland	#N/A	11.35%	4.67%
Sweden	80.75	4.23%	0.00%
Switzerland	86.75	4.23%	0.00%
Taiwan	86.25	5.01%	0.51%
Tajikistan	#N/A	12.64%	5.52%
Tanzania	66.00	10.06%	3.83%
Thailand	68.25	6.30%	1.36%
Togo	60.75	12.64%	5.52%
Trinidad and Tobago	76.00	8.13%	2.56%
Tunisia	63.00	13.94%	6.37%
Turkey	61.00	8.89%	3.06%
Turks and Caicos Islands	#N/A	6.30%	1.36%
Uganda	62.75	12.64%	5.52%
Ukraine	57.00	19.77%	10.20%
United Arab Emirates	80.50	4.87%	0.42%
United Kingdom	76.25	5.01%	0.51%
United States	68.50	4.46%	0.23%
Uruguay	76.50	6.30%	1.36%
Uzbekistan	74.75	8.89%	3.06%
Venezuela	57.00	30.89%	17.50%
Vietnam	72.75	8.13%	2.56%
Zambia	66.25	15.89%	7.65%
Algeria	67.00		3.83%
Brunei	80.75		0.51%
Gambia	65.50		4.67%

Guinea	59.50		7.65%
Guinea-Bissau	60.75		6.37%
Guyana	75.50		1.36%
Haiti	56.75		8.50%
Iran	61.25		6.37%
Korea, D.P.R.	51.00		10.20%
Liberia	57.75		7.65%
Libya	71.50		2.56%
Madagascar	62.50		5.52%
Malawi	56.00		8.50%
Myanmar	54.75		10.20%
Russia	70.25		2.56%
Sierra Leone	62.75		5.52%
Somalia	55.50		8.50%
Sudan	47.75		17.50%
Syria	51.50		10.20%
Yemen, Republic	51.75		10.20%
Zimbabwe	58.50		7.65%

<i>Final ERP</i>	<i>Tax Rate</i>	<i>CRP</i>
4.87%	9.00%	0.64%
8.89%	15.00%	4.66%
6.30%	10.00%	2.07%
12.64%	25.00%	8.41%
13.94%	35.00%	9.71%
8.89%	18.00%	4.66%
7.08%	22.00%	2.85%
4.23%	30.00%	0.00%
4.59%	23.00%	0.36%
7.08%	20.00%	2.85%
10.06%	0.00%	5.83%
11.35%	0.00%	7.12%
11.35%	27.50%	7.12%
11.35%	9.00%	7.12%
30.89%	25.00%	26.66%
5.01%	25.00%	0.78%
13.94%	0.00%	9.71%
10.06%	30.00%	5.83%
5.33%	0.00%	1.10%
19.77%	25.00%	15.54%
12.64%	10.00%	8.41%
6.30%	22.00%	2.07%
7.47%	34.00%	3.24%
6.30%	10.00%	2.07%
13.94%	27.50%	9.71%
11.35%	20.00%	7.12%
13.94%	33.00%	9.71%
4.23%	26.14%	0.00%
11.35%	26.86%	7.12%
5.01%	0.00%	0.78%
5.33%	27.00%	1.10%
5.14%	25.00%	0.91%
7.08%	35.00%	2.85%
12.64%	30.00%	8.41%
15.89%	26.86%	11.66%
10.06%	20.00%	5.83%
8.13%	30.00%	3.90%
8.13%	25.00%	3.90%
5.78%	18.00%	1.55%
19.77%	35.00%	15.54%

More than
0
50.001
55.001
57.001
60.001
62.001
64.001
66.001
68.001
69.001
72.001
74.001
76.001
80.001
82.501
85.001

<i>Rating</i>
Aaa
Aa1
Aa2
Aa3
A1
A2
A3
Baa1
Baa2
Baa3
Ba1
Ba2
Ba3
B1
B2
B3
Caa1
Caa2
Caa3
Ca
C

7.08%	22.00%	2.85%
5.78%	12.50%	1.55%
5.01%	21.00%	0.78%
4.23%	22.00%	0.00%
8.13%	27.00%	3.90%
17.18%	25.00%	12.95%
13.94%	22.50%	9.71%
12.64%	30.00%	8.41%
5.14%	20.00%	0.91%
15.89%	30.00%	11.66%
10.06%	25.00%	5.83%
4.59%	20.00%	0.36%
5.01%	25.83%	0.78%
15.89%	30.00%	11.66%
8.13%	15.00%	3.90%
4.23%	29.93%	0.00%
13.94%	25.00%	9.71%
7.08%	22.00%	2.85%
7.47%	25.00%	3.24%
5.14%	0.00%	0.91%
10.06%	30.00%	5.83%
5.01%	16.50%	0.78%
6.69%	9.00%	2.46%
5.14%	21.00%	0.91%
7.08%	30.00%	2.85%
6.69%	22.00%	2.46%
13.94%	15.00%	9.71%
5.01%	12.50%	0.78%
5.01%	0.00%	0.78%
6.30%	23.00%	2.07%
6.69%	27.81%	2.46%
8.89%	25.00%	4.66%
5.14%	29.74%	0.91%
5.01%	0.00%	0.78%
8.89%	20.00%	4.66%
6.30%	20.00%	2.07%
12.64%	30.00%	8.41%
4.87%	26.40%	0.64%
5.14%	15.00%	0.91%
12.64%	10.00%	8.41%
15.89%	20.00%	11.66%

5.78%	20.00%	1.55%
30.89%	17.00%	26.66%
4.23%	12.50%	0.00%
5.33%	15.00%	1.10%
4.23%	24.94%	0.00%
5.01%	25.00%	0.78%
8.89%	10.00%	4.66%
5.78%	24.00%	1.55%
15.89%	15.00%	11.66%
15.89%	30.00%	11.66%
5.33%	35.00%	1.10%
7.08%	15.00%	2.85%
6.69%	30.00%	2.46%
12.64%	12.00%	8.41%
10.06%	25.00%	5.83%
10.06%	15.00%	5.83%
7.08%	30.00%	2.85%
7.47%	33.00%	3.24%
17.18%	32.00%	12.95%
10.06%	32.00%	5.83%
8.89%	25.00%	4.66%
4.23%	25.80%	0.00%
4.23%	28.00%	0.00%
11.35%	30.00%	7.12%
17.18%	30.00%	12.95%
12.64%	30.00%	8.41%
4.23%	22.00%	0.00%
7.08%	15.00%	2.85%
13.94%	29.00%	9.71%
7.08%	25.00%	2.85%
11.35%	30.00%	7.12%
7.08%	10.00%	2.85%
6.30%	29.50%	2.07%
6.69%	25.00%	2.46%
5.33%	19.00%	1.10%
5.78%	31.50%	1.55%
4.87%	10.00%	0.64%
5.78%	18.94%	1.55%
7.08%	16.00%	2.85%
11.35%	28.00%	7.12%
5.01%	20.00%	0.78%

13.94%	30.00%	9.71%
8.13%	15.00%	3.90%
7.47%	18.94%	3.24%
4.23%	17.00%	0.00%
5.78%	21.00%	1.55%
5.78%	22.00%	1.55%
13.94%	30.00%	9.71%
8.13%	27.00%	3.90%
5.78%	25.00%	1.55%
19.77%	30.00%	15.54%
8.13%	20.00%	3.90%
12.64%	28.00%	8.41%
13.94%	36.00%	9.71%
11.35%	25.00%	7.12%
4.23%	20.60%	0.00%
4.23%	19.61%	0.00%
5.01%	20.00%	0.78%
12.64%	18.00%	8.41%
10.06%	30.00%	5.83%
6.30%	20.00%	2.07%
12.64%	27.00%	8.41%
8.13%	30.00%	3.90%
13.94%	15.00%	9.71%
8.89%	25.00%	4.66%
6.30%	0.00%	2.07%
12.64%	30.00%	8.41%
19.77%	18.00%	15.54%
4.87%	9.00%	0.64%
5.01%	25.00%	0.78%
4.46%	25.00%	0.23%
6.30%	25.00%	2.07%
8.89%	15.00%	4.66%
30.89%	34.00%	26.66%
8.13%	20.00%	3.90%
15.89%	30.00%	11.66%
10.06%	26.00%	5.83%
5.01%	14.68%	0.78%
11.35%	27.00%	7.12%

15.89%	27.59%	11.66%
13.94%	27.59%	9.71%
6.30%	27.13%	2.07%
17.18%	17.71%	12.95%
13.94%	14.68%	9.71%
19.77%	24.05%	15.54%
15.89%	27.59%	11.66%
8.13%	24.00%	3.90%
12.64%	20.00%	8.41%
17.18%	30.00%	12.95%
19.77%	22.00%	15.54%
8.13%	20.00%	3.90%
12.64%	25.00%	8.41%
17.18%	27.59%	12.95%
30.89%	35.00%	26.66%
19.77%	28.00%	15.54%
19.77%	27.59%	15.54%
15.89%	24.72%	11.66%

PRS Score				Country	PRS in June 2025
Less than	ERP	Default Spread	Rating	Albania	74.0
50.000	30.89%	17.50%	Below Ca	Algeria	67.0
55.000	19.77%	10.20%	Ca	Angola	63.5
57.000	17.18%	8.50%	Caa3	Argentina	69.5
60.000	15.89%	7.65%	Caa2	Armenia	66.3
62.000	13.94%	6.37%	Caa1	Australia	79.5
64.000	12.64%	5.52%	B3	Austria	77.3
66.000	11.35%	4.67%	B2	Azerbaijan	74.3
68.000	10.06%	3.83%	B1	Bahamas	77.0
69.000	8.89%	3.06%	Ba3	Bahrain	70.5
72.000	8.13%	2.56%	Ba2	Bangladesh	66.0
74.000	6.69%	1.62%	Baa2	Belarus	64.8
76.000	6.30%	1.36%	Baa1	Belgium	75.3
80.000	5.33%	0.72%	A2	Bolivia	60.5
82.500	5.01%	0.51%	Aa3	Botswana	77.5
85.000	4.59%	0.23%	Aa1	Brazil	69.5
90.000	4.23%	0.00%	Aaa	Brunei	80.8

Updated Default Spread (1/1/25)	
0.00%	
0.23%	
0.42%	
0.51%	
0.60%	
0.72%	
1.02%	
1.36%	
1.62%	
1.87%	
2.13%	
2.56%	
3.06%	
3.83%	
4.67%	
5.52%	
6.37%	
7.65%	
8.50%	
10.20%	
17.50%	

Bulgaria	71.8
Burkina Faso	60.0
Cameroon	61.0
Canada	82.5
Chile	76.0
China, Peoples	71.3
Colombia	64.5
Congo, Dem. F	56.8
Congo, Republ	63.0
Costa Rica	75.5
Cote d'Ivoire	65.0
Croatia	72.8
Cuba	56.8
Cyprus	74.8
Czech Republi	80.5
Denmark	86.8
Dominican Rep	73.5
Ecuador	69.8
Egypt	62.8
El Salvador	72.3
Estonia	69.5
Ethiopia	57.5
Finland	78.5

France	73.5
Gabon	70.3
Gambia	65.5
Germany	80.3
Ghana	69.0
Greece	70.0
Guatemala	71.8
Guinea	59.5
Guinea-Bissau	60.8
Guyana	75.5
Haiti	56.8
Honduras	67.3
Hong Kong	80.0
Hungary	72.0
Iceland	83.5
India	72.0
Indonesia	68.5
Iran	61.3
Iraq	66.3
Ireland	84.5
Israel	74.0
Italy	76.8
Jamaica	76.8
Japan	80.3
Jordan	65.0
Kazakhstan	74.8
Kenya	62.0
Korea, D.P.R.	51.0
Korea, Republi	81.0
Kuwait	80.3
Latvia	72.0
Lebanon	51.5
Liberia	57.8
Libya	71.5
Lithuania	74.0
Luxembourg	84.8
Madagascar	62.5
Malawi	56.0
Malaysia	75.3
Mali	57.8
Malta	76.5

Mexico	69.0
Moldova	64.5
Mongolia	71.3
Morocco	72.3
Mozambique	57.0
Myanmar	54.8
Namibia	70.8
Netherlands	81.5
New Zealand	78.5
Nicaragua	70.3
Niger	56.0
Nigeria	61.3
Norway	87.8
Oman	78.5
Pakistan	58.3
Panama	70.5
Papua New Gu	70.0
Paraguay	70.8
Peru	69.8
Philippines	70.0
Poland	74.0
Portugal	78.3
Qatar	80.3
Romania	67.3
Russia	70.3
Saudi Arabia	81.0
Senegal	60.0
Serbia	66.3
Sierra Leone	62.8
Singapore	86.8
Slovakia	67.0
Slovenia	76.0
Somalia	55.5
South Africa	68.5
Spain	73.5
Sri Lanka	69.0
Sudan	47.8
Suriname	67.5
Sweden	80.8
Switzerland	86.8
Syria	51.5

Taiwan	86.3
Tanzania	66.0
Thailand	68.3
Togo	60.8
Trinidad and T	76.0
Tunisia	63.0
Turkey	61.0
Uganda	62.8
Ukraine	57.0
United Arab Ei	80.5
United Kingdo	76.3
United States	68.5
Uruguay	76.5
Uzbekistan	74.8
Venezuela	57.0
Vietnam	72.8
Yemen, Repub	51.8
Zambia	66.3
Zimbabwe	58.5

Row Labels	Average of Corporate Tax Rate
Africa	27.59%
Asia	24.05%
Australia & New Zealan	26.00%
Caribbean	17.71%
Central and South Ame	27.13%

Eastern Europe & Russi	16.42%
Middle East	14.68%
North America	25.57%
Western Europe	19.94%

Updating Sequence

1. Relative Risk Worksheet
2. Sovereign Ratings
3. CDS Worksheet
4. Default Spreads
5. PRS Worksheet
6. Country GDP
7. Country tax rates
8. Regional Weighted averages